Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 — UNIT 3 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_03 · UNIT 3

1 UE26ME142A · Unit_03: UNIT 3

This unit master organizes source material for Engineering Mechanics 1, Unit_03, officially titled UNIT 3. It includes 12 video transcripts and 14 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26ME142A, Unit_03 (UNIT 3). Bucket folder present: yes. Videos: 12; slide decks: 14.

EV0000001

2 CLASS 33 Introduction to Centroid

This lesson is indexed from the source video titled CLASS 33 Introduction to Centroid. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000002 · EV0000003

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] Hello. In today's session, let's discuss about the topic distributed forces. [00:00:05.892] If you consider any external force in nature, each and every external force is a distributed force because external forces are applied mechanically on a body and they will have a definite area of contact. [00:00:22.816] However small the area might be. That's why each and every external force is a distributed force. I'm going to consider the example of this automobile wheel, which is in contact with the [00:00:37.680] pavement as shown in this figure. Now just observe this contact point. Actually it's not a point, it's the area of the tyre which is in contact with the pavement. You can see in this enlarged weave.

[00:00:53.872] So each and every particle of the taya which are in contact with the pavement will be experiencing the force from the pavement. So it is actually a distributed force, but [00:01:09.412] If you remember the numericals of any automobile example, like a car, when we consider the free body diagram of a car, we are going to consider the reaction from the pavement onto the wheel like this, as a concentrated force. [00:01:26.240] But actually, it is a distributed force as shown in this figure. We are going to assume the same force is a concentrated force because the area of the contact between the tire and the [00:01:41.264] surface or the pavement is very, very small compared to the other dimensions of the car. Example, when we consider the distance between the two wheels, the area of this contact is very small. So we are going to assume [00:01:58.756] It is a concentrated force by considering the resultant of this distributed force. Similarly, if you consider any contact forces, all the contact forces are distributed forces. Even [00:02:16.352] One more best example is when we talk about the earth's gravitational attraction, that is also a distributed force, because each and every particle on the body will be attracted by the earth's. [00:02:32.224] Gravitational force. But when we draw the free body diagram of any body, we consider that as a concentrated force, and we consider the weight W passing through the center of gravity. So we assume that as a concentrated force. [00:02:50.596] These are the few examples of the distributed forces. And in actual sense, concentrated forces doesn't exist in nature. Please remember this point. Now let's [00:03:08.324] Study about the earth's gravitational attraction. In this line, I am going to explain you how to locate the center of gravity of any three-dimensional body practically. So now let us consider this. [00:03:24.036] Body, three dimensional body of any size and shape. Let M be the mass of this body. Now I'm going to consider a point A. Through the point A, the body is suspended with the help of this card. Now [00:03:39.904] If I draw the free body diagram of this body, there are only two external forces acting on the body. The first one is tension in the card, and the one more is the self-weight of this body. Since [00:03:54.320] This body is in equilibrium. These two external forces, one is tension and one more is self-weight. These two should be equal, opposite, and [00:04:05.296] collinear. That's why this W is along the same line as that of this tension in the card, and it is downward because the tension in the card is upward. So we are going to mark this line of action of the cell weight [00:04:22.240] by drilling a small hypothetical hole on this body. This is the case when the body is suspended through point A. [00:04:31.812] Now, let us consider figure two. Now I'm going to consider a point B at the same bridget body. Now the body is suspended through point B. Again, we are going to draw the free body diagram of this body. There are only two external forces. [00:04:49.232] One is tension in the card and one more is the self rate. Now [00:04:55.108] The body is in equilibrium, so these two forces should be equal, opposite and collinear, then only they'll get get cancel each other, then the body will be in a state of equilibrium. That's how we are going to get the line of action of this.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.728] W It will be along the same line as that of the card, but in the opposite direction. And this also we are going to mark by drilling a small hypothetical hole on this rigid body. So we got the line of action for the second case. [00:05:25.412] Now we are going to repeat the experiment by suspending the same body through the point C. If you observe the line of action for the various cases, all the lines of action are concurrent at a particular point. [00:05:40.272] Point. So this is the point through which we will assume the entire weight of the body will be acting vertically downward. That point we call it as the center of gravity. [00:05:52.720] Irrespective of the orientation of the body this point will be

remains the same. So that this is how we are going to get G practically. Then how to [00:06:05.604] locate determine the center of gravity the same g mathematically. So to locate the center of gravity we are going to apply the concept of method of moments. [00:06:18.084] Here, let's consider there is a three-dimensional body of any size and shape. Let g be the center of gravity of this body. Then the coordinates of this g will be always x bar, comma, y bar, comma, z bar. [00:06:32.544] To find the value of X bar Y bar Z bar mathematically, we are going to apply the principle of moments. For this, what we are going to assume is we know each and every body will be having infinite number of [00:06:46.064] Particles. Each and every particle will be acted upon by the earth's gravitational force. So like that, I am going to consider one particle here within this body, and this will be acted upon by the earth's gravitational force, and this line of action of the force will be vertically. [00:07:03.760] Downward and the magnitude of that force is DW. Since the particle is very very small, uh [00:07:11.652] It's just like a point, and the coordinates of this point are x, y, comma, z. Now we are going to apply the principle of moments to find the value of y bar. According to this statement of [00:07:25.892] principle of moments. The moment of the resulting gravitational force W about any axis it can be X, Y, R Z, will be always equal to [00:07:37.600] Some of the moments of the gravitational forces acting on all these particles about the same axis. This is the standard statement of the principle of moments. Let's apply here the moment of the resultant gravitational force W. [00:07:56.388] Then how to find the resultant gravitational force here. The resultant gravitational force W will be equal to sum of the weights of all these particles within the body. So that will be equal to the integral of dw. Next, moment of the [00:08:15.204] This particle about x axis. The moment of this particle about x axis will be equal to force into perpendicular distance. What is force? Forces DW [00:08:28.624] And distance is y. So moment is y into dw. This is the moment of only one particle. Next we want the sum of the moments. So that will be constant integral of y into dw. So this is the RH sub the principle of moments. Now apply this. [00:08:47.524] Moment of the resulting gravitational force that is LHs is [00:08:52.452] forces w into perpendicular distances y bar. So w into y bar will be equals to integral of y into dw. Once we simplify we are going to get the value of y bar. So y bar will be equal to integral of y into dwed by [00:09:08.256] Capital W And this W is total weight of the body. [00:09:14.180] We got the value of this y bar. Similarly, if we apply the principle of moments about y axis and z axis, we are going to get the value of other two coordinates that is x bar and z bar. So once we know this, we [00:09:30.352] Now we can locate the center of gravity of this three-dimensional body mathematically. Now, this is the point through which the entire weight of the body is assumed to be acting vertically downward, irrespective of the orientation of the body. [00:09:49.824] Uh next we know the weight of the body will be always equals to mass into gravity. So w is equals to mg. [00:09:58.096] Similarly, dw will be equals to g into dm. Substitute the value of w and dw here, we are going to get the one more expression for x bar, y bar and z bar. This is independent of the gravity.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.384] This is the point through which we assume the entire mass of the body is concentrated. This is called as center of mass. X bar, y bar, and z bar gives us the coordinates of the center of mass. [00:10:29.792] This is [00:10:35.172] The center of mass and center of gravity will coincide as long as the acceleration due to gravity is constant throughout the body, and at the same time, the forces are parallel to each other. [00:10:52.260] Next, we know dense mass is equal to density into volume. And volume also

we know area into thickness. Now assume there is an object having [00:11:04.704] Uniform density throughout as well as a uniform thickness throughout. In that case in both numerator and denominator, we are going to get [00:11:14.948] Density rho as constant as well as the thickness also as constant. So both numerator and denominator, we are going to cancel rho and thickness, we are going to get the one more expression in the form of area. So x bar and y bar and z bar will give us the. [00:11:34.532] coordinates of the centroid of the lamina. And centroid is the point where we assume the entire area of the body is concentrated. [00:11:46.180] We are going this is the general formula to look at the centroid x bar integral of x into dA by total area, y bar integral of y into d a by total area. But when we are [00:11:59.588] Uh considering any elemental strip within that lamina, then instead of considering the general expression, we are going to be more specific. Uh if you look at this figure, there is a lamina something like this. Then I want to locate the centroid of this lamina. In that case, I am going to consider this elemental strip. [00:12:17.360] strip within this lamina. Now let's see be the centroid of this small strip. Then the coordinates of the centroid of this [00:12:26.980] Point R XC, Y C. So the formula is going to be X coordinate is X C and Y coordinate is Y C. The formula is going to be X bar is equals to integral of X C into DA by total area. Y bar is integral of [00:12:43.376] Yc into DA by total area. So this formula we are going to make use when we are going to consider a small strip within the lamina. [00:12:54.884] Next, there is one more formula. Okay, before that, the numerators is distance into area, distance into area. That's why this card is a first movement of [00:13:07.680] Area. We are going to make use of the concept of first moment of area to locate the center of a given lamina. [00:13:15.460] Next, there is one more formula. X bar is equals to summation of AI Xi by total area. Y bar is summation of AI Yi by total area. We are going to make use of [00:13:28.164] This particular formula in the composite areas. Composite area is nothing but any lamina figure which will be having some basic figures like combination of basic figures like. [00:13:40.964] triangle, rectangle, square, circle, semicircle like that. So in that case we are going to make use of this particular formula. And we can write it as a1 x1 plus a2 x2 plus so on [00:13:58.564] Plus a nxn by total area. That's how we are going to expand this particular formula. [00:14:06.308] Next is axis of symmetry. Axis of symmetry is any line at the axis which divides the given lamina or a figure at any three-dimensional body into two equality. [00:14:19.856] halves. So that one will be the mirror image of the other. You can look at this. Here there is a right circular cone. If you draw [00:14:29.760] vertical line passing through the apex here through this point. Now this left of the section is the mirror image of the right of the section. So the the center of gravity R centroid will be on this because [00:14:44.896] The moment of left of this will be equals to moment of right of this, these two will get cancel each other. So it is going to be center of gravity or center of mass is going to be here. It is applicable even for laminus as well.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.292] This is one more figure. If we draw a line passing through this, this right of the section is the mirror image of the left of the section. So center of gravity will be on this axis. Similarly, here also. [00:15:15.232] So remember in numericals whenever the lamina is given is having the axis of symmetry, then the centroid will be on the axis of symmetry. [00:15:27.108] Next, let's find the y coordinate of central of a triangle of base B and a height H from the method of integration. So I'm going to consider a triangle. You can consider either a right angle triangle or just any triangle, I'm going to consider a triangle like this. [00:15:46.308] Let us consider a triangle A, B, C, Let B be the base of this triangle, let H be the

height of the triangle. Then capital A is the area of this triangle. [00:16:06.468] Let that be equals to A is equals to B into H. Next, I need to find the Y coordinate of the centrate. So I'm going to consider [00:16:19.300] an element. This is before that this is the reference axis, let it be x axis. I'm going to consider one smaller area within this lamina. [00:16:33.348] This is the lamina, this is the element I am going to consider, which is at a distance of y with respect to reference axis. Then [00:16:43.652] The width, the depth of this element need to be dy. This element can be anywhere from this to this point T. So y is varying from 0 to H. And the depth of the element dy is also a variable. It can be [00:17:00.996] Anything from 0 to H. Next, the width of the element is also a variable because when the value of y is equals to 0, the width of this element will be equal to b. As the value of y increases, the width of the element [00:17:17.232] decreases. So it is a variable. That's why I'm going to call it as x. Now we know the formula to find y bar. To find the y bar we have a formula y bar is equal to [00:17:34.532] integral of yc into d a by total area e. Now a is total area that is half into sorry half into base into height then [00:17:49.668] DA is area of this element. The area of the element, since it is in the form of rectangle, almost like a rectangle, d A will be equal to B into H. Here B is X into depth of the element is DY. Next. [00:18:05.872] Y C we should find. The what is YC? This shaded area will have a centrite C [00:18:14.116] Consider assume this is a centroid C. Now the distance of the centroid with respect to reference line that is x-axis, this is Yc. When Dy is very very small, this Yc is approximately equal to Y. So YC can be written as Yc is equal to Y itself. [00:18:33.956] So I know the value of y c d a then a substitute here. This will be equal to integrate integral of y. Yc is y into d is x into dy divided by total area capital E. Next. [00:18:52.548] X is a variable and y is also variable. So there are two variables. So we have to express this x in the form of constant bh as well as a variable y. Because we have the term dy here. So we have to replace this x. To replace the x, I am going to consider the same triangle here. [00:19:14.244] Now, when the value of this d y is very very small, the element will be just like a 9. This is x. And this we know b, and this we know y. And the total height of the rectangle is h. This is also known to us. [00:19:33.860] Next. We can use the concept of similar triangles because these two are parallel to each other. By using the concept of similar triangles, [00:19:43.972] Base of the smaller triangle that is x by base of the bigger triangle that is b will be equals to height of the smaller triangle. The height of the smaller triangle is only this. This will be equals to [00:19:56.784] Total h minus y. So h minus y divided by height of the bigger triangle that is h. Now I can write x as x is equal to b into h minus y divided by h. This is

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:16.196] x in the form of constants b and h and a variable y. So substitute here in this expression. So y bar is equals to we are going to get it as y bar is equals to integral. I'll take 1 by s here integral of [00:20:34.052] Y into X is B into H minus Y by H into D Y. Next. [00:20:47.204] This is 1 by A. As I said before, B and H are constants, so you can take them out. B and H. Y is varying from 0 to H. Then in uh Y into H is H into 1. [00:21:03.172] Minus y square into d y d by outside the bracket. Simplify this, this will be equals to area is b into h by half into base into height. So this will be bh into b by h [00:21:22.244] into this will be integration h is constant integration of y is y square by two so h into y square by two minus integration of y square is y cube by three zero to h. [00:21:43.076] This is y cube by three zero two h. Next, this will be equals to [00:21:54.084] Two H to by uh B B will get canceled here H square into [00:22:03.108] Once we substitute the value of y as h, so h square minus of h cube by 3, and now the next term is going to be 0. So 2 by h square into h cube by 2

minus h cube by 3. [00:22:24.132] This is going to be H cube by six. So two by H squared into H cube by six. This will be equal to two three. [00:22:41.572] One by three H. So Y bar is equals to H bar three. This is the Y coordinate of the central of a triangle with respect to bottom plus member. This is a triangle if we consider. [00:22:58.468] This is x-axis passing to the bottom most member this is where the centroid will be, and this is y bar that is h by three. Similarly, with respect to up x, this will be equal to total h minus h by 3 will be equal to 2 h by 3. [00:23:18.756] So remember if the axis is passing to the bottommost member of a triangle, then the y coordinate of the centroid is h by 3. If the axis is passing to the topmost member, x-axis is passing to the topmost member at the apex of the triangle. [00:23:36.004] In that case, the y-coordinate of the central is 2h by 3. That's it for today's class. Thank you.

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Full transcript: 3368 normalized words, 98 timestamped segments, 1425.152 seconds duration. Low-confidence segments flagged: 0.

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3 CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina

This lesson is indexed from the source video titled CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000022 · EV0000023

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.036] Hello, in today's class I am going to deal with how to locate the centroid for a sector and semicircle and few basic formulas before starting with the numericals. Let's get started. [00:00:15.364] Centred of an area of a circular sector. For this, I am going to consider a sector which is symmetrical about. [00:00:24.100] X axis. Look at this sector. This sector is symmetrical about x axis. The purpose for considering this sector symmetrical about x-axis is [00:00:35.972] The center of this sector will be an x-axis, so its y coordinate will be 0. So no need of finding the y coordinate, we can write it as 0. And in one more way, you can consider a sector which will be symmetrical about [00:00:50.624] Y-axis. In that case, the x coordinate of the centroid will be zero. We need to find only the value of y-coordinate of the centroid. That's the purpose of considering the figure this way. [00:01:02.976] Next, let O be the center of this sector, let capital R be the radius of the sector, then the total angle of this sector, the inscribe the angle let it be 2 alpha. Since the total angle is 2 alpha, this angle [00:01:18.640] with respect to x axis will be alpha and the remaining angle will also be alpha. Next we need to find only the y coordinate of the [00:01:27.984] Sorry, x coordinate of the centroid. To find the x coordinate of the centroid, we have a formula x bar is equal to integral of x into dA divided by total area. To apply this formula, we have to consider an element within this sector. I'm going to consider a element [00:01:45.616] Which is varying along this radial line. It can be anywhere from 0

to capital R. Since it's a variable, I'm going to call it as. [00:01:54.468] The distance of that element with respect to the center wall, let it be R0. Then the radial width is a variable, let it be DR0. Then the [00:02:06.372] Angle made by that element with respect to x axis, let it be theta. Then the inscriptor the angle of that element let it be d theta. D theta will be varying, uh it's a variable, and theta will be varying from minus alpha to [00:02:21.456] Plus alpha and R0 is varying from 0 to capital R. This shaded region is the area which we have considered for the using the formula x bar is equals to integral of x c into d divided by total area. Next, what is [00:02:40.004] XC and what is DA in this? DA is nothing but area of this element. This element is almost like a rectangle. So DA will be the area of a rectangle that is B into H. What is B here? [00:02:54.512] That is the width of the element that is equal to DR0. And what is H? H is nothing but the arc length that will be equal to R0 into inscribed angle D theta. So that is the value of dA B into H. [00:03:10.352] Next, we need to find the value of x is c. X c is the centroidal distance of the element with respect to y axis, that x coordinate of the centroid of that smaller element with respect to. [00:03:25.968] Y-axis. So here this is XC, just drop it perpendicular from this element onto the x-axis. This is the x coordinate of that element, that itself is x. Then how to find this xc? [00:03:38.128] When we need to find this XC, we are going to make one assumption that d theta is very very small. When d theta is very very small, theta plus d theta will be approximately equals to theta, and in that case, this element will be just like a [00:03:53.632] Point. So X will be the base of a right angle triangle and R0 will be the hypotenuse of a right angle triangle. Then in that case, how to find X? That will be equal to R0 into cos theta. We are going to use the cosine rule there. [00:04:11.940] Once we know the value of X C and DA, now we should know the value of capital A. Capital A is the area of the sector. Area of the sector will be equal to half into R squared into inscribed angle, that is [00:04:27.744] to alpha, once we simplify that, we are going to get the value of A as R square into alpha. If you don't remember this formula, just integrate this BA. [00:04:39.168] And R will be varying from 0 to capital R, and theta is varying from minus alpha to plus alpha. If we integrate this by applying the limits, then also we are going to get the value of area of sector. Once we know every term in the equation one. [00:04:56.176] Substituting the equation one x is r naught cos theta, then d a is r naught d theta into dr0 by capital A. Simplify that.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.716] This apply the limits as I said before R0 is varying from 0 to capital R and R theta is varying from minus alpha to plus alpha. So simplify this and now substitute the value of A. [00:05:23.908] Once we substitute the value of A, we are going to get the X coordinate of the centroid of the sector. That will be equal to 2r sin alpha by 3 alpha. And to use this formula, you have to remember the alpha should be in terms of. [00:05:40.368] radiance and try to understand this formula in this way. The distance of the centroid with respect to center on the axis of symmetry that is 2 r sin alpha by [00:05:53.728] Alpha because always the centroid will be on the axis of symmetry for a sector from center to that point centroid on the axis of symmetry is 2 R sine alpha by 3 alpha. [00:06:10.756] So x bar is q r sine alpha by three alpha and y bar is zero. Similarly, if the sector is symmetrical about y axis, then the x coordinate of that [00:06:21.152] centroid will be that is x bar will be 0 and y bar will be 2 r sin alpha by 3 alpha. Again alpha will be radiance and r is the radius of the sector. Now let's locate the centroid of area of a semicircle. [00:06:37.392] For this, I am going to consider a semicircle which is symmetrical about y-axis. If the semicircle is symmetrical about y-axis, then the

x coordinate of the centroid will be 0 because the centroid will be on the axis of symmetry. That's the reason why I have considered the figure like this. If you consider [00:06:55.088] A semicircle which is symmetrical about x-axis, then y-coordinate of the centroid will be 0. You need to find only x-coordinate. Right now, here we need to find only the y coordinate of the centroid of the semicircle. We know. [00:07:10.176] Y bar will be equals to integral of yc into da divided by total area. Then how to find yc and dA and total area? For this we have to consider one element. I'm going to consider a element which is varying along this radial direction. [00:07:27.408] It can be anywhere from this zero to capital R. It is varying in this way. Next. [00:07:34.916] Let the inclination of that element with respect to the reference axis, that is x-axis, let it be theta. Then the inscribed angle let it be d theta. Then the width of the element is also a variable along radial direction. [00:07:50.320] That's why we are going to call it as DR naught. Next, in this formula, DA stands for area of this element, that is shaded area. [00:08:00.592] When d theta is comparatively small, this element will be almost almost like a rectangle. So dA will be the area of a rectangle that is B into H. [00:08:10.832] Here B is DR0, that is width of the element. Next, we need to find the depth of the element that will be equal to arc length, that is r0 into d theta. In this way we can find the value of dA. Next, what is yc? [00:08:27.056] Yc is nothing but Y coordinate of the centroid of the element. So in this when d theta is very very small, this element will be almost like a [00:08:38.656] point. In that case, the y distance of that element will be equal to r naught sin theta. Because here we are going to get a right angle triangle like this. [00:08:52.164] That's why yc will be equal to R0 into sine theta. Then substitute the value of BA, then Y C in the equation 1. Then capital A is area of the semicircle that is I R square by 2. Once we substitute [00:09:10.148] Uh in the equation one, we are going to get it as integral of r naught sin theta. [00:09:16.868] into r0 d theta into dr0 divided by total area. As I said before, r naught is varying from 0 to capital R and theta is varying from 0 to pi. It is a double integration. So integral of 0 to pi, integral of 0 to r, r0 [00:09:35.264] square sine theta into d theta into d R naught. Once we integrate R naught square into with respect to limits from 0 to R it is going to be 2 R cube by 3. At the same time integral of sin theta will be cos v [00:09:52.944] Then simplify this again. So finally we are going to get y bar as 4 r by 3 pi. This is the y coordinate of the centroid, and here the centroid will be on the y axis.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:08.708] Before starting with the numericals, uh let's know about few basic formulas which we need for solving the numericals in Centroid. [00:10:17.124] The first one is let's consider there is a rectangle of base B and height H. Now we need to locate the centroid. For this case, in the first case, where will be the centroid? The centroid will be somewhere here. [00:10:31.808] Then what is X and Y coordinates of the centroid? X bar will be this distance and y bar will be this distance. Since the point is in the first quadrant, both x bar and y bar will be positive. So x bar is b by two and y bar is h by two. Next. [00:10:50.660] Look at this, there is a same rectangle of base V and the height, H. The centroid is here. Since the centroid is in the second quadrant, the x coordinate of the centroid will be negative, and the y coordinate of the centroid will be negative. [00:11:06.608] Positive. So x bar will be minus b by 2 and y bar will be plus h by 2. Next, let's consider a rectangle. Now the rectangle, entire rectangle is in a third quadrant and the centroid of this will be somewhere here. [00:11:22.928] Since the centroid is in the third quadrant, both x and y coordinates of the centroid will be negative. That's why it is minus b by two and y by is also minus h by two. This is how the values

will change depending on the position of the centroid. [00:11:42.436] Next, let us consider there is a triangle of base B and height H. Where will be the centroid here? From the derivation, we know centroid will be at a distance of H by 3 with respect to the base of the triangle. [00:11:59.216] Then what about with respect to apex of the triangle? With respect to apex it will be 2 h by 3. And here, since the x-axis is passing through the base of the triangle, y coordinate of the centroid will be h by 3. [00:12:15.428] Now, let's consider there is a right angle triangle of base B and height H. Now, where will be the centrite? Somewhere here. [00:12:23.492] So centroid will be here. Then what is X bar from the derivation we know X bar is this distance from ear to ear distance. That will be equals to base by 3. And what is Y bar? Y bar is this distance, y coordinate of the centroid. [00:12:40.640] That will be h by 3 because x is passing through the base of the triangle. So y bar is h by 3. And one more, since point is in the first quadrant, both x bar and y bar will be positive. Let's consider one more triangle, which is like this, which is [00:12:59.876] in the third quadrant and uh b is the base of the triangle and h is the height of a triangle. Where will be the centroid? Somewhere here. Then what is X bar? X bar is the X coordinate of this C. [00:13:12.992] This is x bar. Since the point is in the third quadrant, x bar will be negative. Then what will be that value? With respect to this point, it will be 2b by 3. [00:13:24.740] Whereas with respect to this it is B by three. Since y axis is passing to this point it will be two b by three minus two b by three. Then y bar will be. [00:13:35.808] This distance. Since the x axis is passing through the ax of the triangle, then the y coordinate of central will be two h by three. Since the point is in the third quadrant, y bar is minus two h by three. [00:13:50.500] Let's consider one more triangle like this. Let B be the base and H be the height. Where will be the central? Somewhere here. Then what is X bar? X bar is this distance from here to here. What will be that distance? [00:14:05.696] 2b by 3 because from this side it is b by 3, from this side it is 2b by 3. And at the same time, since the centroid is in the second quadrant, x bar will be negative. So it is minus 2b by 3. Next, what is y bar? [00:14:22.512] Why by the y coordinate of the centroid? That will be with respect to base, since x axis is coinciding with the base, it will be h by three. And the point is in the second quadrant, so y coordinate will be positive. So h by three. [00:14:37.540] Next is circle. Let us consider there is a circle of radius r. For circle, we know the centroid will be at the center of the circle. Now, this will be the centroid. Then what is X bar? X bar is this distance that will be equal to. [00:14:56.676] R then what is y bar? That is y coordinate of this point C that will be also equal to R. And remember point is in first quadrant, so both are positive. Next consider a circle.

EV0000030 · EV0000031 · EV0000032 · EV0000033

Transcript block 4 (source interval 000900-001199 seconds): [00:15:11.620] Of radius R, and where is the center of the circle? That is coinciding with the origin here, both X and Y. So both the origin as well as the center of the circle are coinciding. Then where will be the center here itself? So x bar and y bar will be equal to 0. [00:15:32.132] Now let's consider a circle like this. If you look at the circle, the circle is symmetrical about this x-axis and this is the center of the circle. [00:15:42.244] This is on the x-axis. Then what is x bar? X bar is this distance that will be equal to radius. Since the point is [00:15:53.188] On the negative side of the x axis, x bar will be minus minus r. Then what is y bar? Since the point is on the x axis, y bar is equal to 0. Next consider a circle like this. Now, where will be the centroid? [00:16:12.996] center here itself, then what is x bar? Since the point is on the y axis, x bar will be 0. Then what is y bar? Y bar is this distance [00:16:26.148] Since it is on the negative side of the y-axis, y bar will be minus. And what is this distance? Radius itself. So y bar is equal to minus r. Now [00:16:39.556] Let's know about the

semicircle. Let us consider there is a semicircle. Let O be the center of the semicircle. Now [00:16:49.380] Whenever we come across semicircle, remember always we have to start with the axis of symmetry. Where will be the axis of symmetry? Now in this case, just draw a vertical line passing through. This is the axis of symmetry. [00:17:03.940] Then where will be the centroid? On the axis of symmetry, somewhere here like this. Then what is X bar? X coordinate of the centroid. Since the axis of symmetry and y axis are parallel to each other, x bar will be this distance, that is r. And what is y bar? [00:17:21.200] We know the distance of the centroid with respect to centre and the axis of symmetry is 4 r by 3 pi. So y bar is 4 r by 3 pi. [00:17:32.228] Let's consider one more semicircle, which is symmetrical about x-axis. Then look at this, the origin as well as the center of the semicircle are coinciding with each other. Now, where will be the centroid? Centroid will be on the axis of symmetry. [00:17:49.488] So this is the axis of symmetry and this is the centroid. The OC distance is 4R by 3 pi, that we know that is X bar. 4 R by 3 pi. Then what is Y bar? Since the centroid is on the x axis, y bar is equal to 0. [00:18:09.348] Next consider a semicircle something like this again here also it is symmetrical about x axis. Whenever you come across a semicircle, first thing is you have to locate the center of the semicircle. This O is the center of the semicircle. [00:18:24.100] Then draw the axis of symmetry, then the centroid will be on the axis of symmetry. Then the OC distance is 4R by 3 pi, that is from the derivation. But what is X bar? X bar is this distance from the origin till C [00:18:39.728] Then what is this? This will be equal to total that is radius minus O C distance that is 4R by 3 pi. And since the point is in the second quadrant, x bar will be negative. [00:18:53.956] Next, y bar. Y bar, since the point is on the x-axis, y bar will be zero. Next consider a semicircle, something like this. [00:19:05.764] Now, first locate the center of the semicircle. This is the center. Then draw the axis of symmetry. This is the axis of symmetry. Next, where will be the center? Somewhere here. We know from the derivation, this distance OC is 4r by 3 pi. [00:19:23.312] But what is X bar? This is X bar that is equals to 4R radius. Then what is Y bar? Y bar is this distance that will be equals to total R minus 4R by 3 pi. Since the point is in the [00:19:39.812] Fourth quadrant y coordinate will be negative. This is now you have to consider the formulas when we come across semicircle. Next one is sector. We know in this figure. [00:19:55.044] The sector is symmetrical about y-axis. So x-coordinate of this centroid will be 0 because the centroid is on the y-axis itself. And what is y-coordinate of the centroid? We know from the derivation of centroid, the distance of the centroid with respect to center on the axis of symmetries.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:15.268] Two by third of the R sin alpha by alpha. That's why y bar is this. [00:20:22.084] Next, consider a sector which is symmetrical about x-axis. In this case, where will be the centroid? On the axis of symmetry. So this is where the centroid is. Then what is X bar? [00:20:35.780] This distance is x bar, then what is that? That is equal to 2 r sine alpha by 3 alpha. Then what is y bar? Since the point is on the x axis, y bar is equal to 0. That's it for today's class. Thank you.

EV0000038 · EV0000039 · EV0000040 · EV0000041

Full transcript: 3360 normalized words, 86 timestamped segments, 1254.570667 seconds duration. Low-confidence segments flagged: 0.

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4 CLASS 36 Centroid_ Numerical

This lesson is indexed from the source video titled CLASS 36 Centroid_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000042 · EV0000043

Graphs and diagrams

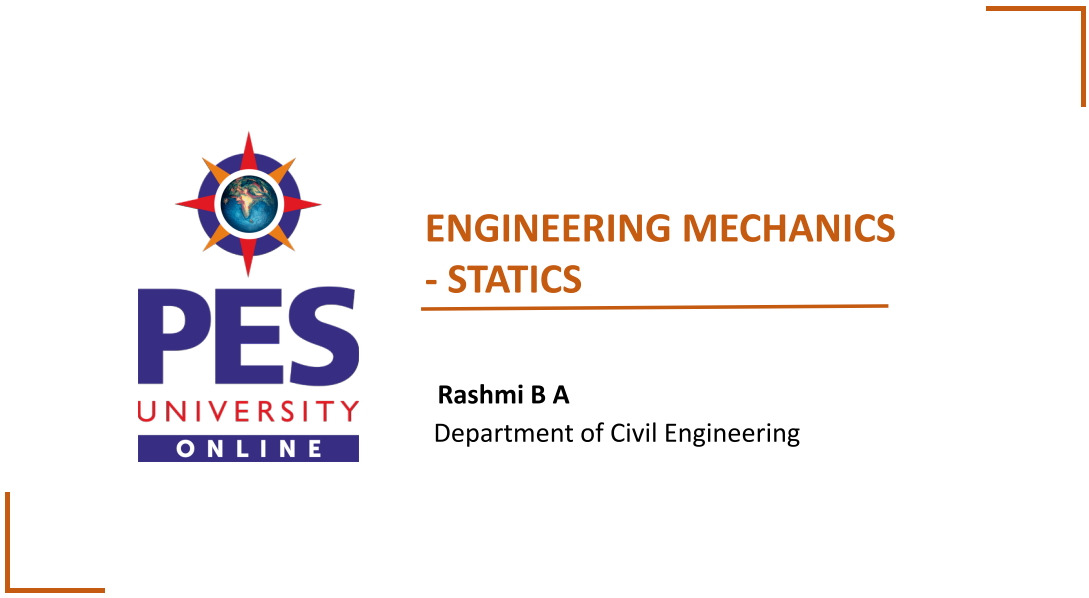


Figure G000049. UE20CV101 UNIT 4 SESSION 3 · source slide page 1.

EV0000313

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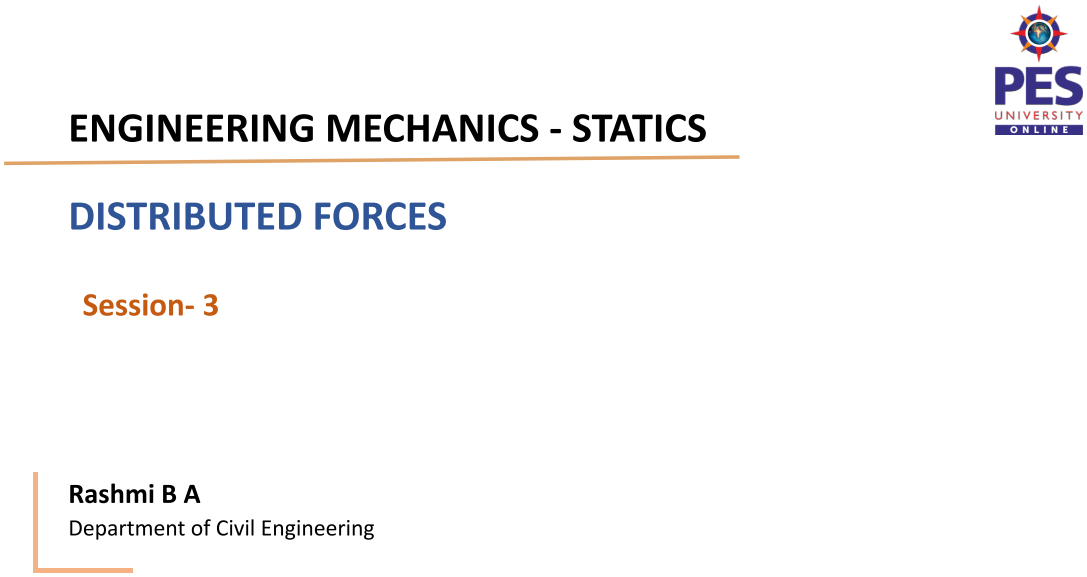


Figure G000050. UE20CV101 UNIT 4 SESSION 3 · source slide page 2.

EV0000316

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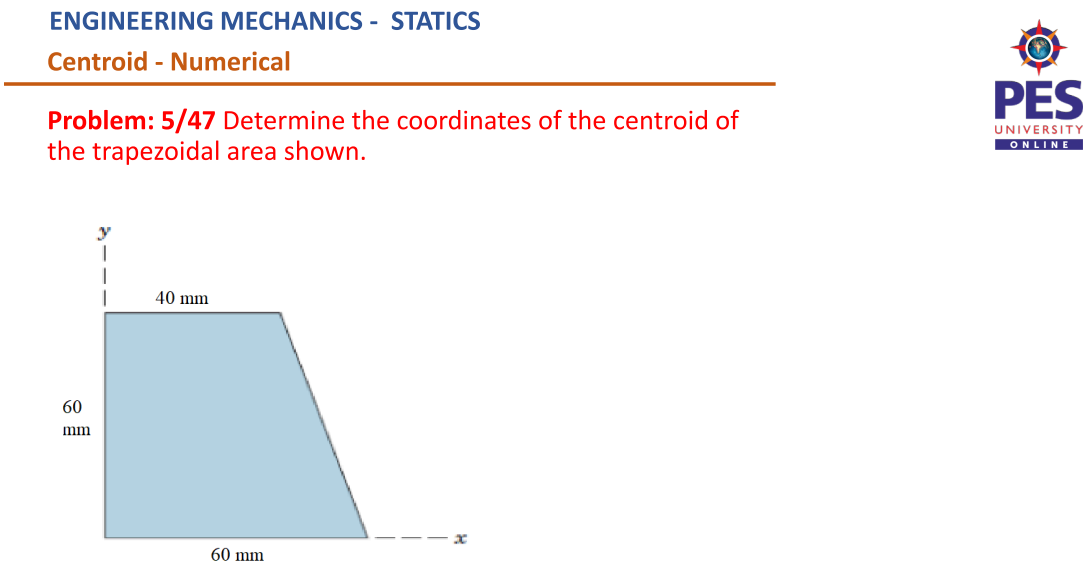


Figure G000051. UE20CV101 UNIT 4 SESSION 3 · source slide page 3.

EV0000319

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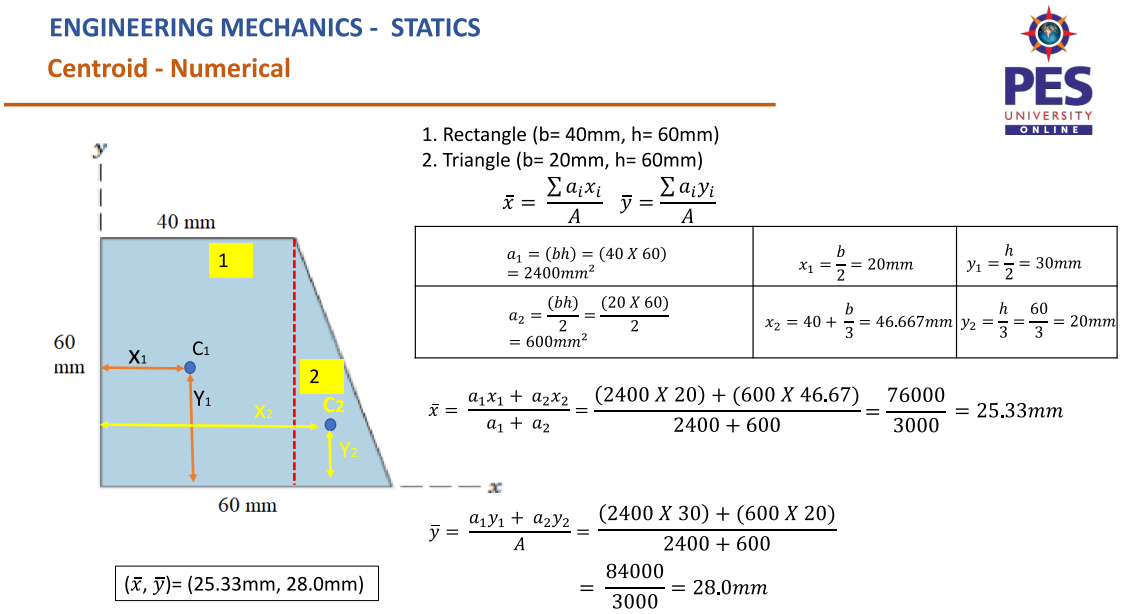


Figure G000052. UE20CV101 UNIT 4 SESSION 3 · source slide page 4.

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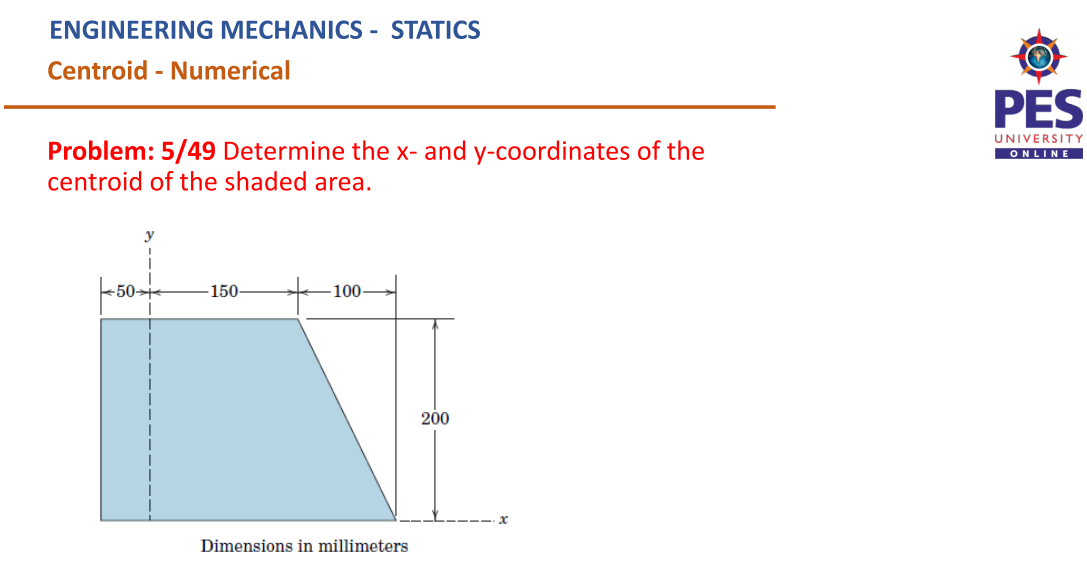


Figure G000053. UE20CV101 UNIT 4 SESSION 3 · source slide page 5.

EV0000325

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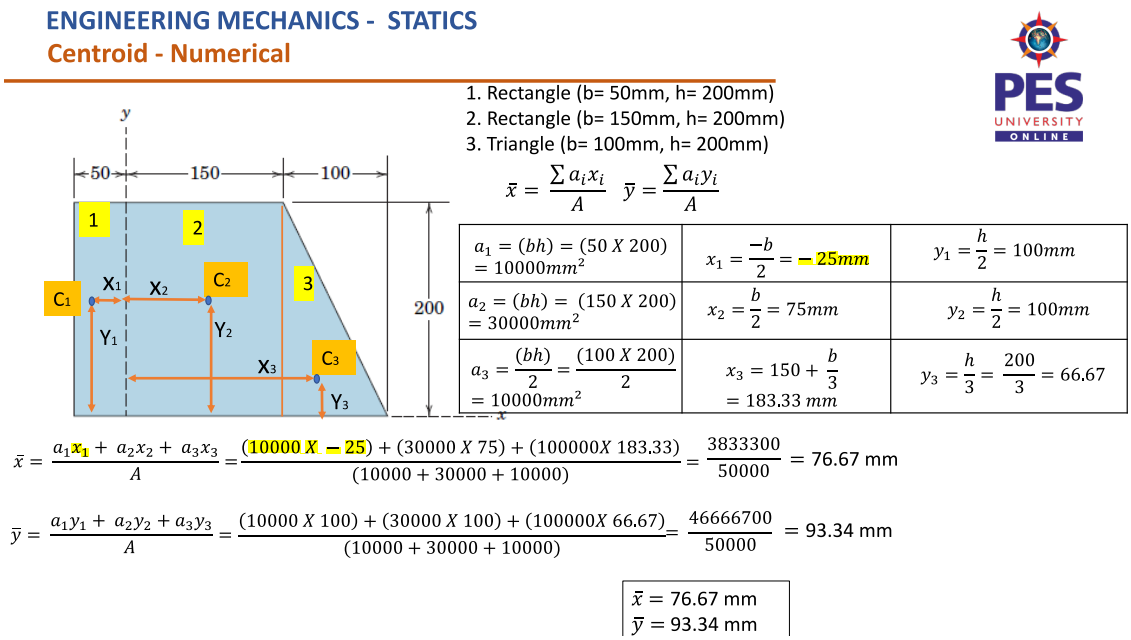


Figure G000054. UE20CV101 UNIT 4 SESSION 3 · source slide page 6.

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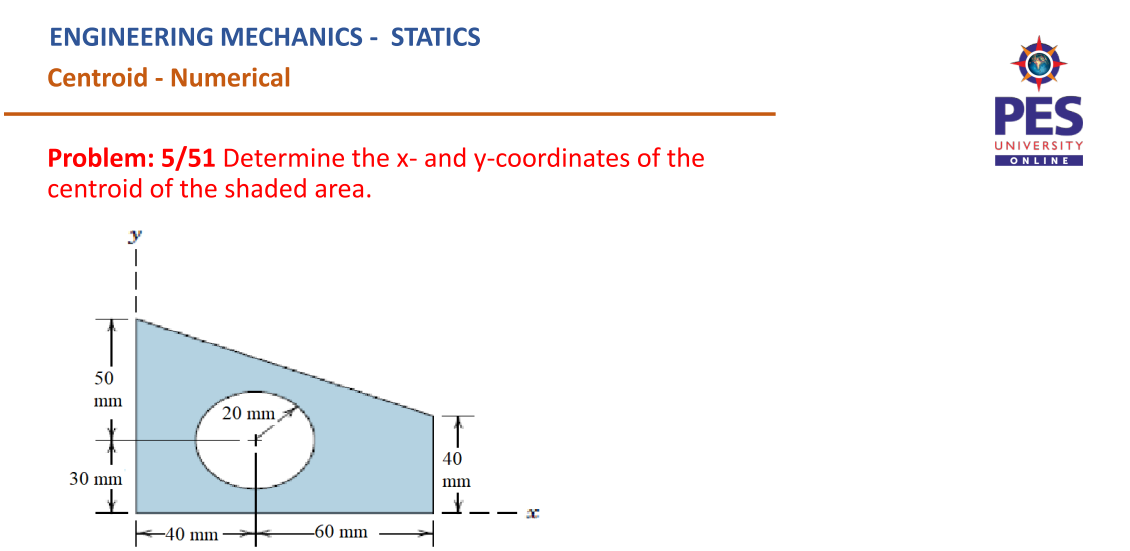


Figure G000055. UE20CV101 UNIT 4 SESSION 3 · source slide page 7.

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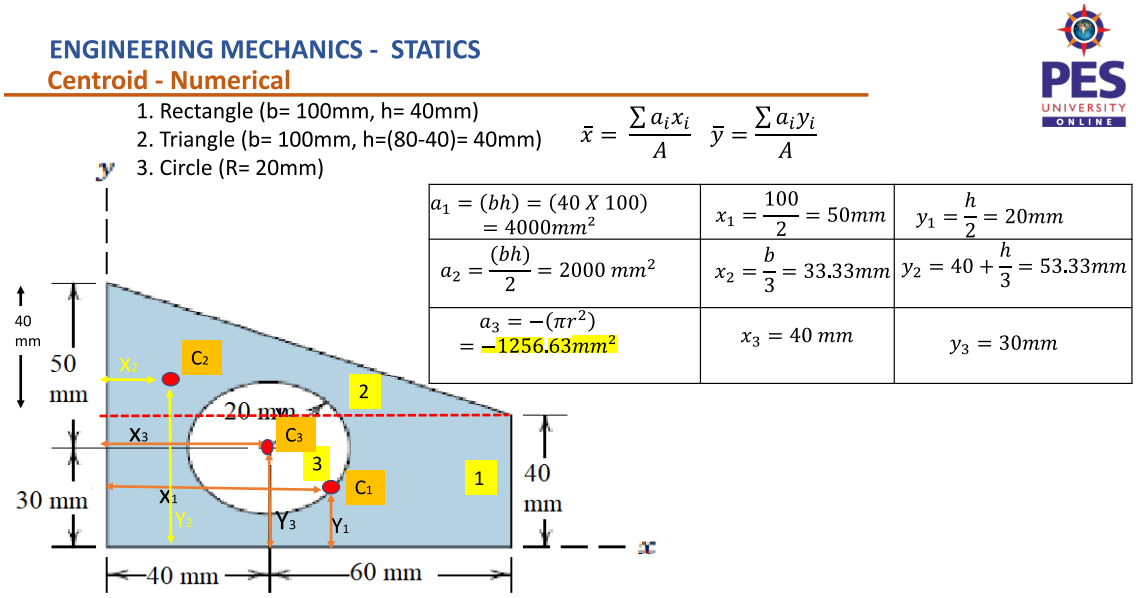


Figure G000056. UE20CV101 UNIT 4 SESSION 3 · source slide page 8.

EV0000334

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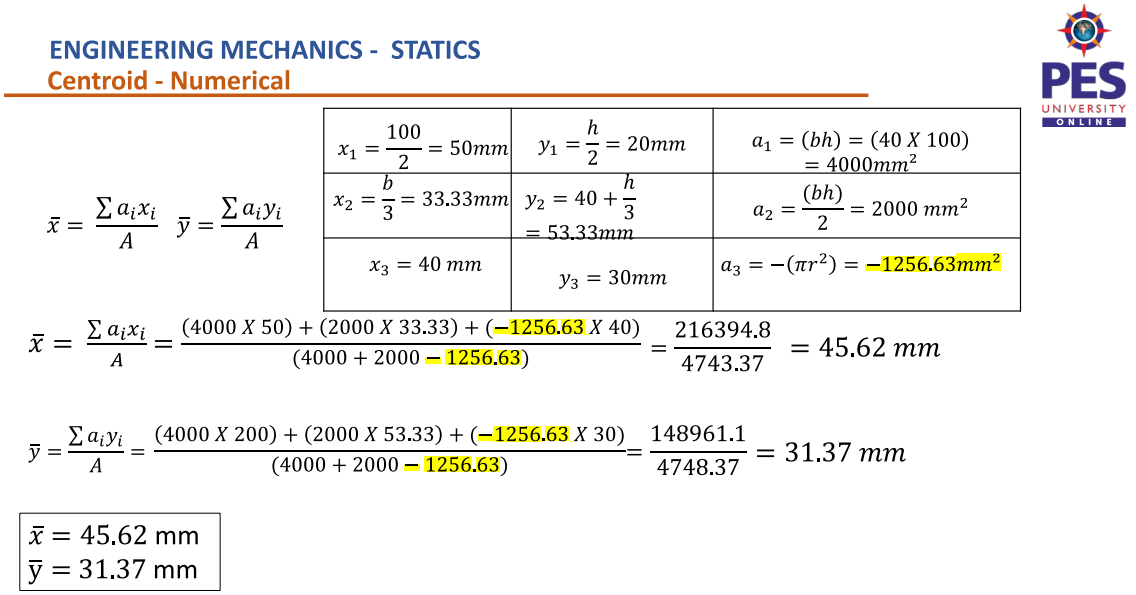


Figure G000057. UE20CV101 UNIT 4 SESSION 3 · source slide page 9.

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Figure G000058. UE20CV101 UNIT 4 SESSION 3 · source slide page 10.

EV0000340

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000312 · EV0000313 · EV0000314
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 3 Rashmi B A Department of Civil Engineering EV0000315 · EV0000316 · EV0000317
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical Problem: 5/47 Determine the coordinates of the centroid of the trapezoidal area shown. EV0000318 · EV0000319 · EV0000320
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical 1. Rectangle (b= 40mm, h= 60mm) 2. Triangle (b= 20mm, h= 60mm) σ $aixi$ σ $aiyi$ ȳx= ϖy= A A $a1$ = bh= 40 X60 b h = = 30mm 1 x1 = = 20mm $y1$ = 2400mm^2 2 2 (bh) (20 X60) b h 60 $a2$ = = 2 2 $x2$ = 40 + = 46.667mm $y2$ = = = 20mm 3 3 3 = 600mm^2 x1 C1 2 Y1 $a1x1$ + $a2x2$ 2400 X20 + (600 X46.67) 76000 ȳx= = = = 25.33mm x2 C2 $a1$ + $a2$ 2400 + 600 3000 Y2 2400 X30 + (600 X20) $a1y1$ + $a2y2$ ϖy= = A 2400 + 600 84000 (ȳx, ϖy)= (25.33mm, 28.0mm) = = 28.0mm 3000 EV0000321 · EV0000322 · EV0000323
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical Problem: 5/49 Determine the x- and y-coordinates of the centroid of the shaded area. EV0000324 · EV0000325 · EV0000326

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical 1. Rectangle (b= 50mm, h= 200mm) 2. Rectangle (b= 150mm, h= 200mm) 3. Triangle (b= 100mm, h= 200mm) σ $aixi$ σ $aiyi$ ıьx= ϖy= A A 1 2 h $a1$ = bh= 50 $X200$ $-b$ $y1$ = = 100mm $x1$ = = $-25mm$ 2 = 10000mm^2 2 x1 C2 3 x2 b h C1 $a2$ = bh= 150 $X200$ $x2$ = = 75mm $y2$ = = 100mm = 30000mm2 2 2 Y2 Y1 C3 (bh) (100 $X200$) b h 200 x3 $a3$ = = $x3$ = 150 + $y3$ = = = 66.67 2 2 3 3 3 Y3 = 10000mm^2 = 183.33 mm $a1x1$ + $a2x2$ + $a3x3$ 10000 $X-25$ + 30000 $X75$ + 100000X183.33 3833300 ıьx= = = = 76.67 mm A (10000 + 30000 + 10000) 50000 $a1y1$ + $a2y2$ + $a3y3$ 10000 $X100$ + 30000 $X100$ + 100000X66.67 46666700ϖy= = = = 93.34 mm A (10000 + 30000 + 10000) 50000 ıьx= 76.67 mm ϖy= 93.34 mm EV0000327 · EV0000328 · EV0000329
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical Problem: 5/51 Determine the x- and y-coordinates of the centroid of the shaded area. EV0000330 · EV0000331 · EV0000332
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical 1. Rectangle (b= 100mm, h= 40mm) σ $aixi$ σ $aiyi$ ıьx= ϖy= 2. Triangle (b= 100mm, h=(80-40)= 40mm) A A 3. Circle (R= 20mm) 100 h $a1$ = bh= 40 $X100$ $x1$ = = 50mm $y1$ = = 20mm = 4000mm^2 2 2 h (bh) b = 40 + = 53.33mm $a2$ = = 2000 mm2 $x2$ = = 33.33mm $y2$ 3 2 3 40 $a3$ = $-\pi r2$ mm $x3$ = 40 mm = $-1256.63mm^2$ $y3$ = 30mm x2 C2 2 Y1 x3 C3 3 1 C1 x1 Y2 Y3 Y1 EV0000333 · EV0000334 · EV0000335
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numerical h 100 $a1$ = bh= 40 $X100$ $y1$ = = 20mm $x1$ = = 50mm 2 2 = 4000mm^2 b h (bh) $x2$ = = 33.33mm $y2$ = 40 + = = 2000 mm2 σ $aixi$ σ $aiyi$ 3 3 $a2$ ıьx= ϖy= 2 = 53.33mm A A $x3$ = 40 mm $a3$ = $-\pi r2$ = $-1256.63mm^2$ $y3$ = 30mm σ $aixi$ 4000 $X50$ + 2000 $X33.33$ + (-1256.63 $X40$) 216394.8 ıьx= = = = 45.62 mm A (4000 + 2000 -1256.63) 4743.37 σ $aiyi$ 4000 $X200$ + 2000 $X53.33$ + (-1256.63 $X30$) 148961.1 ϖy= = = = 31.37 mm A (4000 + 2000 -1256.63) 4748.37 ıьx= 45.62 mm ϖy = 31.37 mm EV0000336 · EV0000337 · EV0000338
10. Source slide page 10 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000339 · EV0000340 · EV0000341

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.836] Hello. In today's class let's solve few problems in centroid. [00:00:06.820] This is the first question in today's class. Problem 5/47. Determine the coordinates of the centroid of the trapezoidal area as shown in the figure. This is the given lamina. We have been asked to locate the centroid of this lamina. [00:00:22.544] Now remember we don't have a formula for trapezium like X bar and Y bar formula. So what we can do here is we can use the method of approximation and we can locate the centroid of this lamina. To use the method of approximation, we know the formula x bar is equals to summation of EI Xi by. [00:00:40.208] Total area y by summation of a i yi divided by total area. To use that formula, we have to divide this into elemental areas. [00:00:48.864] Now let's divide this into elementary areas. How to divide this? Just draw a vertical line passing through this point. We are going to get one rectangle and one triangle. Together we will get a trapezium. [00:01:02.736] So now instead of one trapezium we have a rectangle and a triangle. The first one is rectangle. And what will uh [00:01:12.656] Let's know about the dimensions of this rectangle. The width of the

rectangle is 14 and the depth of the rectangle is 60. Take that. Then the second elemental area is. [00:01:24.080] triangle here base of the triangle will be total 60 minus 40. So we will get it as 20. After that the height of the triangle will be the height of the rectangle that is 60 ml. [00:01:38.160] Next, we know the formulas x bar and y bar. Remember, since we have your two elemental areas, the formula is going to be a1 x1 plus a2 x2 divided by total area that is a1 plus a2. Similarly, y bar formula will be a1 y1 plus a2 y2 divided. [00:01:55.792] it by total area. Now let's find out what is a1, what is x1, what is y1, next a2, x2, y2. That's what we have to find it out to find the value of x bar and y bar. Let's start with the first term a1. [00:02:13.860] A1 is nothing but area of the first elemental area. For time being, forget about this triangle. Just consider this rectangle. So area of the rectangle is A1. That will be equal to B into H. What is B from here? 40. What is H? 60. Substitute. [00:02:31.632] Here you will get it as 2400 mm square. Next, we need X1 and Y1. What is X1 and Y1? If you look at this rectangle, again it is a lamina. Each and every lamina will have a centroid. [00:02:49.168] That will be this centroid. C1 stands for the centroid of this rectangular lamina. The coordinates of the C1 are X1, Y1. Remember that. X1 is X coordinate of C1, Y1 is Y coordinate of the C1. [00:03:06.592] Then what is this x1? We know for a rectangle from here the centroidal distance will be b by 2. So b by 2. Here b is 40. So 40 by 2 you will get it as 20 millimeter. Next y1. What is y1? [00:03:23.456] From ear to ear distance that will be total will be sixty, sixty by two will be y one, that is h by two. [00:03:30.916] You will get it as 30 mm. So we are done with the first elemental area. Next come to the second one. A2 will be the area of this triangle. What is A2? Half into base into height. [00:03:45.328] Base is 20 and the height is 60. Substitute here, you will get the value of triangle A2. After that, we should know the value of X2 and Y2. X2 and Y2 are nothing but the coordinates of the centroid of the second lamina. [00:04:02.960] Since this is a triangle here it will be having a centroid that is C2. The x coordinate of C2 is X2, y coordinate of C2 is Y2. So this is X2 and this is [00:04:15.616] Y2. Then how to find X2? Remember the triangular formula. From this point to here, it will be B by 3. But what is X2? X2 is from the y axis. So that will be equal to 40 plus b by 3. [00:04:33.732] That is X2. What is base here? Base is nothing but base of the triangle, that is 20. Substitute here, and we will get the value of X2 as 46.67. Next, we should know the value of Y2. [00:04:48.976] We know the standard formula with respect to base of the triangle, the y coordinate of the centroid will be h by 3. So h by 3. [00:04:59.264] H is 60. So 60 by 3 you'll get it as 20 mm. Now we know every terms in x bar and y bar. So just expand this. The formula is going to be like this: substitute the values of a1, x1, a2, x2 in this, and you'll get the value of x bar.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:19.044] So x bar you will get it as 25.33 millimeter. Similarly, y bar, since there are only two elemental areas, the terms are going to be a1 y1 plus a2 y2 divided by total area that is a1 plus a2. [00:05:36.228] Substitute the values of a1, y one, a2, y2, you'll get the value of y bar. Simplify this. You'll get test twenty-eight point. [00:05:45.552] 0 millimeter. Now finally we got the coordinates of this trapezoidal area. The coordinates a centroid. The coordinates of the centroid of the trapezoidal area are 25.33 comma 28. [00:06:01.600] millimeter. So that's the point where the entire area of this trapezium is assumed to be concentrated. That is what the meaning. [00:06:15.716] Next question 5 slash 47. Determine X and Y coordinates of the center of the shaded area. Now look at this. [00:06:27.632] This is the given lamina area area. We have been asked to locate the centroid of this lamina. And look at this, the y-axis is here at a distance of 50 with respect to this member of a rectangle. Now [00:06:42.944]

Again, we don't have a standard formula for this figure. We have to use the method of approximation. To use the method of approximation, we have to divide this into elemental areas. Let's divide this into elemental areas as rectangle and triangle. [00:07:00.036] Yeah, what you can do here is either you can go for only one rectangle of base 200 and you can consider this is one rectangle of 50 and one more rectangle of 150 and one triangle. That's up to you. [00:07:16.772] Here I have taken it as three elemental areas. One is this rectangle, one more is this rectangle, and one more is a triangle. Let's start with the first one. The first one is a rectangle, here width of the rectangle is 50, and the depth of the rectangle is 200. [00:07:35.744] Second one is again a rectangle of width 150 and the height or a depth 200. Next third one is a triangle of base 100 and the height 200. [00:07:49.872] So we have three elemental areas. Now we know the formula in method of approximation. X bar is summation of AI Xi by total area, y bar is summation of AI yi divided by total area. Since here we have three elemental areas, the formula is going to be [00:08:07.536] be a one x one plus a two x two plus a three x three by total area that is a one plus a two plus a three. Similarly [00:08:17.360] A1 Y1 plus A2 Y2 plus A3 Y3 divided by total area that is A1 plus A2 plus A3. Now let's know about what is A1, A2, A3. First A1 is nothing but AF the first. [00:08:32.592] Element since it is in the form of rectangle, area of this rectangle will be B into H. So B is 50, we know from here, H is 200, substitute you will get the value of A1. Next, what is X1 and Y1? Uh [00:08:48.800] This lamina will have a centroid. For time being, forget about this second rectangle and this triangle. Just consider this rectangle. [00:08:59.760] This rectangle is the area so it will have a centroid C here. Some are here C one we will call it as. Then the X and Y coordinates of this centroid are nothing but X1, Y1. What is X1? [00:09:15.696] The distance of this C1 with respect to y axis and y1 is the distance of the c1 with respect to x axis. So since this c1 is in the second quadrant, look at this. Where is the c1? It is in the second quadrant. So x will be negative and y1. [00:09:33.456] Will be positive. So x is negative. That's the reason. And x will be equals to b by 2. We know the value of b 50 from here, substitute here, 50 by 2 will be equals to 25. Since it is in second quadrant, x coordinate will be negative. And y1 will be h by 2. [00:09:52.224] That will be hundred. Then come to the second one. Area of this will be equals to B into H. Here B is one fifty and H is two hundred. Substitute that you will get the value of A2. Okay, now for time being forget about first elemental area and the third one, this triangle.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.872] Just consider about this rectangle. This rectangle is again an elemental area, so it will have a centroid that is C2. X2, Y2 are the coordinates of this C2. So this is X2 and this is Y2. Then [00:10:25.284] Then what is X2? X2 will be B by 2. B we know 150. So 150 by 2 we will get it as 75. And Y2, we know that is Y2 is equal to H by 2. Here H is 200. So 200 by 2 we will get it as 100. [00:10:44.068] We are done with the first two areas. Now let's come to this triangle. A3 stands for area of the triangle that will be half into base into height. Base is 100 and the height is 200. Substitute you will get the value of A3. Then [00:11:01.444] Again, it's a lamina, it will have a centroid that will be C3. The coordinates of this C3 will be X3 from year to ear distance and Y3 will be this distance. Then how to find this X3 year? We know from the standard formulas the distance. [00:11:21.476] Central with respect to this line will be B by 3. Please remember the formulas. But X3 is measured with respect to y-axis. So that will be equal to 150. From year to year it is 150 plus b by 3. [00:11:36.100] So x rays 150 plus b by three. And y three we know for a triangle the y the distance of the centroid with respect to its base will be h by three. So h by three you will get the value of y three. [00:11:51.264]

Now we know all the terms in the given formula and please rem uh remember here x1 is negative so take it take that as minus itself. [00:12:02.208] Plus into minus you will get this entire term as negative, whereas the second term is positive, third term is also positive. Just simplify this, you will get the value of x1. Then y1 is [00:12:18.916] If you simplify you'll get the value of y1 as 93.34 mm. Now we'll get the coordinates of this entire of this even area. The coordinates of this android are 76.67 millimeter and 93.34 millimeter. [00:12:37.636] That's the point where we assume the entire area of the lamina is concentrated. [00:12:45.924] This is the next question 5 slash 51. Determine X and Y coordinates of the centroid of the shaded area. Now look at the shaded area. [00:12:58.020] What is shaded here? It's not like a trapezium. In this trapezium, the circle has been removed. Then how to locate the center end of only this remaining area. [00:13:08.740] Again, we don't have a formula, standard formula, so we have to follow the method of approximation. To follow the method of approximation, we have to divide this into elemental areas. Think how to get this shaded area. [00:13:25.540] Earlier in one of the previous problem we just divide the trapezium into rectangle and triangle. But here if we divide this as triangle and rectangle, will we get this shaded area? [00:13:38.468] No. So what we can do here is we can divide this into triangle and rectangle. Then in that what do we need to detect? We need to detect this circle. [00:13:49.924] So two additions like we have to add rectangle, triangle in that direct circle, you will get the remaining area that is shaded area. [00:13:59.428] Now let's start with method of approximation. I am going to draw a horizontal line passing through this point for Tm. Now there is this rectangle plus triangle, in that we have to deduct this circle. Then only we will get this shaded area. [00:14:19.332] Now, uh in that case what would be the height of this triangle? Will it be fifteen? [00:14:25.284] No, because here it is 40. This is 40, these two are paralleline, so this height will be 40. Then what will be the height of the triangle? Total minus 40. Total. From ear to ear it is 80. 80 minus 40 will. [00:14:42.916] Give us 40 mm. So don't get confused. Don't take it as 50. Now let's start with the first one. Here the first one is a rectangle of width 100 mm and height 40 mm. [00:14:57.732] So rectangle 100 mm 40 mm. Second one is this triangle, base of the triangle will be 100 and the height of the triangle will be 40.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.372] Consider that. Third one is deduction, remember that. And it's the circle of radius. So yes, radius twenty. Consider that. [00:15:21.668] Now let's start with the method of approximation. The formula remains the same summation of AI xi by total area, summation of AI divided by total area. Let's start with the first elemental area. A1 stands for area of the first elemental. [00:15:37.860] Since it's the rectangle, area will be b into h. Consider that. Next, we need to find the value of x1 y1. [00:15:48.932] Now for time being forget about this circle as well as the triangle. Just consider this rectangle. Since it's the lamina, it will have a centroid that will be C1. C1 stands for centroid of this rectangle. [00:16:04.868] Then x1, y1 are the x and y coordinates of this c1. Then what is x1? It will be since it's a rectangle, we have the y axis here, so it will be equal to b by 2 and y1 will be equal to h by 2. So take x1, y1 value. [00:16:24.772] Next, we have to start with the second area. Here the second one is a triangle. So area will be base into height by 2. [00:16:35.972] Next, we know the value of base and we know the height of the triangle. Substitute here you will get the value of area of the triangle. When we talk about [00:16:45.520] The triangle we should forget about other areas. Just consider this triangle. This triangle is again a laminary area, it will have a centroid that is C2. [00:16:56.384] X2, Y2 are nothing but X and Y coordinates of this C2. Then what is this X2? We

know in a for triangle the formula this x coordinate with respect to this side will be B by 3. [00:17:11.008] So x2 is equals to b by 3. Then what is y2? Y2 is this distance total. But in the standard formula, we know this distance, that is the distance of the centroid with respect to its basis H by 3. [00:17:24.656] That is from this to this is h by 3. But what is y2 total. So it will be equal to 40 plus h by 3. [00:17:33.732] So you get the value of y2. Then the last one is circle. Since area we have to deduct here, area doesn't exist here, it should be taken as negative. As being removed from this trapezium, so area itself is negative. [00:17:50.800] Uh area is minus pi r square, you will get less 1256.63 mm square. [00:17:58.500] It will have a centroid that will be C3, then the coordinates of that will be X3, Y3. X3 and please remember you should not consider X3Y3 as negative because this point is in the first quadrant. [00:18:14.756] The sign of X and Y coordinates of the centroid depends upon where is the centroid, in which quadrant the centroid is. Since the centroid is in first quadrant, both X3 and Y3 will be positive. [00:18:29.124] Then what is X3 from year to year we know 40 given in the figure itself, take that. Then what is Y3 here? It is given as thirty. Consider that. [00:18:39.076] Now substitute all the values. Just substitute the formulas. The formula is going to be a1x1 plus a2x2 plus a3x3 by a1 plus a2 plus a3. That is the general formula. [00:18:51.396] I don't think you will get any doubts in first two terms. When you come to the last term, since this is negative, so consider that within the bracket A3 minus, and here this x is positive. [00:19:08.356] So plus into minus this entire term will become negative. Similarly here it is going to be negative. Once you simplify you take the value of x bar as 45.62 mm. Similarly y bar summation of a i y i divided by total area. [00:19:25.344] uh substitute all the values from this tablet column and then [00:19:32.868] You'll get the value of y bar. Y bar is thirty one point three seven. Finally we got the coordinates of the centroid of the given laminar shaded area forty five point six to thirty one point three seven. That's the point where we assume the entire area is concentrated. [00:19:52.516] Expir by bar. That's about today's class. Try to solve few more problems given in the assignment as well as in the question bank. If you come across any doubt, you can post your queries in Pesum forum. Thank you.

EV0000054 · EV0000055 · EV0000056 · EV0000057

Full transcript: 3058 normalized words, 84 timestamped segments, 1209.6 seconds duration. Low-confidence segments flagged: 0.

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5 CLASS 35 Centroid_ Numerical

This lesson is indexed from the source video titled CLASS 35 Centroid_ Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

Graphs and diagrams

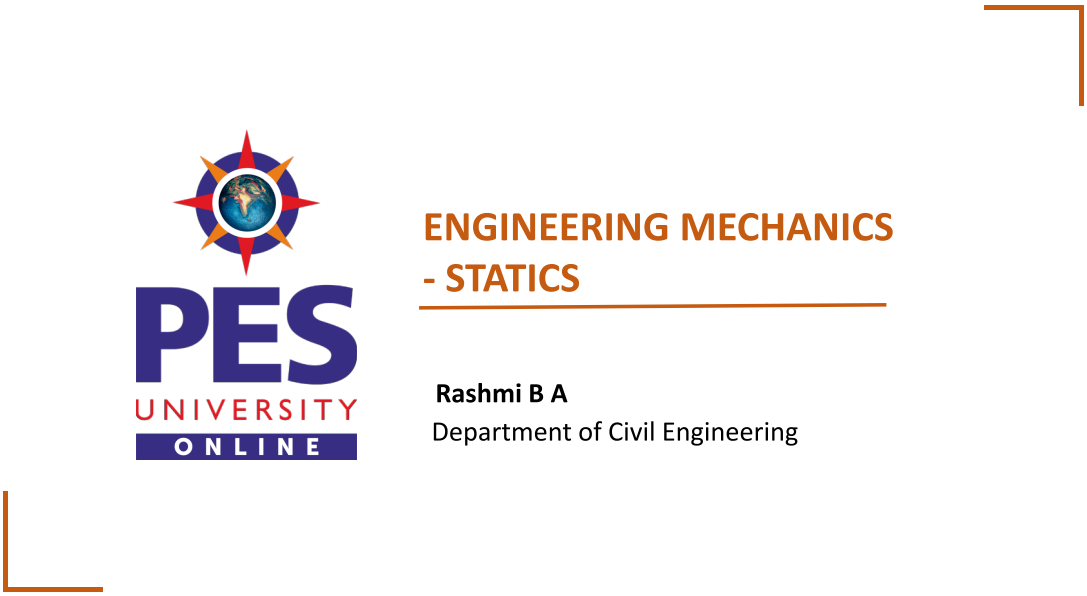


Figure G000059. UE20CV101 UNIT 4 SESSION 4 · source slide page 1.

EV0000343

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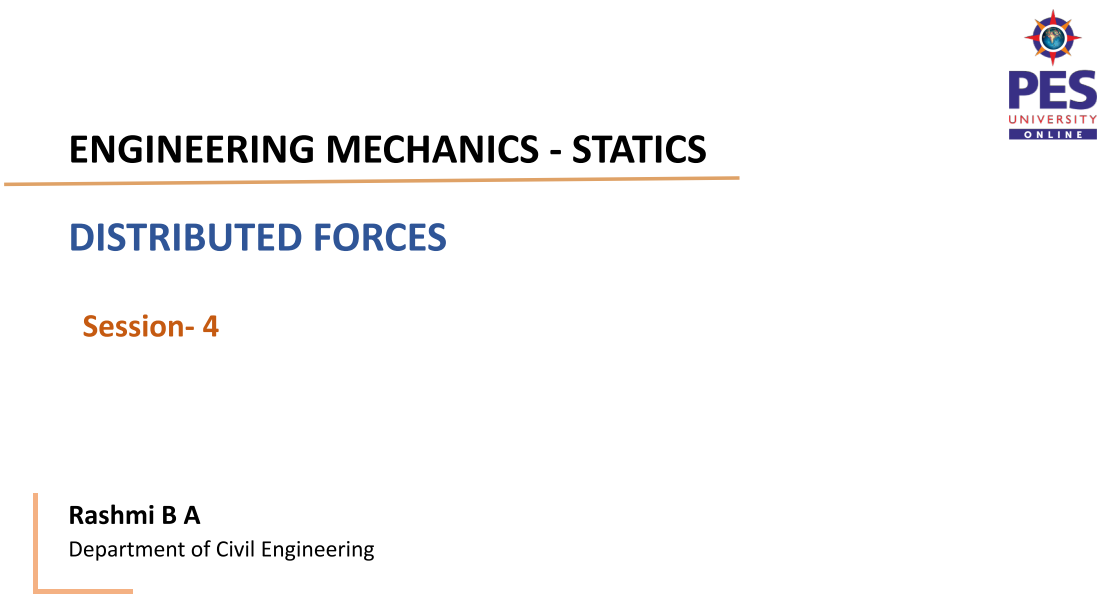


Figure G000060. UE20CV101 UNIT 4 SESSION 4 · source slide page 2.

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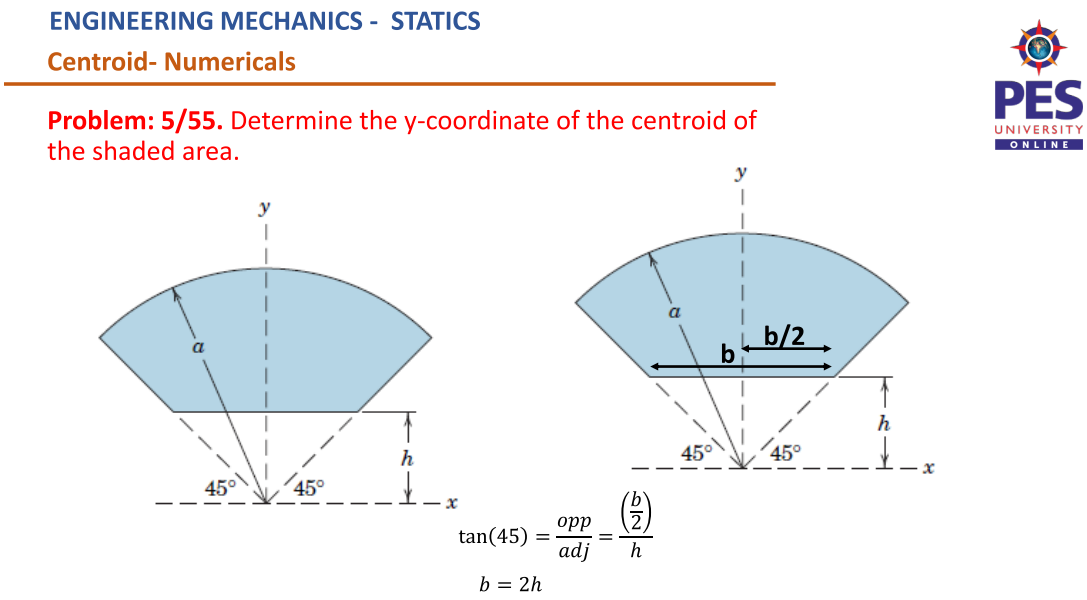


Figure G000061. UE20CV101 UNIT 4 SESSION 4 · source slide page 3.

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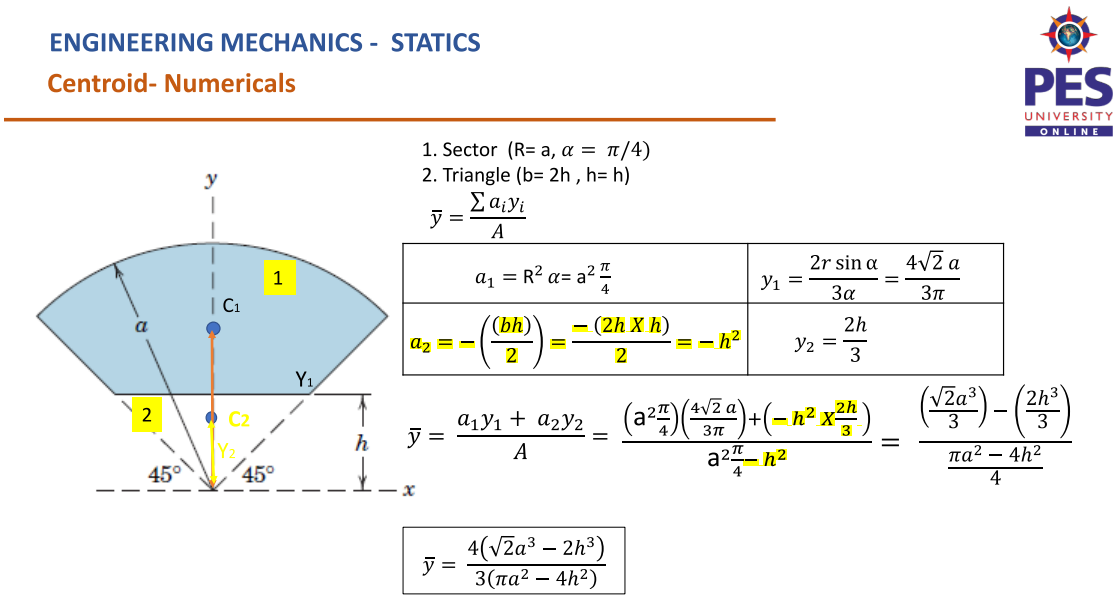


Figure G000062. UE20CV101 UNIT 4 SESSION 4 · source slide page 4.

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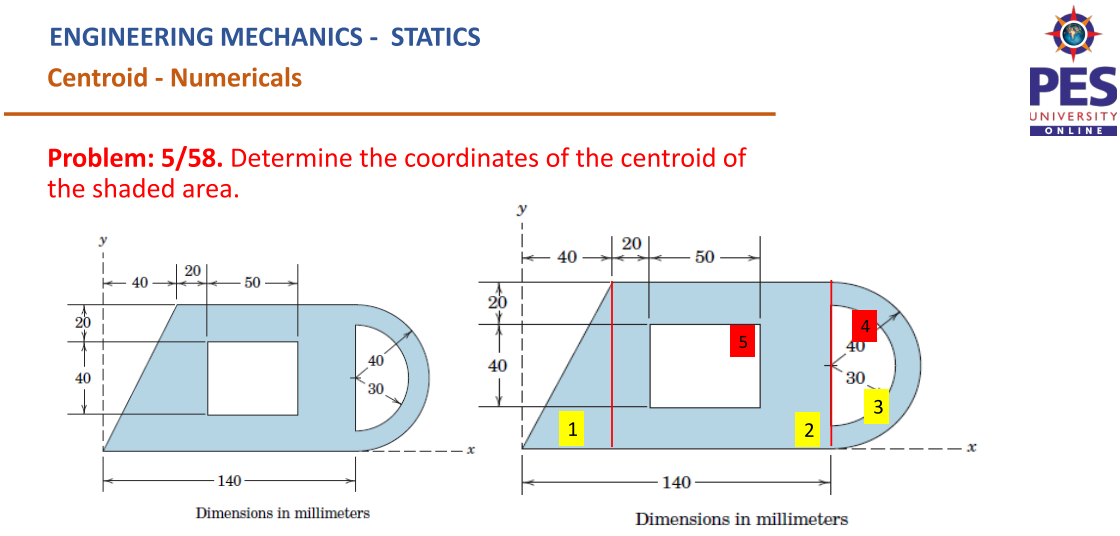
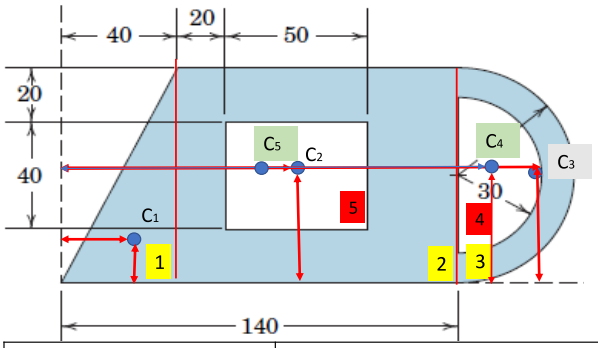


Figure G000063. UE20CV101 UNIT 4 SESSION 4 · source slide page 5.

EV0000355

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$a_1 = \frac{bh}{2} = \frac{40(80)}{2} = 1600$	$x_1 = \frac{2b}{3} = \frac{2(40)}{3} = 26.67$	$y_1 = \frac{h}{3} = \frac{80}{3} = 26.67$
$a_2 = bh = 100\ X80 = 8000$	$x_2 = 40 + \frac{b}{2} = 40 + \frac{(140-40)}{2} = 90$	$y_2 = \frac{h}{2} = \frac{80}{2} = 40$
$a_3 = \frac{\pi R^2}{2} = \frac{\pi 40^2}{2} = 800\pi$	$x_3 = 140 + \frac{4R}{3\pi} = 140 + \frac{4(40)}{3\pi} = 157$	$y_3 = R = 40$
$a_4 = -\left(\frac{\pi R^2}{2}\right) = \frac{\pi 30^2}{2} = -450\,\pi$	$x_4 = 140 + \frac{4R}{3\pi} = 140 + \frac{4(30)}{3\pi} = 152.7$	$y_4 = R = 40$
$a_5 = -(bh) = -(50\ X\ 40) = -2000$	$x_5 = 40 + 20 + \frac{b}{2} = 60 + \frac{50}{2} = 85$	$y_5 = \frac{h}{2} = \frac{80}{2} = 40$

Figure G000064. UE20CV101 UNIT 4 SESSION 4 · source slide page 6.

EV0000358

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ENGINEERING MECHANICS - STATICS
Centroid – Numerical

$x_1 = \frac{2b}{3} = \frac{2(40)}{3} = 26.67$	$y_1 = \frac{h}{3} = \frac{80}{3} = 26.67$	$a_1 = \frac{bh}{2} = \frac{40(80)}{2} = 1600$
$x_2 = 40 + \frac{b}{2} = 40 + \frac{(140-40)}{2} = 90$	$y_2 = \frac{h}{2} = \frac{80}{2} = 40$	$a_2 = bh = 100\ X80 = 8000$
$x_3 = 140 + \frac{4R}{3\pi} = 140 + \frac{4(40)}{3\pi} = 157$	$y_3 = R = 40$	$a_3 = \frac{\pi R^2}{2} = \frac{\pi 40^2}{2} = 800\pi$
$x_4 = 140 + \frac{4R}{3\pi} = 140 + \frac{4(30)}{3\pi} = 152.7$	$y_4 = R = 40$	$a_4 = -\left(\frac{\pi R^2}{2}\right) = \frac{\pi 30^2}{2} = -450\,\pi$
$x_5 = 40 + 20 + \frac{b}{2} = 60 + \frac{50}{2} = 85$	$y_5 = \frac{h}{2} = \frac{80}{2} = 40$	$a_5 = -(bh) = -(50\ X\ 40) = -2000$

$$\bar{x} = \frac{\sum a_i x_i}{\sum a_i} = \frac{(1600\ X\ 26.67) + (8000\ X\ 90) + (800\pi\ X\ 157) + (-450\pi\ X\ 152.7) + (-2000\ X\ 85)}{1600 + 8000 + 800\pi - 450\pi - 2000} = \frac{771381.5}{8699.557} = 88.67\ mm$$

$$\bar{y} = \frac{\sum a_i y_i}{\sum a_i} = \frac{(1600\ X\ 26.67) + (8000\ X\ 90) + (800\pi\ X\ 157) + (-450\pi\ X\ 152.7) + (-2000\ X\ 85)}{1600 + 8000 + 800\pi - 450\pi - 2000} = \frac{326654.3}{8699.557} = 37.55\ mm$$

$$\bar{x} = 88.67mm, \bar{y} = 37.55mm$$

Figure G000065. UE20CV101 UNIT 4 SESSION 4 · source slide page 7.

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Figure G000066. UE20CV101 UNIT 4 SESSION 4 · source slide page 8.

EV0000364

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000342 · EV0000343 · EV0000344
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 4 Rashmi B A Department of Civil Engineering EV0000345 · EV0000346 · EV0000347
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Centroid- Numericals Problem: 5/55. Determine the y-coordinate of the centroid of the shaded area. b/2 b b opp 2 tan 45 = adj= h b= $2h$ EV0000348 · EV0000349 · EV0000350
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Centroid- Numericals 1. Sector (R= a, α= π/4) 2. Triangle (b= 2h , h= h) σ $aiyi$ ϖy= A π 2$r$sin α 4 2 a 1 $a1$ = R2 α= a2 $y1$ = = 4 3α 3π C1 bh $-2hXh$ 2h $a2$ = − = = $-h2$ $y2$ = 2 2 3 Y1 2a3 4 2 a $-2h3$ + $-h2$ $X2h$ 3 3 2 C2 $a1y1$ + $a2y2$ a2π ϖy= = 4 3π 3 = Y2 A a2π $\pi a2$ $-4h2$ 4$-h2$ 4 4 2a3 $-2h3$ ϖy= 3 $\pi a2$ $-4h2$ EV0000351 · EV0000352 · EV0000353
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Centroid - Numericals Problem: 5/58. Determine the coordinates of the centroid of the shaded area. 4 5 3 1 2 EV0000354 · EV0000355 · EV0000356

6. Source slide page 6 (extracted text): C5 C4 C2 C3 5 C1 4 1 2 3 h 80 bh 40(80) 2b 2(40) $a1$ = = = 1600 $x1$ = = = 26.67 $y1$ = = = 26.67 2 2 3 3 3 3 b (140 −40) h 80 $a2$ = bh= 100 $X80$ = 8000 $x2$ = 40 + = 40 + = 90 $y2$ = = = 40 2 2 2 2 $\pi R2$ $\pi 402$ $4R$ 4(40) $y3$ = R= 40 $a3$ = = = 800π $x3$ = 140 + = 140 + = 157 2 2 3π 3π $\pi 302 a4$ = $-\pi R2$ = = −450 π $4R$ 4(30) $y4$ = R= 40 2 2 $x4$ = 140 + = 140 + = 152.7 3π 3π b 50 h 80 $a5$ = $-bh$= −50 $X40$ $x5$ = 40 + 20 + = 60 + = 85 $y5$ = = = 40 = −2000 2 2 2 2 EV0000357 · EV0000358 · EV0000359
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Centroid – Numerical 2b 2(40) h 80 bh 40(80) $x1$ = = = 26.67 $y1$ = = = 26.67 $a1$ = = = 1600 3 3 3 3 2 2 b (140 −40) h 80 $a2$ = bh= 100 $X80$ = 8000$x2$ = 40 + = 40 + = 90 $y2$ = = = 40 2 2 2 2 $4R$ 4(40) $\pi R2$ $\pi 402$ $y3$ = R= 40 $x3$ = 140 + = 140 + = 157 $a3$ = = = 800π 3π 3π 2 2 $4R$ 4(30) $\pi 302$ $-\pi R2$ = = −450 $\pi x4$ = 140 + = 140 + = 152.7 $y4$ = R= 40 $a4$ = 3π 3π 2 2 b 50 h 80 $a5$ = $-bh$= −50 $X40$ $x5$ = 40 + 20 + = 60 + = 85 $y5$ = = = 40 2 2 2 2 = −2000 σ $aixi$ 1600 $X26.67$ + 8000 $X90$ + 800$\pi X157$ + −450$\pi X152.7$ + (−2000 $X85$) 771381.5 $\text{ц}_\text{ь}x$= = = = 88.67 mm σ ai 1600 + 8000 + 800π−450π−2000 8699.557 σ $aiyi$ 1600 $X26.67$ + 8000 $X90$ + 800$\pi X157$ + −450$\pi X152.7$ + (−2000 $X85$) 326654.3 ϖy= = = = 37.55 mm σ ai 1600 + 8000 + 800π−450π−2000 8699.557 $\text{ц}_\text{ь}x$= 88.67mm, ϖy= 37.55mm EV0000360 · EV0000361 · EV0000362
8. Source slide page 8 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000363 · EV0000364 · EV0000365

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello. In today's session, let's solve few problems and centroid. How to locate the centroid of the given figure. This is the question 5/55. Locate the y coordinate of the centroid of the shaded area. Look at this figure. [00:00:18.500] Here, they have given a sector. In the sector, they have removed this triangle, and we have been asked to locate the centroid of only this shaded area, and that to only the y-coordinate. They have not asked about the x-coordinate. If they ask about x coordinate, then where will be the centroid? [00:00:38.564] Look at the figure. Due to symmetry, the centroid will be on the y axis so x coordinate of the centroid will be 0. Since they have not asked, it's not necessary to mention about x bar. We need to find only the value of y bar. [00:00:55.268] Next, how to get this shaded area? To get the shaded area, we have to use the method of approximation. We have to divide this into elemental areas. We can divide this as a sector and remove this triangle from the, then you will get this shaded area. [00:01:14.628] Then, before getting into the method of approximation, let's know about the dimensions of the sector as well as the triangle. There is a sector, the inscribed angle will be 90 degrees, nothing but pi by 2. Then radius is A. That's about sector. [00:01:31.248] Then come to this triangle. Here the height of the triangle will be H. Then what will be this base? This one. What will be this base of the triangle that we don't know? I am going to call the basis B since B is the term which I introduced in the [00:01:49.284] Question, I should know the value of B in the form of H. Then how to express B in the form of H. For that I am going to consider only this part of the triangle, that is right angle triangle. [00:02:00.368] If this is 45 degrees, this angle will also be 45. Then due to symmetry, if total is B then this will be B by 2. So we know two sides, opposite side I know B by 2, then adjacent side I know as H. [00:02:14.368] So since the angle is also known to us, use the tan relationship. Tan 45 is equals to b by 2 by H. Now we will get the value of H. Sorry, we will get the value of B. B is equals to 2 into H. [00:02:29.188] Next, let's move on to the method of approximation. Let's divide this into elemental areas. First

one, let it be a sector of radius A, and the angle alpha will be here will be equal to this will be alpha. Remember. [00:02:44.608] In the sector derivation total angle was 2 alpha. Here also 2 alpha is pi by 2, that is 90 degree. So alpha will be equals to pi by four. Next triangle, second one. [00:02:58.724] Base of the triangle will be 2 into h and the height of the triangle is h itself. And we know the formula for y bar, summation of a i yi divided by total area. So we need to find a1 y1 then a2 y2. Let's start with the first one. [00:03:16.400] area of the first elemental element. So A1 stands for area of the sector. What is the formula to find area of a sector? R square into alpha. Remember alpha should be in terms of radians. [00:03:31.076] So R's R is A, so it is going to be A squared into alpha is pi by 4. So A squared pi by 4 is the value of. [00:03:39.300] This A1 area of this sector. Then we should do the value of Y1. What is the meaning of Y1? This sector will have a centroid that is C1, then the Y coordinate of that C1 will be Y1. [00:03:54.500] Since the sector is symmetrical about y-axis, the centroid of the sector will also be on the y-axis. Now, this is the centroid of the sector. Then this is y1. How to find this y1? [00:04:09.636] To find the Y1, we know the distance of the centroid of this sector with respect to its center and the axis of symmetry is 2 R sin alpha by 3 alpha. This is the formula standard formula. [00:04:26.916] Now, radius we know that is a consider that and alpha we know pi by 4, substitute here, you will get the value of y1 is 4 root 2 into a divided by 3 pi. [00:04:42.692] Next is triangle. Since this triangle is removed or it's not shaded, area should be taken as negative. So minus half into base into height. What is base [00:04:56.624] Two times the dark h and the height is h itself. Substitute here, you will get the value of A2S minus h square. After that, we should know y2. What is y2?

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.104] This triangle will have a centroid that will be C2. The y coordinate of C2 is Y2. So this distance is Y2. [00:05:18.912] Look at this axis, x-axis is passing through the ax of the triangle. So, what is the distance of the centroid with respect to its apex in a triangle? That will be 2H by 3. So, Y2 is equal to 2H by 3. [00:05:35.908] Now we got the values of a1, a2, y1, y2, substituting the formula. The general formula is a1 y1 plus a2 y2 by a1 plus a2. [00:05:48.768] But since the value of A2 is negative, substitute that as it is here. That's why you will get the second term as negative here as well as in the denominator also. [00:06:01.540] Once you simplify you will get the value of y bar. This is the value of y bar. That's the point where the entire area of the shaded region is assumed to be concentrated. [00:06:16.292] Let's move on to the next problem 5 slash 58. Determine the coordinates of the centroid of the shaded area. [00:06:25.764] Now, look at this. This is the shaded area. We have been asked to locate the centrate of only shaded area. Then how to locate the centr of only shaded area? [00:06:36.868] Here also we have to use the method of approximation, divide this into elemental areas, then use the formula of x bar and y bar, find out x bar and y bar. [00:06:47.844] How to divide this into elemental areas. First if you draw a vertical line passing through this point, this will be triangle, then [00:06:58.816] This is semicircle, so the remaining will be rectangle here, then semicircle. So we will get the solid area. In that we have to deduct this rectangle, then semicircle of radius 30. Only then we will get the shaded area. [00:07:15.396] Let's start with the first one. So this is the first one triangle. Second one is this rectangle. [00:07:23.204] And the third one is this bigger semicircle of radius 40. In that we have to deduct fourth one, that is semicircle of radius 30. Fifth one is also deduction, that is a rectangle of width 50 and the height. [00:07:40.048] Fourty. Then only we will get this shaded area. Okay, let's start with the first one. [00:07:48.868] We know the formula x bar and y bar a1 y1 plus a2 y2, so on divided by total area.

Similarly, y bar is a1 y 1 plus a2 by 2, so 1 divided by total area. So let's start with the first one. First one is the triangle of base 40 and the height. [00:08:07.204] Height will be the this that will be diameter of this bigger semicircle A T. So B into H by 2 will give us 1600 mm square. That is area of the triangle. Then we need to find X1 Y1. [00:08:24.708] X1, Y1 are the coordinates of the centroid of this triangle. The centroid of the triangle will be somewhere here, that will be C1. The x coordinate is this, that is x1. Then how to find this x1? [00:08:39.392] We know the standard formula, this x1 will be equals to with respect to this side, it will be two times the of base by three. [00:08:48.100] So, x1 is equal to 2b by 3, your base is 40, substitute that by 3, you will get the value of x1 as 26.67. Next, we should know the value of y1. [00:08:59.632] With respect to base of the triangle, y coordinate will be h by 3, here h is this AT. So AT by 3 will give us the value of y1. We are done with the first elemental area. [00:09:14.320] Let's come to the second area that is this bigger rectangle. The width of the rectangle will be how much? Total will be 140 minus 40, because this 40 is included in the triangle. So 100 is the width of the [00:09:30.416] rectangle, then the height of the rectangle is 80. So A2 will be equal to B into H that is 100 into 80. Next we should know X2 Y2. [00:09:41.796] Ah, this rectangle will have a centrite somewhere here at the center, C2, x coordinate of this, that is the distance from y axis to c2, that is x2. [00:09:54.308] This is x2. Then how to find this x2? From the derivation we know from ear to ear distance will be b by 2. But x2 is from y axis, so we have to add this 40. So 40 plus b by 2.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.788] So, 40 plus b by 2, you will get the value of x2. Then y2 is the y coordinate of this c2, this distance, that will be equal to total 80, 80 by 2 will be 40. [00:10:24.260] We are done with the second lamina. Come to the third elemental area. The third area is semicircle of radius forty. So A three is [00:10:34.628] I R square by 2 R is 40. You will get the value of area of this bigger semicircle. Next, this bigger semicircle will have a central [00:10:44.768] Where will be that centroid? On the axis of symmetry, somewhere here. If you draw a horizontal line passing through the center, this will be the axis of symmetry. So C3 will be somewhere on the axis of symmetry. Then what is X3? X3 is this distance. [00:11:01.956] From y axis to C3. Then what will be this total distance that will be equals to from year to year 140 is given here. Plus what is this distance? That will be 4 r by 3 pi. [00:11:17.092] So that will be equals to 140 plus 4r by 3 pi. Since it's a we are talking about bigger semicircle, radius is 40. Next y 3. Y3 is y coordinate of this C3 that will be equals to the di radius itself that will be 40. [00:11:36.516] Next is the deduction part, that is fourth one, we have to deduct this smaller semicircle of radius 30. So, area will be negative minus pi r square by 2, you will get it as minus 450 into pi. [00:11:54.272] Next, where will be the centroid of this smaller semicircle? Again, that will be on the axis of symmetry somewhere here, C4. Then what is X4? Ear to ear distances X4. [00:12:08.804] Then how to find the 64? From year to year we know 140 plus center to centroid will be 4r by 3 pi. Since C4 is in the first quadrant, it will be positive. [00:12:22.020] 140 plus 4 R by 3 pi ERR is 30, you will get the value of X4. Then what is Y4? Y4 will be the diameter of sorry radius of this circle that will be 40. [00:12:37.220] Next, the last is the deduction of this rectangle. So area is negative, that will be equal to minus b by 4. [00:12:47.488] So b minus b into h. What is b here? 50 and what is the height of this rectangle? 40. You'll get it as minus 2000. Then we should know x phi and y phi. X phi and y phi are the [00:13:01.712] Coordinates of the centroid of this C phi. Then where will be what is this x phi? This distance is x phi. Then how to find that? That is 40 from here to here 40 plus 20 plus what is this? 50 by 2. [00:13:19.748] So that is the value of x

phi, 40 plus 20 plus b by 2. We will get it as 85. Then we should know y2, sorry, y phi, y phi will be equals to [00:13:34.500] This 80 by 240. We got the value of all the terms in the equation. [00:13:42.628] In this formula, so substitute here a1 x1 then plus a2 x2 plus a three x three plus. Then you should open the bracket, since it is minus it should be. [00:13:58.528] Minus 450 pi into plus of 152.7. Similar next plus of minus 2000 into 85 divided by total area will be a one plus a2 plus a three. [00:14:13.488] Since it is minus it will be minus four fifty pi 450 into pi, then minus two thousand. Once you simplify it, get the value of x bar. Similarly, if you s substitute all the values of [00:14:27.216] A1 Y1 A2 Y2 A3 Y 3 A4 Y 4 A5 Y 5 You'll get the value of Y bar. The Y bar is 37.55. [00:14:39.204] Millimeter. This is the point 88.67, 37.55 will give us the coordinate of the centroid of the shaded area. That's the point where we will assume the entire area is concentrated. [00:14:57.700] That's about today's session. Uh try to solve the problems given in the assignment as well as in the cushion bank. If you come across any doubts, you can post your queries in the PESO forum. Thank you.

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Full transcript: 2306 normalized words, 64 timestamped segments, 911.082667 seconds duration. Low-confidence segments flagged: 0.

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6 CLASS 29 Introduction to Beams

This lesson is indexed from the source video titled CLASS 29 Introduction to Beams. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000070 · EV0000071

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] Hello, in today's session I'm going to explain beams. [00:00:07.428] When you are in the basement of any multi-story buildings like schools, apartments, or even in Mars, you can see some part of the building will be projected below the roofing, just like this, like a horizontal member. [00:00:22.752] You can see in this picture these horizontal members, those projections, what you can see in the basement, we call them as beams. These beams play very important role in all the structures. [00:00:35.332] Then how to define these beams? We are going to define beam as a structural member which offer resistance to bending at the rotation due to the applied loads. And these beams are usually long prismatic bars and the loads are applied normal to the axis of the bar. [00:00:52.228] We have different before moving on to the different types of beams, let's know about different types of supports. The first one is roller support. This is the roller support here. You can see the beam is resting on this roller support. [00:01:10.384] Ah when we draw the free body diagram in the numericals, uh if there is a roller support, usually we consider there will be only one reaction from the roller support. The reason is uh if you think of applying a [00:01:23.600] Vertical load here, we can't move the beam along y direction because the roller will restrict the movement of this beam

along the y direction. Whereas if you think you are going to apply a horizontal force on this beam like this, definitely it is going to move along the horizontal direction. [00:01:41.264] Because there won't be any restriction from the support onto the movement of the beam along X direction. If you think of applying a couple at this free end, it is free to rotate about this roller support. That's why. [00:01:57.328] Remember whenever we have a roller support, roller will offer resistance for the motion only along y direction, that's why we will be having only one reaction normal to the surface of the contact. This is about roller support. [00:02:12.272] Next one is fixed support. In fixed support, the beam will be like this. It is represented in this way. Now, if you think of applying a vertical load here anywhere on the beam, this support will restrict the moment of the [00:02:28.960] Beam along y direction. At the same time, if you think of applying a horizontal force on this beam, again the [00:02:37.120] Support will offer restriction for the motion of the beam along X direction. That's why it will be having two reactions. One will be along X direction, one more will be along Y direction. In the same way, if you think you are going to apply a couple here at the free end R due to the application of the force. [00:02:54.864] Do you think the beam will start to rotate like this about the axis of the beam? No, it won't, because [00:03:05.380] This will have a resisting couple. So that can be either clockwise or it can be counterclockwise. So please remember when we come across a fixed support, there will be [00:03:19.456] Three reactions to our forces, one along X, one more along Y, and one more will be the resisting couple like this. Next one is spin arranged support. [00:03:33.252] This support will offer resistance for the motion of the beam in both along x direction as well as along the y direction. That's why two reactions, one along x direction and one more along y direction. This slide is very important to solve the numericals. [00:03:53.156] Next is different types of beams. We are going to broadly classify the beams as statically determinate beams and statically indeterminate beams. Statically determinate beams. [00:04:07.616] We can find the support reactions using the conditions of equilibrium. And in one more way, if the beam is having just sufficient number of [00:04:20.036] support reactions which are enough for the beam to be in equilibrium. That is also called as statically determinate beam. If the beam is having more [00:04:30.864] Number of support reactions that necessary for the stable equilibrium configuration that is called as statically indeterminate beam. Let's talk about statically determinate beams. [00:04:45.924] The first one is simple beam. If one end of the beam is on a pin connection or a hinged connection, and if one end is an roller, that we call it as a simple beam. [00:04:57.248] How many number of support reactions here? If you consider the V me it's AB, then since it is a pin, there will be two reactions, one along X direction, one more along Y direction. So AX AY.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.224] Then at B it's roller support, so one reaction that is RB reaction at B. So total we have three reactions, we have three conditions of equilibrium. [00:05:25.680] We can find all the three reactions using three conditions of equilibrium. That's why the name statically determinate B. This is the first one simple beam. [00:05:37.860] The second one is cantilever beam. If one end of the beam is fixed like this and the other end is free, it is not resting on any of the support. It is called as a cantilever beam. [00:05:51.056] Let's count the number of support reactions. Since it is a fixed support, there'll be two forces, one along its direction, [00:06:01.232] One more in a Y direction and one more is the resting momentum. You can consider it as either clockwise or counterclockwise, no need to worry about the direction. If you consider clockwise. [00:06:14.544] And if the answer is positive, our assumption is wrong. If you consider counterclockwise, if the answer is positive, our assumption is

correct. [00:06:23.620] The last one is combination. This is also called as overhanging beam because this part of the beam is extended beyond the support, hence it is called as overhanging beam. Let's count the number of support reactions. It is pined connection. [00:06:41.264] So two reactions, one along X, one more along Y. Then it's a roller, only one reaction along Y direction. So total we have three reactions, hence it is statically determinate. These three are the statically determinate beams. [00:07:00.132] Next, we have statically indeterminate beams. Remember, in this we can't find all the support reactions using the three conditions of equilibrium. The first one is continuous beam. [00:07:16.144] If the beam is having more than three support reactions, three are more than three support reactions, it is called as continuous beam. Now look at this. [00:07:28.960] Let's count the number of support reactions. It is a pinned connection, so two reactions, one along X and one more along Y, AX, AY. Next at B it's a roller support, so only one reaction. At C again it's a roller support, so one more reaction. [00:07:46.640] Total we have four reactions Ax, AY, RC, and RB. Hence it is statically indeterminate B. The next one is fixed B. [00:07:59.940] If both ends of the beam are fixed, it is called as fixed beam. See here it is fixed here and here also it is fixed support. Hence it is called as fixed beam. How many number of support reactions? Fixed support will offer [00:08:16.432] Three reactions, two forces and one more is moment, and here also two reactions and one more is moment. As I said before, the direction depends upon us how we are and we can assume the direction of the forces as well as the moment. [00:08:35.300] So this is about fixed beam. Next one is end supported cantilever. This is also called as propped cantilever. Nothing but if one end of the beam is fixed, the other end is resting on a roller, it is called as end supported cantilever, a propped cantilever. [00:08:52.960] Let's count the number of support reactions. Since it is a fixed support, three reactions and at B it's a roller support, so one reaction RB. Total we have four reactions, hence it is statically indeterminate B. [00:09:11.716] This is about different types of beams. Next, the different types of loading on beams. The first one is point load. This is also called as concentrated load. As the name itself indicates, if the force acting on a beam is acting at a particular point. [00:09:31.504] are the area over which the force is acting is very very small compared to the dimension of the beam. We call that beam as a point load or concentrated load. Look at this figure there is a simple beam and [00:09:46.992] The load is acting at a particular point here, hence it is called as a point load or concentrated load. When we solve the numericals, we want all the loads to be point loads for our calculation purpose.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.672] And this point load can be either inclined, vertical, or horizontal. Right now it is vertical. Few times it can be horizontal. A few times it can be inclined. Whenever they give the inclined load, remember we know how to resolve the load along. [00:10:20.368] Force along X and Y. Same concept here also we are going to apply and find the components of that load. Nothing but force along X and Y direction. The second one is UDL. This is also called as rectangular loading, uniformly distributed loading. [00:10:40.292] We are going to represent the loading like this. The blue color is the beam, and this is the load that is acting on this beam. [00:10:49.280] The intensity, this is the assume this is the starting point of the load and this is the end point of the load. From ear to ear, the intensity is uniform throughout, hence the name uniformly distributed load or rectangular loading, and here W stands for the height of this rectangle. [00:11:06.960] And L stands for the base, nothing but the span of the UDL. Then, as I said before, we want all the loads to be point load or concentrated loads. Then how to convert this UDL into a point load. Remember? [00:11:23.248] That point load

should produce the same effect as that of this UDL on a beam. There should not be any net change on the [00:11:34.016] Body. Then how to get that? For that to find the resultant of this UDL, there is a formula. Since it is in the form of rectangle, resultant will be equal to W into L nothing but base into height of the rectangle. [00:11:47.940] And intensity of this loading can be either given in the form of kilonewton per meter or newton per millimeter, and L can be either in millimeter or meter, and L is the span of the UDL. Next. Once you find the resultant, this is the [00:12:04.592] Point load R is equals to W into L. Where exactly this point load will be acting because UDL is acting over a span of L meters. Then wave wheels are acting. Remember, since it's the rectangle from the concept of centroid, it will be exactly at the midpoint L by 2 from here. [00:12:22.256] L by 2 from here. So look at this figure. Here this R which is acting at a distance of L by 2 will produce the same effect as that of the UDL acting over a span of L meters. That's what the meaning. Next is [00:12:41.764] U VL that is uniformly varying load. This is also called as triangular load. [00:12:48.996] Look at this figure, there is a beam. This is a beam and there is a UVL. The intensity of the loading is varying from 0 to W over a span of L meters. And this W will be either in kilonewton per meter or Newton per millimeter. [00:13:07.204] Next, as I said before, we want all the loads to be point or concentrated loads. Then how to find the resultant of this load? Remember, since it is in the form of triangle, we are going to use the formula of [00:13:22.304] Triangle half into base into height. The resultant of this will be half into base into height, that is W into L by 2. [00:13:31.108] You will get the magnitude of the resultant of this loading. Next, where exactly that will be acting? From this side, it will be 2 times that of L by 3, and from here L by 3. That you should take it. [00:13:45.476] So please remember this 2L by 3 and L by 3. If you are measuring from this side, it will be L by 3 from here to L by 3. [00:13:54.416] Three. So the R which is acting at this distance will produce the same effect as that of the UVL acting over span of L meters. Fourth one is combined loading. It's also called as trapezoidal loading. [00:14:09.956] It's a combination of UDL and UVL, hence it is called as combined loading. Look at the figure here. There is a beam and [00:14:20.164] Ah there is a loading. At one hand the height of the loading is W1, at the other end it is W2, and the span over which the load is acting is L. [00:14:31.716] Now, how to find the resultant of this trapezoidal loading or combined loading. We know the formula for rectangular loading and triangular loading. So what we can do is the easiest way is just divide this into rectangle and triangle. [00:14:49.412] So we are going to get one resultant from rectangle and one more resultant from the triangle. Now I am going to draw a horizontal line like this. Just assume we are going to draw a horizontal line like this. Now this is rectangle whereas this is triangle.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.364] For the time being, forget about the triangle. Just consider the rectangle. How to find the resultant of the rectangle? Base into height. What is height here? This is W1, so this is the height. Then what is the base? [00:15:22.288] L so R1 stands for R1. One is nothing but we are going to assume the first loading is a rectangle. So R1 is the resultant of the rectangle. R1 is equal to W1 into L. We got the magnitude of the resultant of the rectangle. [00:15:39.460] Next thing is where exactly this R1 will be acting, we know from the concept of centroid from ear to ear, L by 2. Similarly, from ear to ear also L by 2. So we got both magnitude as well as the point of application of the [00:15:55.296] First resultant. Now let's move on to the triangle. For time being forget about the rectangle. If you consider triangle, to find the resultant of the triangle, we know the formula half into base into

height. What is base here? L. It is the same. [00:16:12.480] Then what about the height of this triangle? W2 is not the height because W2 is the height from Ear to B. Total height is W2. Then how to find the height of the triangle that will be equal to total W2 minus this one. [00:16:29.728] W1 that will be the height. So resultant two stands for resultant of the rectangle. [00:16:36.704] Triangle R2 is equals to half into W2 minus W1 into L. Next we should know the point of application or the action of this. So from this side it will be 2 L by 3, whereas from here it will be L by 3. So this way we can find the resultant of the trapezoidal loading. [00:16:56.388] The last one is general loading. In general loading, the intensity of the loading will be varying. And in the syllabus we have the numericals on point load, UDL, UVL, and combined loading. [00:17:14.404] That's it for today's class. Try to solve the questions from the question bank and go through the assignment questions. If you come across any queries, you can post it in the PESU forum. Thank you.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Full transcript: 2633 normalized words, 71 timestamped segments, 1050.033333 seconds duration. Low-confidence segments flagged: 0.

EV0000070 · EV0000071 · EV0000072 · EV0000073

7 CLASS 30 Beams_Numerical

This lesson is indexed from the source video titled CLASS 30 Beams_Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000086 · EV0000087

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.356] Hello, in today's class let's discuss few problems and beams. This is question 5/101. Question is determine the reactions at A and B for the beam subjected to the uniform load distribution. [00:00:19.908] Look at this question. There is a beam AB. The name of the beam is simple beam because one end of the beam is having pin connection, other end is having roller, and we need to find the support reactions at A and B. [00:00:39.172] Since it is a pin connection, there will be two reactions, one along x direction, one more along y direction. We need to find what is ax and what is a y. [00:00:49.988] Next, at B, since it's a roller, there will be only one reaction that is reaction at B R B that we have to calculate. These three reactions will be because of this loading on the beam. What is the name of this loading? [00:01:04.772] Remember it's UDL uniformly distributed loading that we call it as rectangular loading also. Let's get into the solution part. [00:01:17.092] First thing is always we have to convert this UDL into a point load because in our free body diagram we want all the forces to be [00:01:27.524] point load or concentrated loads. Then how to convert this UDL into a point load? What is the formula? Do you remember? The formula is the resultant of this UDL R will be equal to W into L. What is W intensity of this loading [00:01:44.996] And L is span over which it is acting. So R is equals to W into L. Here what is W? W is intensity that is 6 kilo Newton per meter

and what is L? Span that is 300 meter millimeter. [00:02:00.932] Since the load is given per meter and this is in millimeter, convert from meter to millimeter or millimeter to meter. Just take care of the conversion. So I have converted 300 millimeter to uh [00:02:17.764] meter so meter meter will get cancelled resultant of this will be equal to 1.8 kilonewton. Once we know the resultant of the loading let's start with the free body diagram of the bean A B. Remember [00:02:31.824] Whenever we want to draw the free body diagram, the first step is we need to project all these important points vertically downward just like this. Draw the uh consider point here, draw a vertical line passing through this. [00:02:46.000] This is the point where the load is ending. Just draw a vertical line, then draw a vertical line passing through B. Now draw a horizontal line, this will be the beam AB in the free body diagram. Now [00:02:57.568] Either start from left end of the beam or from the right end of the beam so that you will not miss any of the forces in the free body diagram. I am going to start from this end of the beam and I'm going towards right side. [00:03:11.216] At EA. Look at the beam here. At EA it is having a pinned connection. Pinned connection means two reactions. So consider those two reactions AX and AY. Remember? [00:03:22.816] We are going to consider the direction of the R uh R's arbitrarily because we don't know whether Ax is right or left, AY upward or downward. [00:03:32.292] This is the assumption we are going to make. Remember, if we get the answer as positive, our assumption is correct. If you are getting the answer as negative, then we have to change the direction of the arms. That's it. No need to worry. Now we are done with the [00:03:49.252] Point A. Next we have UDL here. Now in the free body diagram we are going to replace UDL by a point load. The intensity of the point load is 1.8 kilonewton, so 1.8 kilonewton here. [00:04:02.800] But where exactly this will be acting? Remember the formula for U UDL it will be at a distance of L by 2 from here and L by 2 from here. So what is that L by 2? Since L is 3000 it will be equal to 150 mm and 150 mm. [00:04:19.824] So we are done with this UDL. Next we don't have any loading from in between this three hundred millimeters, so no need to do anything here. Then come to B at B since it's the roller support. The roller will offer only one reaction that is. [00:04:35.680] Normal, so RB you have to consider vertically upward. Now this is the complete free body diagram of a beam. Once we are done with the free body diagram of a beam, we have to apply the conditions of equilibrium. Look at the forces here. [00:04:54.692] They constitute general force system. So we have three conditions of equilibrium. Let's start with the first one, sigma Fx 0. There is only one horizontal reaction that is Ax. We don't have any other horizontal force or reaction.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.856] So a x will be equals to zero. So we are done with the first one. Next, if you consider second condition sigma f y 0, we have two unknowns along y direction. One is a y [00:05:27.620] This one and one more is RB this one. So to avoid we will get the equation. To avoid that I'll go I'll start with the third condition of equilibrium that is moment about any point zero. So I am going to apply that about here. [00:05:44.612] Remember this is sum of the moment of all the forces about A should be 0. If you consider Ax and AY, these two are acting at the point A itself. So moment of these two will be 0. So don't consider these two forces. Next, move in this direction towards right. [00:06:03.588] After this, the next force is 1.8 kilonewton. Apply the right hand thumb rule. Once you apply the right hand thumb rule here, your finger thumb is going inward. If the thumb is going inward, that indicates it is [00:06:19.780] Clockwise and in one more way if you are moving in this direction to reach here, you have to take this turn. This is the shortest rule. So this is clockwise. Clockwise

means negative. So minus 1.8 into perpendicular distance will be 150. That's what is written here. Next. [00:06:40.004] We are done with the moment of 1.8 kilonewtons. From here to here we don't have any forces. Then come to RB. RB is acting in the counterclockwise direction. So it is counterclockwise positive. Consider it as plus. [00:06:56.640] forces R B into perpendicular distance. What is the perpendicular distance from point B to A? It is 150 plus 150 plus 300. So consider that that should be equal to 0. Simplify this [00:07:13.316] Once you simplify this, you are going to get the value of RBS positive plus 0.45 kN. What the plus sign indicates our assumption is correct. We assume the force RB is upward. So that is correct. [00:07:30.384] That is what this plus sign indicates. We are done with the reactionate roller B. Next we are left with only one more unknown that is AY. Then how to find AY? We have one more condition of equilibrium that is sigma f of y is equal to 0. [00:07:49.476] Apply this sigma f y is equals to 0. Now ax is horizontal, so you can't consider ax when we are talking about the forces along y direction. [00:08:00.964] AY is upward. Consider plus AY. Then 1.8 is downward. So minus 1.8. RB is upward. So plus RB. That should be equal to 0. We know the value of RB that is 0. [00:08:17.584] 45 do the substitution here. So a1 minus 1.8 plus 0.45 is equal to 0. Then you simplify, you'll get the value of a1. [00:08:29.696] You'll get the value of Ay as 1.35 kN and the answer is positive. That indicates our assumption is correct. So we assume AY as upward force that is correct. So replace that plus sign by upward. [00:08:46.948] So we got all the three reactions. Remember, since it is a pin connection, what we can do here is we can write the reaction at A as under root of ax square plus a y square. Here [00:09:00.976] Ax is 0, so reaction at A will be equals to directly AY. That's it. This is the next question. 5 slash 103. Calculate the reactions at A and B for the beam loaded as shown. [00:09:18.500] First, let's look into the beam. Do you remember the name of this beam? It is called as combination or overhanging beam. [00:09:32.164] Combination is because it is having pin connection here and it's a roller. Overhanging is this part of the beam, this one meter is extended beyond the support B roller support B. Hence it is called as overhanging beam. Now come to this loading. What is the name of this loading? [00:09:52.068] It is the combination of triangular and rectangular loading. Then how to divide this loading? We know if we draw a line ear, horizontal line passing through this, it will be a rectangle down and a triangle ear.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.740] Let's see how we can draw the free body diagram for this. First thing is we have to divide this into rectangle and triangle. To divide that into rectangle and triangle, just draw a horizontal line passing through this. Now look at this. For the time being, forget about this triangle. [00:10:31.856] Just consider this rectangle. Is it nothing but rectangular loading are UDL uniformly distributed loading? Yes. Then how to find the resultant of UDL? For that we know the formula W into L. [00:10:47.408] W stands for intensity of the loading and L stands for the span over which the load is acting. Here W is right now 400 because that is the height of this rectangle. [00:11:01.568] And the span L is nothing but the distance over which it is acting. 6 plus 1 plus 1 that is 8 meters. Now 400 into 8. Since this is also in meter and the distance is also in meter, no need to worry about the conversion part. So total the resentment of this first loading is 3.2 kilonewtons. [00:11:21.220] Next. The second one is triangle. For when you consider a triangle, you no need to consider the rectangle. Just forget about the rectangle. [00:11:31.364] Now, how to find the resultant of this triangle? We know the formula R is equals to half into base into height. Base is here A 6 meters because the triangle is having a base steel here, that's it. [00:11:46.724] So half into six into height. What is the

height of this triangle? Total is eight hundred. Then how to find this height of a triangle? Total eight hundred minus height of the rectangle. You will get this height. So use the formula. [00:12:06.148] This is W L by 2. So it will be 1.2 kN. And one more thing. When you talk about R1, that is the resultant of this rectangular loading, where exactly will that be active? At the [00:12:20.832] L by 2 distance with respect to a and L by 2 with respect to this end. Nothing but 8 by 2 meters. Then where will be the resultant of this triangle with respect to this side we know L by 3. If you are talking from this side, it will be 2L by 3. That's why I have written it as next. [00:12:40.772] L by 3 with respect to A. 6 by 3 is equal to 2 meters. Next, we can now we can start with the free body diagram of the beam. Whenever we [00:12:53.732] Need to draw the free body diagram. Remember we need to project all these points vertically downward. Now draw a horizontal line. This will be the beam AB. Start from any one of the end, either from this end, or from this end. [00:13:09.604] Now I am going to start with the end. A. What is the name of this connection? It is a pin connection. So two reactions, one along x direction, one more along y direction. This is up to you the direction. Right, left, upward or downward, that's up to you. Next. [00:13:28.336] We are done with this. Next what do we have? We have R2 from here 1.2 kilonewton at a distance of 2 meters. So consider R2 1.2 kilonewton at a distance of 2 meters with respect to E. We are done with the resultant of the second one. [00:13:44.080] Where is the resultant of the first one with respect to 4 meters. 4 meters means after R2 we are going to get that. So at a distance of 4 meters, R1 is equal to 3.2 kilonewt. So this is how we are going to get R1 and R2 here. [00:14:01.220] After this, we are done with the resultant of the loadings. Next move in this direction. We don't have any other loading, but here at a distance of 7 meters with respect to A, we have a roller support. [00:14:15.520] So the roller support will be having only one reaction RB vertically upward. So now we are left with one meter here. Consider that as it is one meter. We are done with the free body diagram of the beam. [00:14:30.384] Once we are done with the free body diagram of the beam, next thing is we need to apply the conditions of equilibrium. What is the first condition of equilibrium? Sigma F is equal to 0. Since there is no horizontal force acting on the beam, Ax will be straight away. [00:14:48.548] it will be equal to 0. Next. I'll start with the moment equation because if you apply sigma f y 0, we have two unknowns a y then r b. Just to avoid that I am going to apply moment about E as 0. So apply moment about E as 0.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.800] Once you apply moment about E as 0, look at this at A we have two forces Ax and AY. Since these two are acting at the same point, moment of these two forces will be zero. That's the reason why I'm not considering Ax and AY. Next. [00:15:22.080] More in this direction, which is the next force we have? 1.2 kilonewton downward. Don't look whether it is downward or upward. When we are talking about moment you should think whether it is [00:15:32.848] Making clockwise rotation or anti-clockwise rotation with respect to point under consideration. Right now 1.2 is clockwise with respect to apply the right hand thumb roll. If your thumb is going inward, so it is clockwise. [00:15:47.876] Since it is clockwise it should be written as minus into what is force here? 1.2. What is the perpendicular distance? 2 meters. So that's what minus 1.2 into 2 meters. Next comes the second force that is 3.2 kilonewton. [00:16:04.160] First think about the direction clockwise or counterclockwise. It is clockwise with respect to A, so minus forces three point two into distances four meter. [00:16:16.452] That's what minus 3 point two into 4. We are done with the moment of this force. Next, from ear to ear, we don't have any other force. Lastly, we have RB. RB, if you apply right

hand thumb rule, the uh thumb is coming outward. So R B is [00:16:33.968] Counterclockwise plus what is force? R B. Then what is distance from B to A? What is the distance? Counted three plus four seven. [00:16:45.316] Plus Rb into 7 is equal to 0. Once you simplify this, you will get the value of RB. So Rb is equal to 2.17 kN and the answer is positive, that indicates R B is act upward. [00:17:02.916] Now we are left with only one force that is AY. [00:17:07.012] To find a y we are left with one more equation sigma f y is equal to 0. Apply that. Again start either from descent or from descent. Now don't consider ax, remember that a y is upward, so plus a y [00:17:23.936] Then minus 1.2 because it is downward. 3.2 is also downward, so minus 3.2 RB is upward, so plus RB should be equal to 0. So we'll get this equation. [00:17:38.180] Since we know the value of RB, that is 2.17, substitute that here, find the value of Ay. AY you will get it as 2.23 kilonewton. The answer is positive. That indicates [00:17:53.124] Our assumption is correct, so a y is acting in the upward direction. Once you are done with this, remember reaction at A will be always equal to under root of a x squared plus a y squared. [00:18:06.608] Since ax is 0, reaction at a will be a f itself and that will be on a y direction. [00:18:16.932] Okay, by this I'll conclude today's topic. You can go through the assignment questions, try to solve the assignment questions, and go through the question pack. If you come across any doubts, you can post your questions and queries in PESU forum. Thank you.

EV0000098 · EV0000099 · EV0000100 · EV0000101

Full transcript: 2868 normalized words, 72 timestamped segments, 1115.968 seconds duration. Low-confidence segments flagged: 0.

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8 CLASS 31 Beams_Numerical

This lesson is indexed from the source video titled CLASS 31 Beams_Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000102 · EV0000103

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.388] Hello, in today's class let's solve your problems on beams. Question 5/105. Find the reactions at A due to the uniform loading and the applied couple. Look at this beam. First, let's start with the beam. The name of the beam is [00:00:19.396] Cantilever beam because one end is fixed and the other end is free. Hence it is called as cantilever beam. [00:00:27.620] The name of the support here is fixed support. It will have three reactions, two forces, one along X, one more along Y, and one more will be the resisting moment. That can be either taken as clockwise or it can be taken as clockwise or [00:00:43.440] Counterclockwise. They have asked us to find the reaction at A, nothing but Ax, A by then resting moment at A we have to calculate. Before that let's discuss about the type of loading. [00:00:54.496] Uh if you look at this, this is UDL, uniformly distributed loading. When we talk about the free body diagram, we need to convert this UDL into a point load. How to do that?

We know the formula W into L. W will be equal to 2 kN and L will be [00:01:11.856] Three meters. Where exactly that will be acting? Midpoint. Midpoint is nothing but three by [00:01:16.976] Two, that will be one point five meters. And this is moment of a couple it's a free vector. When we consider sigma f and sigma f y we should not consider this because this is moment of a couple. Well sigma f and sigma f y are the sum of the components of the forces along X and Y. So don't [00:01:34.608] Consider this when we talk about sigma f x and sigma f y. Let's get into the solution part. For discussion you need to start with the feedbody diagram. Before that, there is a [00:01:48.256] Rectangular loading. So we need to find the resultant of this rectangular loading. We know the formula to find the resultant R will be equals to W into L. What is W density of the loading that is 2 and L is span over which it is acting three meters. So this will be equals to 6 kilo. [00:02:05.936] Where exactly the six will be acting? Since it's a UBN, it will be acting at a limit point. So consider that six kilometer vertically downward at a distance of 1.5 meter from here. [00:02:22.276] and only one point five meter from here. Next uh let's start with N A. Here it's a fixed connection. Two reactions one three reactions to our forces A then AY There will be one more resting moment M A [00:02:42.020] Next, six we are done with this, then there's no loading and here twelve kilonewton meter. That moment is acting. Consider that. So this is the complete free body diagram of the beam, A B. Start with the conditions of equilibrium. So the first one is sigma above. [00:02:59.792] is equals to zero. If you have this sigma if x is equal to zero, there is only one hard certain four. That is e x. So e x will be equals to zero. Next apply the second condition sigma f of [00:03:14.832] Y is equals to zero. If you have a sign if y is equals to zero, consider upward forces positive or downward negative. Since a is upward, so plus a y, then six is downward minus six. This is equal to zero. Then a y is equal to six kilon. [00:03:32.528] Remember when we talk about forces you are not supposed to consider the moment. [00:03:38.148] The third condition is sum of the moment of all the forces as well as the couple above the point is zero taking counterquest positive. Then moment of these two forces is zero, but MA is nothing but the resisting moment because of the support. That is different sum of the [00:03:55.920] [ASR low confidence; check audio] So, the moments about is different. Since I have considered here counterclockwise, so plus. [00:04:01.952] Yum vector don't multiply the distance. Next is six kilonute, update the right hand thumb rule. So it is going this direction. It is thumb is going inward clockwise. So minus force into distance. [00:04:17.408] One point five. So one point five. Then next if you go like this, lastly we have twelve kilometer meter. Since it is in a counterclockwise direction, it should be taken as plus twelve. [00:04:31.392] It's already forced into distance, so don't multiply the distance. It's really vector. This should be bigger than zero. Now if you simplify this, you'll get the value of MED as minus. [00:04:46.768] Three kilonewton meter. What this minus sign indicates? We assume the existing moment here at A is in the counterclockwise direction, but actually that is in the clockwise direction. Three kilonewton meter clockwise.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.324] So we've got the value of A, A, then movement about E. These are the reactions at this about E. [00:05:12.484] This is the next question 5/116. For the beam and the loading shown, determine the magnitude of force F for which the vertical reaction at A and B are equal. With this value of F, compute the magnitude of the pin reaction at A. [00:05:28.516] So they have asked us to find the vertical reactions at A and B for which value of F, these two will be equal. Nothing but Ax will be equal to R B or one more

way R B will be equal to Ax. Then [00:05:45.988] Ah F is also a unknown. After that there's one more unknown that is Ax. And in this particular question, Ax will not be equal to 0 because this is the entire beam. [00:05:57.824] Horizontal and vertical. This is a single beam. So here we have a horizontal loading on a beam. So Ax will not be equal to 0. If you don't consider the given condition a y is equal to r b, this will become statically indeterminate because ax a y r. [00:06:15.568] R B as well as F will be unknowns. So four unknowns means statically indeterminate. Remember we have to consider if I is equals to RB. Let's get into the solution part. Let's start with the solution. [00:06:31.300] Yo what is vertical reaction at D is equals to reaction at the end. [00:06:35.952] B. So if is equals to R for this. For what value of this F, if A is equal to R B we have to calculate. Just project all these points vertically downward and start with the B. Uh we have this is E and this is B. Next, uh [00:06:53.504] Only the points load in the free body line, so I'm going to call it as one. If this is what I need to find the result end of that loading, I'll call it as R1. R1 is equal to since it is in the form of triangle, it will be half into base into height. W by two. [00:07:09.280] Half into what is W intensity of 2 into length. Span over exactly UEL acting is 2. So this will be equal to 2 kilonewtons. Next way exactly it will be acting from here to L by three from here. [00:07:26.480] L by 3, nothing but, I am going to consider here, [00:07:31.300] R1 that is equal to 2 kilonewt at a distance of from here 2 by 3. From here 2 times of 2 by 3. So do the simplification you'll get it as 2 by 3 will be equals. [00:07:47.652] So from this side it will be L by three, from here it will be two L by three. So this will be four by three and the remaining will be L by three minutes two by three. So we are done with the distance. Next, here we have two repezoidal loading. [00:08:05.616] One wall span of four meters and one more one more wall span of three meters. First consider this up is the loading of span four meters. Then how to divide? Just draw a horizontal line here. So it is a combination of rectangle and triangle. So the second one is [00:08:23.344] rectangle and the third one is triangle. R2 stands for the resultant point load of the rectangle. Formula will be W into L. What is W intensity? Descent is given as two. So two into what is span? Four meters. So four. It will be eight kilonewton. [00:08:40.912] Where exactly this way? 8 kilonewton will be at the midpoint. Since it is four meter, 4 by 2 will be equal to 2 meters, that will be less than the point A. So this is R2 eight kilo Newton. Span will be 2 meters. [00:08:58.000] This is two meters from here to here. Okay. Next, R three stands for the result of the triangle. Half into base. [00:09:10.240] That so that will be four span into height will be equal to five. Total is five. So five minus two. That you'll get us three. [00:09:22.116] twelve kilon. Where exactly this tool twelve kilonewton will be here? From here two times the deaf L by three from here. L by three. How much will get? [00:09:37.316] That will be left side of the point. T by point three three. Two times the tough. From this side, two times the duff. Four by three. So two into four by three. This is the distance. So eight by [00:09:54.564] Three you will get it. Then from here eight by three. From here eight by three means to the calculation, you'll get it as two point double six seven. So from here to here it will be the two point double six seven. So two is here. Two point double six seven means in between eight and

EV0000106 · EV0000107 · EV0000108 · EV0000109

Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.052] Point E we'll get this slowly. This is twelve kilometer. Remember it is between eight and eight. The distances from here to here it will be [00:10:27.876] 2.67. Next, A we have a pin correction, two reactions, AY, then one more will be horizontal. This will be A. I have taken it left side because anyhow here it is that to balance that A should be towards left. Next [00:10:47.780] One meter we don't have any loading,

next comes the trap side loading. Again draw descent line passing through the this will be the fourth one and this will be the fifth one. Descendant of fourth one fourth will be because since it is a rectangle, W into L will be the descendant. What is W three. [00:11:06.916] Because this is the height of the rectangle and L is three. So this will be equal to 9 kilon. Where exactly this 9 will be acting? That we have to calculate midpoint. Nothing but from this point, 1.5 meter. So from here to here this is 1 meter. [00:11:26.788] Then from here to here, one point five meter we have this R4 of nine kilo Newton. Next, after this, we need to find the resultant R5 will be equals to [00:11:40.784] Triangle so half into base into height base is three and the height will be total five five minus three is the height of the triangle. That will be two. So three kilonut. Where exactly this three will be acting from here L by three from here to L by three. L by three means four. [00:11:57.604] L by three. L A ye is three. Three by three will be equal to one meter. Nothing but it will be left side of this nine kilonewtons. This is three kilonewt and the distance will be one meter. Next we have to add this three meters. [00:12:17.668] We don't have any loading for the next two meters, take that as it is. Next year it's roller connection so only by the reaction RB. Yes, take it as it is. We are done with the free body diagram. Remember we have to write AY is equal to RB. [00:12:31.792] So is equals to A by S we it will become statically indeterminate. Next we have to apply the conditions of equilibrium. Start with the first condition sigma f of x is equals to zero. So sigma f of x is equal to zero if you apply. [00:12:48.496] 2 kilonewt is right side, so plus 2, and this is left side, AX. So minus Ax is equal to 0. AX will be equals to 2 kilonewt. Our assumption is correct. We assume AX is towards left. That is correct. Next is moment equation. [00:13:06.080] I will for moment about point is zero. Taking counterclockwise is positive. [00:13:13.348] Okay. Uh now start and I'll be moving towards side. Start with this two kilonut. You are calculating moment about E, so consider this part of your palm RTA, and you have to apply the right hand thumb rule. RTA. So it is going to be the [00:13:29.984] Thump is coming outward because the force is acting in this direction, so curl your fingers in this direction. It is counterclockwise, so plus two into perpendicular distance will be this is the line of action of the force. And this will be the perpendicular distance that is vertical by three. [00:13:49.028] We are then the moment of this next is this A kilo Newtons. It is acting in the counterclockwise direction, so plus yet. And what will be the distance? The distance from here to here is two meters. So what is the remaining for A? Three minus two will use one meter distance. [00:14:08.900] Next is twelve kilon, that is counterclockwise. Plus twelve distances from year to year it is three. T minus two point double six seven. So three minus two point six six seven. Next, yes. Then [00:14:28.772] Three kilonewton is acting in a clockwise direction. So minus three into perpendicular distance will be from here to here one plus this distance that is again one. So two. Next nine kilon counterclockwise. So [00:14:45.744] Minus nine so nine kilometer clockwise. So minus nine into distance will be one point five from here to your one meter. So plus one. Then comes RB. RB is in a counterclockwise direction. So plus RB into perpendicular distance will be.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.132] From here to here it will be two plus three five five plus one six. Then F is in a clockwise direction. So minus F into perpendicular distances. [00:15:19.140] This will be 3 plus 25. 5 plus 3 8. 8 plus 1 9. This should be equal to 0. Simplify this, you'll get the value of you'll give one equation 6 RB minus 9 F is equal to 15. [00:15:39.236] You'll get this expression condition of equilibrium that is sigma f of y is equal to zero. If you apply this [00:15:49.956] Since it is horizontal, only to consider this. 8 is acting

downward, so minus 8, then 12 is acting downward minus 1, 3 is acting downward minus 3, then 9 downward, minus 9, a y I left, so plus a y. Then plus RB. [00:16:07.108] Minus F is equal to zero. Simplify this. Since E is equal to RB. [00:16:17.284] So substitute this, I'll call this equation one. Then you'll get two RB minus F is equal to 26. This is the equation two. Once you solve one and two, from one and two, you'll get the value of RPS. [00:16:35.940] One eight point one kilo Newton. Then F as one zero three six. One zero point three six kilo Newton. This is the value of [00:16:53.796] AY that is one eighteen point one eight is with this I am going to conclude today's session. Uh go through the assignment questions. If you come across any queries you can post your queries in PSU forum. Thank you.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Full transcript: 2619 normalized words, 67 timestamped segments, 1030.250667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:03:55.920

EV0000102 · EV0000103 · EV0000104 · EV0000105

9 CLASS 41.1 Beams_Numerical

This lesson is indexed from the source video titled CLASS 41.1 Beams_Numerical. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000118 · EV0000119

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.668] Hello, in today's class let's solve few problems on beams. This is the first question for today's class. 5/107. Determine the reactions at A and B for the beam loaded as shown. First, let's look into this beam. There is a beam AB. [00:00:21.604] What's the name of this beam? Do you remember? [00:00:24.836] This part of the beam, this 4 meter is extended behind the support, hence it is called as woven hanging beam. It is also called as combination because it's a combination of pin, support, and roller reaction. Now let's get into the loading. This [00:00:44.388] A beam sorry This load 18 kilon per meter it is a UDL spread it over a span of four meters. So this is UDL and this is also it [00:00:58.948] Couple 15 kilon meter and it's a free vector. Remember this it's a free vector, even though they have specified some point like two meter here with respect to N with respect to B. It doesn't matter where it acts because it's a free vector. The moment of a couple is a free vector. [00:01:16.720] The magnitude is 15 kilon meter. Now the first is to find the support reactions AX AY then reaction at B. [00:01:26.096] So the first thing is we have to start with the free body diagram. See, before that we have to convert this UDL into a point load in the free body diagram. How to convert UDL into a point load? We know the formula, the resultant of the UDLR will be equal to [00:01:41.680] W into L A F rectangle. What is W here eighteen kilonewton per meter and what is L span over which UDL is acting four meter. So you'll get four. [00:01:53.700] The various 72 kilo newton. Next, where exactly the 72 kilo newton will be acting? Midpoint, nothing but with respect to B, L by 2, and with respect to the free end, L by 2. [00:02:07.940] Let's start with a free

body diagram of the beam. Remember we have to project all these points vertically downward, then draw a horizontal line, this will be the beam AB. [00:02:19.332] So we got the beam AB. Next start from any one of the end, either from A or from this free end. I am going to start with the end A. Since it is a pin connection, two reactions, one along X direction, one more Y direction. Ax, AY. Next moving this way. [00:02:36.848] Look at this, you have to move in this direction. Then after two meter, the mentioned the moment of a couple fifteen kilonewton meter, take that as it is. Next, after two meter we don't have any loading here. Then at B it's a roller support. [00:02:51.328] uh that will be having only one reaction normal to the surface of contact so RB R B will be vertically upward. Next next four meters we have a UDL we have to consider the result end of that UDL that is 72 kilonewton [00:03:08.304] That way exactly that will be acting. As I said before, midpoint, L by 2 then L by 2. Now are we done with all the forces acting on the beam? Yes, we have considered each and every force acting on the beam. [00:03:22.596] How to find AXA by R B? Since it is a general force system, we have to apply the three conditions of equilibrium, sigma fx, 0, f y 0, then momentum about any point, 0. Let's start with the first one. [00:03:37.604] Sigma F x 0 if you consider there is only one horizontal reaction that is Ax. So Ax will be atomically 0. [00:03:46.436] That's it. Next y direction we have two react two unknowns A by then R B. So I'm not going to consider sigma if 0. Before that I'll apply the moment equation. Moment about AS 0. [00:04:00.688] This is the second condition of equilibrium. Look at this A. At A we have two reactions, but moment of these two will be zero. So don't consider Ax and A bar. Then moving the [00:04:12.304] Right hand side direction like this. There is a moment of a couple 15 kN meter. Look at the direction, is it clockwise or counterclockwise? It is clockwise, so it should be taken as minus 15. And do we need to multiply the distance? [00:04:28.960] Since it's kilonewtan meter, already the distance is there, it's a free vector. So just consider if it is clockwise, take it as negative, if it is counterclockwise, take it as positive. Don't multiply the distance per moment of a couple. So minus 15. [00:04:46.020] Minus 15, we are done with the moment of this. Then come to this RB, RB is in the counterclockwise direction. Apply the right hand top rule. R B is in anti clockwise direction. So plus R B into distance will be 4 meters.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.288] Next, which is an X load 72 kilonton. It is acting in a clockwise direction like this. Clockwise with respect to A. So minus 72. What is the perpendicular distance from this point to A? It is 2 plus 2. 4 plus 2. 6. [00:05:21.892] So this is how you get the equation that should be equal to zero. There is only one unknown in this equation that is RP. If you simplify you'll get the value of R B as positive. [00:05:34.436] 111.75 kilo Newton. This plus sign indicates that assumption is correct. We assume the RB is up, but force that's correct. Next we are left with only one more unknown that is AY. [00:05:46.308] Apply the sigma FY 0. So AY is upward, so plus AY don't consider this because this is moment of a couple, it's not a force. So next is the RB plus R B because it is upward then minus 72 is equal to 0. [00:06:02.848] Since we know the value of RB, substitute RB, you will get the value of Ay. So AY is 39.75 kN and AY you will get it as negative. Please make sure so AY is actually acting in the downward direction. [00:06:23.204] So this is about discussion. Determine the reactions at A, B, and D for the pair of beams connected by ideal pin at C and subjected to the concentrated and distributed loads. Look at this figure. There are two beams A, C, and C D. These two are connected by IDL pin at C. [00:06:41.892] If you think of finding the support reactions A, C, Y, R B, then [00:06:47.408] R D by considering beam A and D, we have more number of support

reactions, then it will become statically indeterminate because we can't find all the reactions using the conditions of equilibrium. Then what we can do here is [00:07:02.352] We can divide this into two games, A C then C D. When you think of AC, consider the reactions. Here two reactions at A, then one reaction at B. Then when you consider this AC separately, you have to consider the reactions at P at C [00:07:18.880] So two reactions. Today we'll have like five support reactions when we talk about being AC in usually. Now so we can't start with being AC. Next option is being C D. If you think of C D, so two reactions at C, then [00:07:36.752] Only one reaction at D. So better start with beam C D, know the pain reactions, then when you consider AC, you already know Cx and C. So beam A C will become statically determinate and you can find the support reactions at A and B. This is how you have to solve this particular question. [00:07:56.836] Now let's get into the part C D. First one is considered being C D. Now we have to start with a free body diagram. Before that we have to find the resultant of this U V L in your point. [00:08:10.116] What is the formula to find the resultant of UVL? We know the formula R is equals to half into base into height. So consider that. [00:08:18.692] Half into base into height. What is height here? 5 kilon meter. Consider that 5. Then what is the base here span over which the load is acting? 4. Consider that. So you get the result in us. [00:08:32.016] 10 kilometer. Next, where exactly this 10 will be acting from this end, nothing but from C, uh it will be at a distance of L by 3, or from this end it will be at a distance of 2 L by 3. [00:08:46.820] Since we know this C here, with respect to C I am going to consider it as L by 3. Since we know the result end of UVL, let's start with the feet body delta of D beam C D. [00:08:59.172] First, project all the points vertically downward, draw a horizontal line, this will be the beam C D. Then start from any one of the end. I am going to start from this end, C. At C it's a pin. [00:09:13.360] So two reactions, one along H one more along one. Next uh we have move in this direction towards right, we have UVL. Now in the feedbody diagram we have to consider the resentment of UVL. [00:09:28.320] What is the results of UV Lenkil Newton? Where exactly is that it is acting? L by 3 with respect to C. So 4 by 3 will be equal to 1.33. Resultant is 10 kilometers. Then what is the remaining distance from year to year? 4 minus 1. [00:09:46.096] 3T that will be equal to 2.67. Then after 1 meter we have a roller support. So consider reactionate roller. This is the complete free body diagram of being CD. We are done with the free body diagram. Let's apply the conditions of equilibrium.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.764] Start with the first one sigma fx 0, there is only one horizontal reaction, CS, so CS will be 0. Next I'll go for moment equation because y direction we have two unknowns. [00:10:21.476] Consider moment about C0. If when you consider moment about C0, don't consider C S and C Y because they are acting at the same point, moment will be 0. Then go in this direction towards right. The next force we will get is [00:10:36.336] 10 kilometer. How is the direction of 10 kilometer? Look at this. It is clockwise with respect to C. So it should be taken as [00:10:44.848] Minus minus forces 10 into distances 1 point three three. So minus of 10 into 1.33 is the moment of 10 kilo into this direction. The next forces R D it is look at this the direction [00:11:01.572] Counterclockwise so plus Rd into distances, total distance that is 1 plus 4, 5 meters. So this is the equation you get for momentum of Cs 0. We have only one unknown that is Rd. Simply 5 this you get the value of R d s 2 point double six seven kilo Newton. [00:11:20.868] Next, we are left with one more unknown C by. So consider sigma FI 0 apply. Once you apply that, plus C by because upward, downwards, or minus 10, upwards or plus R D will be equals to 0. Since we know the value of Rd, substitute here, you will get the value of C

by S. [00:11:39.716] 7.334 kilogram and our assumption is correct because the answer is positive. CY we are taking it as upward. [00:11:48.356] Now we have to go for the beam AC. But remember when we consider the beam AC here, from the previous thing we got C by S 1.7334 kN upward. But when we consider beam AC, [00:12:05.168] At C we have to consider it as a downward force because it is in a state of equilibrium. The joint C is in equilibrium, so whatever the vertical reaction exerted in this beam C D. [00:12:19.588] C D CY is upward. That should be balanced by the reaction from the beam AC. So here it should be taken as downward. Then only upward and downward C will get cancelled, the joint C will be in a state of equilibrium. [00:12:35.396] Let's start with the B A C. Before that we have to convert this UV L into a point, or again we know the formula R is equal to W into L by W L by 2. W here is we know from here it is 3 kilon 5 kilonewton. So consider that 5 kilo newton here. [00:12:52.752] Then span is 2 plus 2 that is 4 by 2 you will get the value of 10 kilometer. Next way exactly this will be acting with respect to C, with respect to C L by 3, whereas with respect to this point 2 L by 3. [00:13:08.548] Now let's get into the free body diagram, project all the points vertically, downward, draw a horizontal line, this will be the A C. After that at A we have two reactions, horizontal, vertical. Consider those two, then [00:13:23.428] Next, till 3 meter we don't have any loading, solid as it is, take consider 3 meter, then we have a vertically downward force of 25 kN. Consider that as it is. Next 3 meters we don't have any loading. Then we have this. [00:13:38.500] Uh roller support, so consider roller reaction R B. Next we have the resultant of this UV, it is at a distance of L by 3 with respect to C. So total is 4, 4 by 3 will be equal to 1.33 with respect to C. So consider that 1.33 with respect to C. [00:13:58.276] If it is 1.33, then with respect to this point it is going to be 4 minus 1.33 kilonewt. Then at C we have to consider Cx and C by. Cx does not matter because Cx is 0, whereas in when we consider beam C D. [00:14:14.992] C by was upward to balance that when we consider being A C C by should be taken as downward because joint C is in a state of equilibrium. Remember this. Already we know the value of C y here, that is 7.834. Consider that. So we are left with only [00:14:32.304] Three announce ax a y then r b. Start with the first condition of equilibrium sigma fx zero. When you consider sigma fx is equal to zero, ax will be zero because cx is equal to zero. Next consider the next condition, moment about d sero. [00:14:49.168] Moment because of Ax and AY will be zero, don't consider these two. Then go in this direction. The first course is twenty five. Look at the direction up in the right hand thumb rule. It is in the clockwise direction. So it is minus minus twenty five into what is the perpendicular distance? Three. Then

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.636] Just go in this direction. Next course is R B. Look at the direction up here right hand rule. It is counterclockwise direction. So plus R B into what is the perpendicular distance from B to A, it is 3 plus 3 plus 2. [00:15:25.412] So that will be equal to 8. Next we have 10 kN. Look at the direction. It is clockwise. So minus 10 into the distance will be from this point to A. [00:15:37.860] That will be equals to from year to year 2.67 plus 3 plus 3. That is a perpendicular distance for 10. Then look at C y it is clockwise so minus Cy into distances from year to year 4 plus 3 plus 3. [00:15:55.392] And no need to consider CS because it is Z or it is on the same line of action. So once you substitute this, you are going to get the value of RD. So the reactionate B you will get it. Next. [00:16:12.548] Is last condition sigma f y 0. If you consider sigma f y 0, it is upwards, so plus a y then minus 25, plus b minus 10, then minus c y will be equal to 0. So simplify you'll get the value of a y. [00:16:28.720] twelve point nine six kilonewton. So we are done with all the support reactions. Remember reaction

at A will be equal to the root of A X square plus AY square. Since Ax is zero reaction at A is A by the answers are tableture. [00:16:48.644] Okay, finally you are going to get the support reactions. With this, I am going to conclude today's session. Go to the assignment questions, then question bank questions. If you come across any doubts, you can post your queries in PESO forum. Thank you.

EV0000130 · EV0000131 · EV0000132 · EV0000133

Full transcript: 2788 normalized words, 67 timestamped segments, 1027.221333 seconds duration. Low-confidence segments flagged: 0.

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10 UE25CV141A_Unit 3_Class 53_AVSummary

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Graphs and diagrams



Figure G000100. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 1.

EV0000454

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams external Effects

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Figure G000101. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

Beams – External Effects

Shear, Bending, and Torsion

In addition to supporting tension or compression, a beam can resist shear, bending, and torsion. These three effects are illustrated in Fig. The force V is called the shear force, the couple M is called the bending moment, and the couple T is called a torsional moment.

These effects represent the vector components of the resultant of the forces acting on a transverse section of the beam as shown in the lower part of the figure.

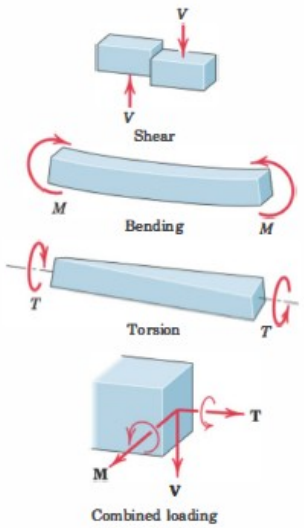


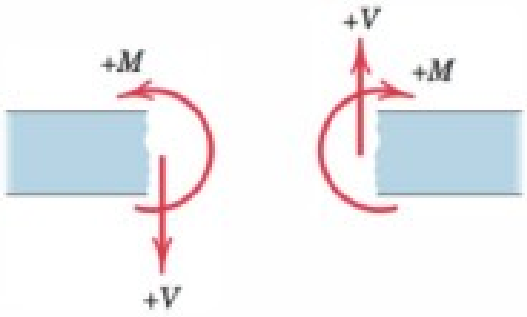
Figure G000102. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 3.

EV0000460

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ENGINEERING MECHANICS STATICS

The conventions for positive values of shear V and bending moment M shown in Fig. are the ones generally used.



S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

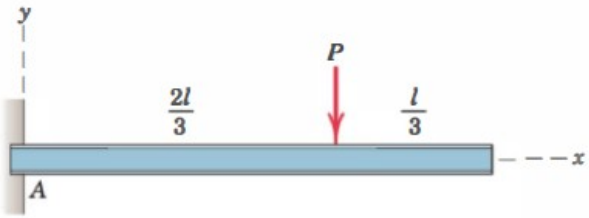
Figure G000103. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS

5/125 Determine the shear-force and bending-moment distributions produced in the beam by the concentrated load. What are the values of the shear force and bending moment at x = l/2?



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Figure G000104. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 5.

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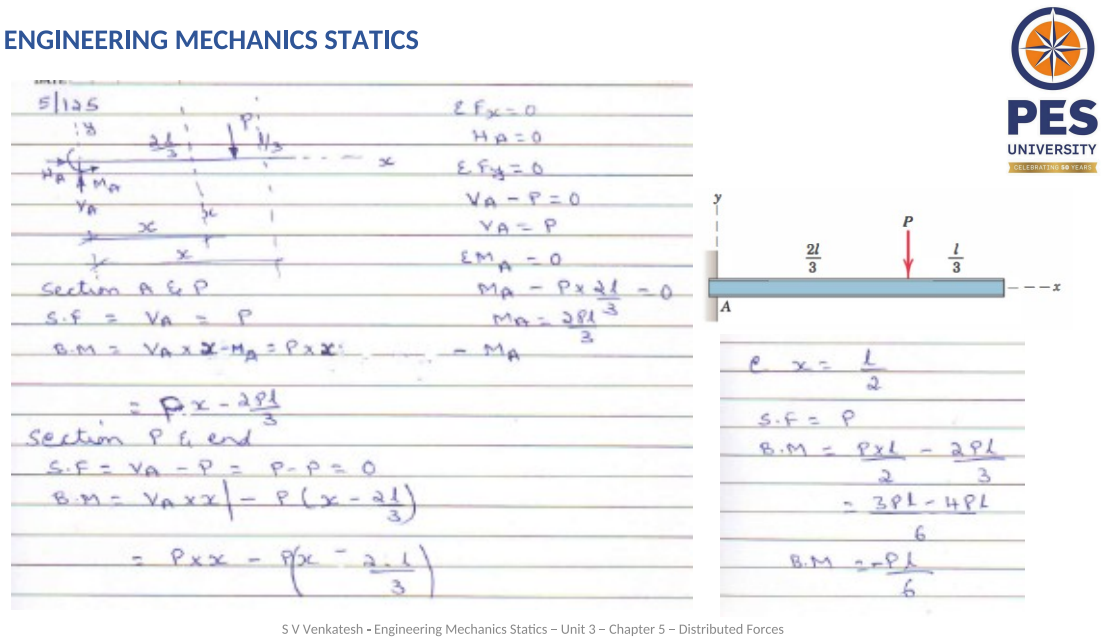


Figure G000105. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 6.

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Figure G000106. UE25CV141A Unit 3 Class 53 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000453 · EV0000454 · EV0000455
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams external Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000456 · EV0000457 · EV0000458
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Beams – External Effects Shear, Bending, and Torsion In addition to supporting tension or compression, a beam can resist shear, bending, and torsion. These three effects are illustrated in Fig. The force V is called the shear force, the couple M is called the bending moment, and the couple T is called a torsional moment.These effects represent the vector components of the resultant of the forces acting on a transverse section of the beam as shown in the lower part of the figure. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000459 · EV0000460 · EV0000461
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS The conventions for positive values of shear V and bending moment M shown in Fig. are the ones generally used. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000462 · EV0000463 · EV0000464
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 5/125 Determine the shear-force and bending-moment distributions produced in the beam by the concentrated load. What are the values of the shear force and bending moment at x = l/2? S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 6 (OCR) ENGINEERING MECHANICS STATICS PES H凸二口 UNIVERSITY CELEBRATING50YEARS 23 3 中 5 2PL-4PL B.M S VVenkatesh-Engineering Mechanics Statics-Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000465
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000466
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000467

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.708] Dear students, welcome to Engineering Mechanics Statics class. My name is Dr Sv Vingitish from Civil Engineering under the Department of Science and Humanities, PS University. This is class number fifty-three. [00:00:19.012] We are going to be starting the new topic in unit three under distributed forces, that is Teams internal effects. So from the engineer mechanics statics book. [00:00:35.492] Written by Miriam and Craig. It's the eighth edition and SI Mosh. So as I told you, this is from Unit Three, from Distributed Forces. Chapter five, Beams Internal Effect is the [00:00:52.900] Now we have discussed about beams earlier, where we have uh discussed about the different types of beams, that is [00:01:11.332] Determinate and indeterminate based on the type of support and the number of supports that is going to be acting on the beam. [00:01:21.380] If it is equal to the number of

equations of equilibrium or less than the number of equations of equilibrium, it is determined B. If the number of supports, external supports, are more than the available number of equations of equilibrium. [00:01:36.900] Then they are treated as indeterminate beams. However, the indeterminate beams can be determined or can be the support reactions can be determined using other equations. Now in this topic we will be discussing about the internal effect. [00:01:55.044] The external effect or the support reactions which we have found or which we know how to find, but now we are trying to find what are the internal forces that is going to be acting in the beam because of the application of the external forces and as well as the external support reactions. [00:02:12.516] So, the internal effects on the beams are three. They are shear force, bending movement, and torsion. Simply called a shear bending untorsion. Now, what is this shear bending and torsion? In addition to the [00:02:31.204] Tension and compression that the beam is going to take. A beam has to resist all these three forces that is shear bending and torsion. So let's look at each one of them. You can see the different diagram that is given. [00:02:49.220] The first one indicates the shear force. Now if you take any section, okay this is the section that it is considered [00:03:00.356] There could be forces acting whatever is indicated here. This will make that section two separate out or it gets sheared off. [00:03:09.860] Okay, so that's why this is called as shear force. If the forces are acting like this, then it is going to get sheared off or the beam is going to get separated. That is what is called as shear force. Next coming to bending moment, if we apply movement like this, okay. [00:03:28.644] Then the beam is going to bend because of this bending, it is called as bending movement. Now this bending moment is not the same as whatever movement that we used to find earlier. Earlier we used to find moment because of all the forces that is acting on a particular beam. [00:03:46.660] But the bending moment is we are going to find out the moment acting only to the left side of the section or to the right side of the section. [00:03:56.772] Okay, considering all the forces that is acting to the left side of the section or to the right side of the section, whatever value that we get will be called as the bending moment. So there's a clear difference between what is bending moment and what is moment. Moment at any particular point is equal to zero as the one of the equations of equilibrium. [00:04:16.580] That is going to be applied when we take entire beam as one, including the support reactions, the applied forces, everything put together, what is the moment that is acting at a particular point should be equal to zero. [00:04:30.148] But whereas bending moment, we are taking only the left side of the portion and whatever is the load that is acting in the left side of the portion, because of that whatever is the moment that is going to be coming, that is treated as bending moment. Or instead of taking left side, we can also take to the right side. [00:04:47.972] For a section, whatever is the force that is acting to the right side of the section on the beam, if we take the movement of all those forces at that section, then whatever value that we are going to get will be called as the bending moment. Okay. Now if you look at this.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.932] This beam is subjected to two movements. One is clockwise, the other one is anti-clockwise, at the two ends. Okay, which makes the beam to bend like a [00:05:16.772] Electrical wire between the two cables, something like this, it is going to bend. Okay, this this is going to bend like this, okay? Just like your half moon. Okay. So when a beam bends like that. [00:05:31.620] The top zone or the top fiber, the extreme top fiber would undergo a compression and it will be under reduction in the length and it will be under compression. Whereas the bottom one, the extreme bottom layer, if you take, it will go, it is

going to elongate and that would be under tension. [00:05:51.332] Now if you move from top to bottom, somewhere in between you will have a layer which will neither elongate nor reduced, which is called as the neutral layer, or the axis along which the neutral layer lies. [00:06:06.928] Is called as a neutral axis. The zone above the neutral axis or the neutral layer is under compression, and that is called as a compressive zone, whereas the zone which is below the neutral axis or the neutral layer is called as a tensile zone. Okay. [00:06:23.748] Next, we have this that is the torsion. If the movement that is acting because of the application of the load along the axis of the beam, that is the longitudinal axis of the beam, that movement is referred to as torsion. So at any particular point in a beam. [00:06:43.428] The internal forces that is going to be acting are the combination of all these three, which is indicated here. So you can see there is shear force V. [00:06:52.708] There is bending moment M as well as there is torsion P torsional moment T acting along the longitudinal axis of the beam. Okay So this is what is the internal effect of a beam [00:07:05.380] Unlike the external effect where we used to have the supports, replace the support by its reaction, determine those reactions by using the equations of the curve. Here, when we have to find out the shear force of bending moment. [00:07:18.404] We have to use a different sign convention. So let's look at what is a sign convention that we will be using to determine the shear force and bending moment. Now, if I want to find out the shear force and bending moment, say along this axis, okay. [00:07:36.420] Then I will consider as I have told you earlier either to the left side or to the right side. So in this case, I will consider the right side of the beam from this section. If there is a point lot that is acting downwards like this, what is indicated, we consider that as positive. [00:07:57.028] Whereas if it is opposite to this, that is acting upwards, then we will consider it as negative. This is the sign convention that needs to be followed for shear. Similarly, if I take a section and take the beam to the left side of the section, [00:08:13.892] If there is a point load acting upwards, then we will take it as positive. If it is acting downwards, we will take it as negative. I hope this is clear. [00:08:22.212] So, based on where the shear force or the point load is acting, whether it is to the left side of the section or right side of the section or left side of the section, the second convention defers. If it is to the right side of the section and if it is acting downwards, [00:08:38.308] If it is acting downwards, then we will take it as positive. If it is to the uh left side of the section, and if it is acting upwards, then we will take it as positive. Vice versa to this has to be taken as negative. I hope this is clear. Now coming to the bending movement. [00:08:58.692] If I take a section here and if I take the portion of the beam acting to the right side of the section, if it is anti-clockwise, that moment whatever we are going to act, okay, will bear make the beam bend like a [00:09:17.700] Electrical wire between the two poles, half of it. Half of the electrical wire between the two poles. If you take only one pole and half of the electrical wire, it is going to bend like this, bend like this, bend like this. [00:09:31.748] Okay. So this you can see that it is like a half of the cable, electrical cable. So that we call it as sagging. If the bending moment causes the beam to sag, then we take it as positive. [00:09:47.428] Okay, if it is in the opposite direction, if this movement would have been in the clockwise direction, the beam would bend like this. You can see that this is like half of the uh

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.060] Rainbow, right? So if it bends like a rainbow, or if the movement acting on the beam makes the beam to bend like a rainbow, we call that as hogging and we take that as negative. Now next coming to this section, if you take a section and if you consider the left side portion of the beam. [00:10:21.188] If the

movement is clockwise, it is going to make the beam bend like a sagging, that is like a half moon, okay, or an electrical cable between the two uh poles. So that we will take it as positive, which is sagging. [00:10:38.596] If it is on the opposite direction, that is if the movement here, if this movement would have been in the counterclockwise direction, then the beam would have been like this, which is similar to your half rainbow. So that is considered as hogging and that could be taken as negative. I hope [00:10:58.564] You have cleared with the sign convention what needs to be used for shear force and many movement. However, we had three internal effects, but we'll be concentrating only on the shear force and many movement because torsional is not going to be coming here, but you should know what are the three forces that is acting. So, however. [00:11:17.348] To understand what we need to do in this section is we need to draw the shear force diagram separately across the length of the beam. Similarly, we need to draw the bending movement diagram across the length of the beam. Now, what is the advantage of drawing the shear force and bending movement diagram is [00:11:37.540] At any point along the length of the beam, if I have the diagram, I can find out what are the values of the shear force and bending movement. The design of the beam will be based on the values of the shear force and bending movement. [00:11:53.604] The bending movement will actually talk about the longitudinal reinforcement that we need to be provided in a reinforcement concrete beam. [00:12:02.948] Whereas the shear force will determine what are the stir-ups that is to be provided in the beam for designing that beam or to carry the load that is coming on that. Okay. Now what we are going to do in this section is only draw the shear force reminiscement diagram or to determine what is the value of the shear force reminiscement at a particular section. [00:12:23.076] So let's take a simple example to understand how do we draw the shear force and diagram. I've chosen a problem 5 r 125. The statement of the problem goes like this. [00:12:35.268] Determine the shear force and bending movement distribution. Distribution means across the length of the beam, what is the distribution of the shear force and bending moment. So we have to draw the shear force and bend time. Produced in the beam by the concentrated load. So there's a concentrated load that is acting. [00:12:52.612] And they have also asked you what are the values of the shear force and bending moment at x is equal to L by 2. This is what is asked. Now, if you read the diagram that is given, this is a cantilever beam because you have a fixed support at A. [00:13:10.180] The other end of the beam is free, there is no support, so this is a cantilever beam. And there is a point load that is acting at a length of 2l pi 3 from A. Right? [00:13:22.212] Now, the remaining length of the beam is L pi 3, which is from the point where the point load is acting to the point where the support is there. Okay. So now [00:13:35.780] To draw the shear force unbending moment diagram, we should consider a section in the portion where there is no variation in the load. That is from A to this point where the point load is acting, there is no load. So we can consider a section here. [00:13:49.520] And then we can get an equation for shear force and minimum diagram and we can substitute the value. [00:13:56.804] of the point from A, whatever is the length, then we can get the shear force amenement diagram or shear force amendment at that point. Similarly we can take one more section between the point where the point load is acting to the end of the beam. And we can also get one more equation. So there will be two equations for this because there is only one point load. [00:14:14.948] And substituting the value for in that equation, we will get the value of the sheer monument diagram at a particular point. [00:14:25.828] Okay now let's look at the solution. This is a solution for drawing the bending mode and cheap for diagram. First thing that we need to do is we have to draw a free body diagram. Free body diagram is free of all the supports. The

supports have been removed and replaced it by the respective reactions. [00:14:42.896] Then we have the point load, distance is also mentioned, the beam is mentioned with respect to a horizontal line because the depth of the beam is not going to affect the shear force embedding moment. Now we need to first find out the [00:14:57.060] Support reactions for which we will use the equations of equilibrium. This is very well known to all of you. You have already solved several problems on external effects of the beam. So we will get the values of the vertical reaction, horizontal reaction, and movement.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:14.084] Horizontal reaction is zero, vertical reaction is P. Moment we will get as 2PL by 3, which will be acting in the counterclockwise direction. Now to find out what will be the equation for shear for something. [00:15:27.716] Diagram or the bending moment. I will take a section here that is X. The distance from A to this section is considered as X itself. Now I'll write the equation for shear force. So as I as we have discussed, we should take either to the left side of the [00:15:41.892] Section the beam or to the right side, if you take it to the left side, what we get for shear forces we have VA which is equal to P, that is what we have got it from here. Now VA is acting to the left side and upwards is positive, so we will get a positive value for VA. [00:15:58.788] Similarly, when we take the bending moment at this point, also make some of all the movements that is acting to the left. So we have the vertical force here VA into the distance X, which will produce uh or which will make the beam to bend like a sagging, so that is positive. [00:16:17.316] Then if I take the movement MA, it is acting in the counterclockwise direction. So if I apply that movement on a beam, it will bend like it will bend like a rainbow, half of the rainbow. So that would be negative minus MA. [00:16:31.876] So substituting the value of MAR and substituting the value of V A, so it becomes P into X minus because this MA, we know that it is Augging, so it is negative 2 P L by 3. This is the equation for shear force that is VA equals P, and the bending moment equation is P x. [00:16:49.312] Minus 2PL by 3 is the equation for many limited. Now if I take a section here and take the distance from A to this section as X, again we will write the equation for shear force. So here we have VA acting to the left side and upwards positive. We also have this point load P. [00:17:08.516] Now that is also acting to the left side of the beam and it is acting downwards, so it is negative. So we get shear force at this section as VA minus P. Okay. So VA is also P, so P minus P will be 0. So shear force. [00:17:23.088] In this zone, that is from the point where the point load is acting to the end of the beam, shear force is zero. Next to take the bending moment, again we will take the moment, so we will get VA into hex distance. Okay. And [00:17:38.756] You have the moment MA which will be acting or making the beam to bend like a hogging, so it is negative, whereas VA is sagging, make the beam to bend like sagging, so it is positive. [00:17:50.820] So minus P into the distance, this is P, the distance that we need to take is this, which will be X minus 2L by 3, which is the length of the point load acting from A. So simplifying that, this is the equation that we get, which is the bending moment equation. [00:18:08.612] Now we have got a distribution equation for the shear force and bending movement. Using this, we can draw the shear force and bending movement diagram. But however, in the problem they have asked you to find out [00:18:21.476] The value of the shear force admitting moment at a point where x is equal to L by 2. So we take x is equal to L by 2, when we take x is equal to L by 2, it is going to be falling in this zone that is between A and the point where the point load is acting. So substituting the value in this equation. [00:18:40.388] So we will get shear force as P, so that is what is indicated. Now coming to bending moment, it is

P into X, X is L by 2, so P L by 2 minus we will get 2 P L by 2. Simplifying that we will get the value for the shear force it is. [00:18:57.152] Minus P L by 6. This is the value that we are going to get. Okay. I hope all of you have understood how we can get the internal effects or what are the internal effects that is acting on a beam. How or what is the sign convention that should be used for shear force and bending movement diagram. And [00:19:16.100] Also, you have understood how do we get the distribution of the shear force and bending moment diagram between the zones or the points where there is no variation in the load, and how to get the value of the shear force and bending moment bending moment from the distribution. Okay. [00:19:33.700] So we will stop today's section here. We'll be continuing in the next class. This was class number 53. In the next class, we will be solving a few more problems on how do we draw the bending moment and GFO diagram and how do we determine [00:19:49.360] the shear force and bending movement diagram or shear force and bending movement values at a particular point. Okay. So thank you very much. My name is Dr. Sv Vengtesh. I am from the civil engineering department of science and humanities PS University. Thank you.

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Full transcript: 3406 normalized words, 76 timestamped segments, 1205.973 seconds duration. Low-confidence segments flagged: 0.

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11 UE25CV141A_Unit 3_Class 54_AV Summary

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Graphs and diagrams



Figure G000107. UE25CV141A Unit 3 Class 54 Lecture PPT · source slide page 1.

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Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams Internal Effects

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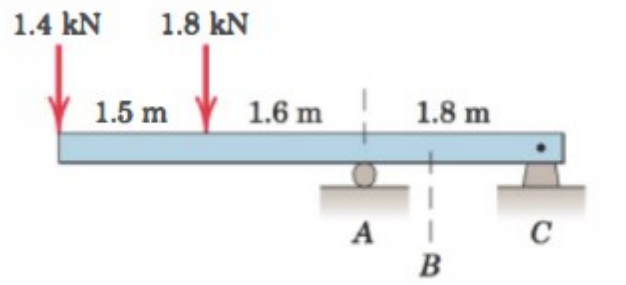
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ENGINEERING MECHANICS STATICS

5/126 Determine the shear force V at a section B between A and C and the bending moment M at the support A.



Problem 5/126

Figure G000109. UE25CV141A Unit 3 Class 54 Lecture PPT · source slide page 3.

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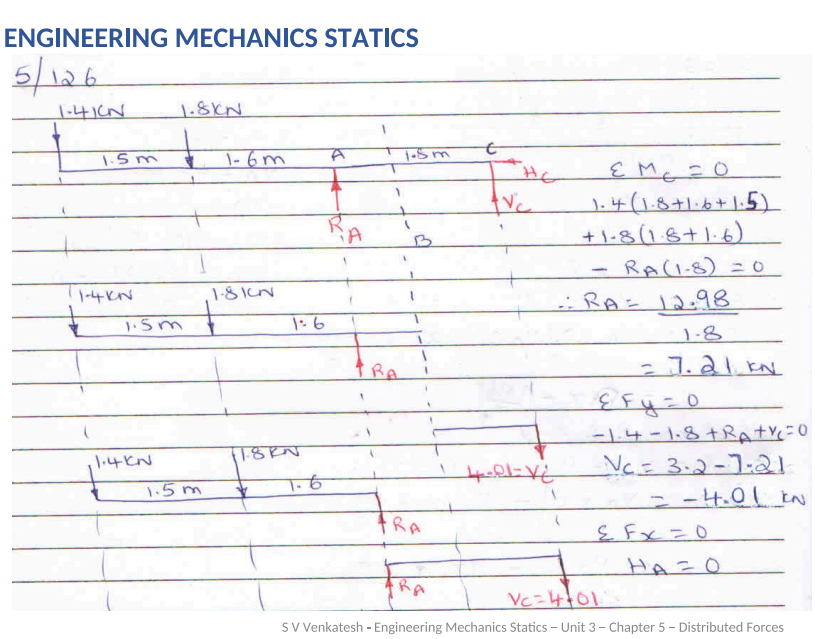


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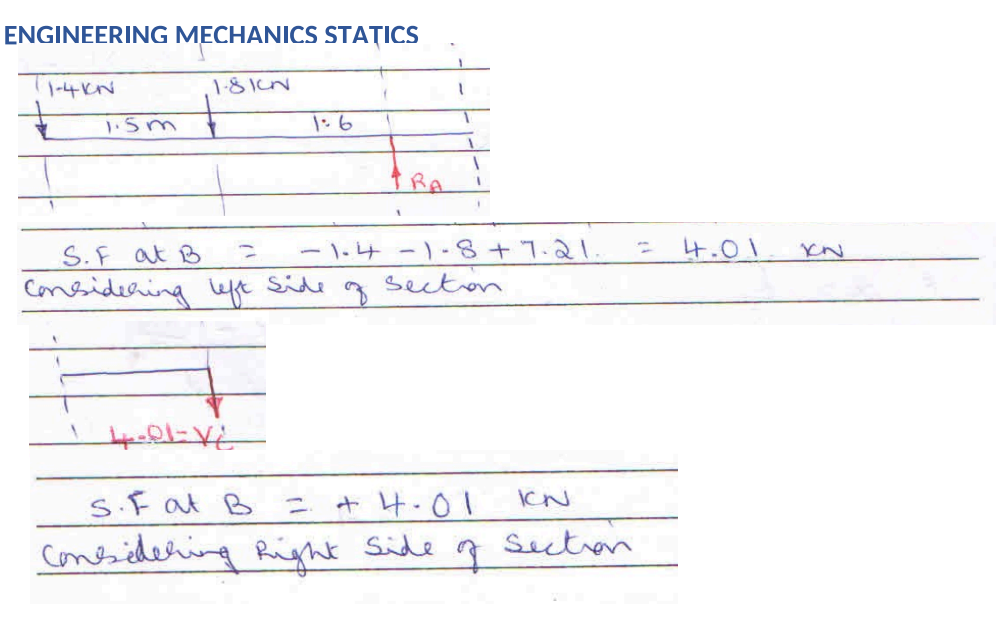


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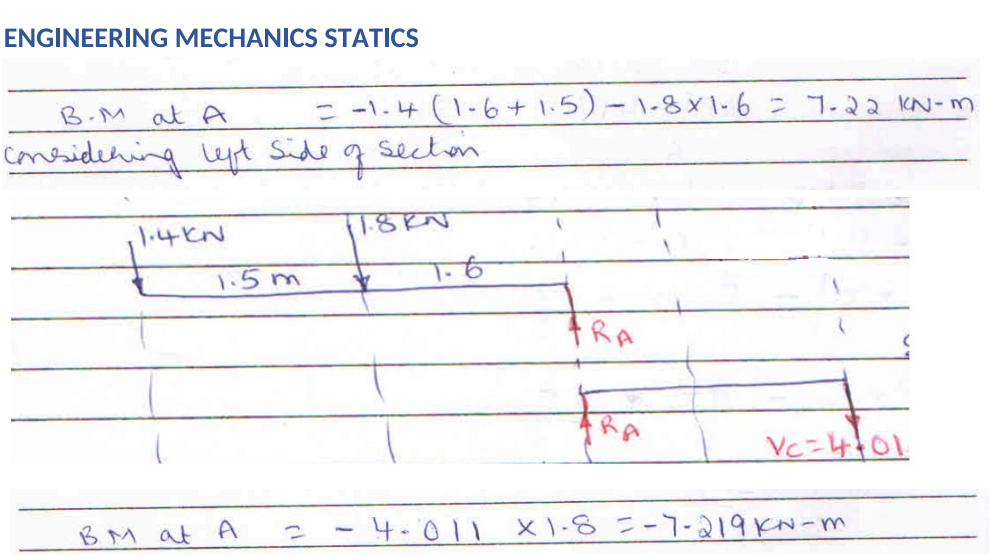


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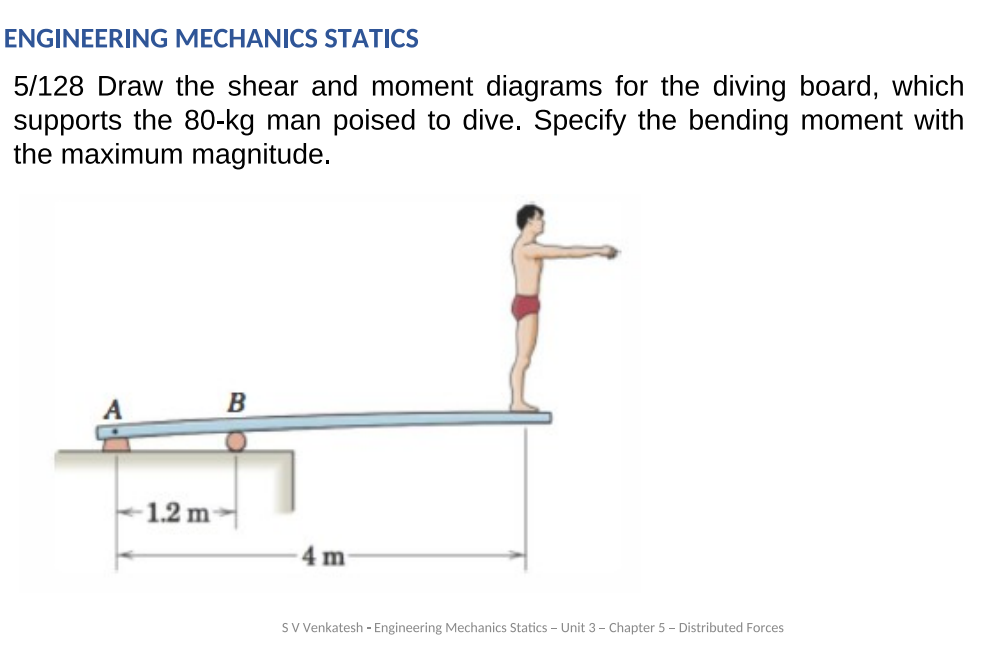


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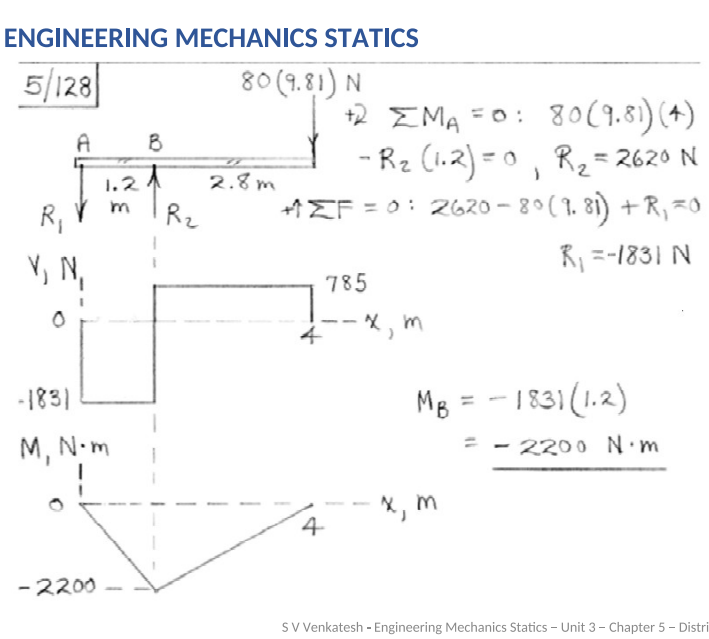


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Figure G000115. UE25CV141A Unit 3 Class 54 Lecture PPT · source slide page 9.

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Important points and why

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2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams Internal Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000471 · EV0000472 · EV0000473
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 5/126 Determine the shear force V at a section B between A and C and the bending moment M at the support A. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 4 (OCR) ENGINEERINGMECHANICSSTATICS 1Q 6 1.8KN PES CN UNIVERSITY 1.5m m CELEBRATING5OYEARS 1. 4(1.8+1.6+1.5) +1.8(1 8+1.6) RyC8) 二0 -462 18102 RA-1.98 S3 18 2 7. a1 rN -14-18+B+y=0 18N 14EN Nc32-1:Q1 1. 6 1.5m -40LEn Vc-4101 S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 5 (OCR) ENGINEERING MECHANICS STATICS -462 18102 PES S3 1.6 UNIVERSITY CELEBRATING5OYEARS 114-18+1Q一. F.01 S.FatB SLu Sect c s←2B=+千O1 CZ SLU Stn Cmrderla S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 6 (OCR) ENGINEERING MECHANICS STATICS (1-6 +1.5)--1-8x1-6 = 7. Qa 二-14 z-3 PES BMNA S& o Sectm CmsiduraUyr UNIVERSITY CELEBRATING5OYEARS 18P2 1.4EN 1.5m RA Vc-4101 x1.8 -7-Q19Kn-m BMaA S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000474 · EV0000475 · EV0000476
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6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000481
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS 5/128 Draw the shear and moment diagrams for the diving board, which supports the 80-kg man poised to dive. Specify the bending moment with the maximum magnitude. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000482
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 9 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000483
9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000484

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.060] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. SUV Vinkitish from Civil Engineering under the Department of Science and Humanities, PES University. This is class number 54.

[00:00:18.340] We are in Unit three, Distributed Forces. And chapter is Distributed Forces. Topic is beams internal effect. [00:00:33.572] In the last class, that is class number 53, we had discussed about a problem that is phi bar 125. So we will be going ahead and solving few more problems in today's class. [00:00:51.524] So first problem that I've chosen for today's class is phi bar 126. The statement of the problem goes like this. Determine the shear force V at a section B between A and C. [00:01:11.364] And the bending movement at the support A. The diagram, if you read the diagram, what is given in the diagram is there is a point load that is acting. [00:01:26.276] The point load is acting at the free end of the beam of magnitude 1.4 kilonewton and it is acting downwards as you can see. Then we have one more point load, which is acting at a distance of 1.5 meters from the free end, that is from the left end. [00:01:46.052] of magnitude 1.8 kilonewtons. Then we have a roller support at point A, which is at a distance of 1.6 meters from the lower point 1.8 kilonewtons. And you have one more support at point C [00:02:03.652] Which is at a distance of 1.8 meters from support A. If you observe the support A, as I have told you, it is a roller support, and the support at C is a hinge or a pin support. [00:02:16.804] So, first thing that we have to do here is to determine the support reactions as we have to determine the shear force at B. Whether we take to the left side of the section or to the right side of the section, B, we need to determine the support reaction at A. [00:02:37.316] And C to find out or to take the section to the right side of the beam. So to first to find out the support reaction, as we have explained in beam's internal effect, we have to use the three equations of equilibrium. [00:02:54.468] So first let us draw the free body diagram. The free body diagram looks like this. What do we have in the free body diagram is the point load of one point four kilometers is acting downwards at the free end, that is the left side end. [00:03:08.612] Then one more load that is of one point eight kilonewtons, which is acting at a distance of one point five meters from the left end. [00:03:16.804] Then we have the support at A which is being replaced by its reaction R A, since it is a roller support, the reaction will be only one and it is acting upwards and it is denoted as RA. [00:03:29.444] Then we have a support at C which is a pin support or a hint support, so we will have two reactions, one vertical, which is denoted as V C and one horizontal, which is denoted as H C. So first let's find out the values of RA, V C and H C. [00:03:47.716] For which we will use the equations of equilibrium. So we will take first equation of equilibrium as summation of the moment to be equal to 0, and I have taken the moment at point C because we have two reactions acting at this point. [00:04:03.076] So it would be easier for us to determine the other reaction that is acting at A. So when I take moment about C, what do we get? We get this point load. [00:04:13.028] Which is 1.4 kilometer into the perpendicular distance is totally equal to 1.5 plus 1.6 plus 1.8. That is what is indicated here, or you can take it from C, which is 1.8 plus 1.6 plus 1.5, and that's what is indicated here. And this [00:04:33.124] Point load that is one point four kilometer will produce an anti clockwise movement, so we will take it as positive. [00:04:38.948] Similarly, this point load of 1.8 kilonewtons will also produce an anti-clockwise movement, so it is 1.8 into the perpendicular distance is 1.8 plus 1.6, that is 1.8 into one plus 1.6 into the load as 1.8. [00:04:57.380] Which is also a positive because it is acting in the anti-clockwise direction. Then we have one more point load that is acting because of the support at A, which is upwards, which will create a clockwise moment at C.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.596] So this Re into the perpendicular distance of 1.8 meters is what is written here with a negative sign. And that is what we have the forces acting on this beam. So that should be equal to zero. [00:05:24.356] So from this

we will be able to get the value of RA which is equal to seven point one kilometers, and since it positive, RA is now acting upwards itself. Okay [00:05:36.772] Next, to find out the other support that is V C, we will take all the summation of all the forces, algebraic sum of all the forces acting along the y direction to be equal to 0. So we have 1.4 kilonoton acting downwards, so it is negative, 1.8 kilonotons acting downwards, it is also negative. [00:05:56.036] But one R A which is equal to 7.2 kilon, which is acting upwards, which will be positive. So considering all that and the reaction at C which is V C, we have considered it to be acting upwards, so positive to be equal to 0. So from this equation we will get VA. [00:06:15.172] Which is equal to minus 4.01 kilonewt, which indicates that since there is a negative sign here. [00:06:23.652] That indicates that the reaction at C is not acting upwards, it is acting downwards. And that is possible because we have a support at C as an heat support, so we can have either upwards reaction or downwards reaction. So VC is now acting not upwards, it is acting downwards. [00:06:41.892] I hope that is clear. Next we will take the other equation of equilibrium that is algebraic sum of all the forces acting along the x direction to be equal to 0. So we will get HA to be equal to 0 because there is no horizontal force acting on the beam, so H will be equal to 0. Okay. Now [00:07:00.484] What is asked in the problem is to determine the shear force at a section B B, that is what is indicated here, which is just to the right side of A. Okay, this is what is the section B. [00:07:17.924] Now to determine the shear force at a section B, we can take or consider the beam to the left side of the section or to the right side of the section and then we can solve it. [00:07:29.764] However, here in this problem I have shown you by both, whether you take to the left or to take to the right, using the sign convention that was mentioned in the last class, you will get the same value for the shear force. That is what I've indicated and that is why [00:07:44.944] You can see the free body diagram that is written here, where I have taken only the beam acting to the left side of the section beam. This is what is the beam body diagram. Similarly, I am also gonna take the section of the beam which is acting to the right side of the section, that is what is indicated here. [00:08:04.740] So we have the point load of one point four kilowattons, one point eight kilowattons, and re reaction R reacting upwards when we consider to the left side of the section. When we consider to the right side of the section. [00:08:18.724] We have only VC and what we have got for V C is negative, which indicates that it is acting downwards, so that is what is indicated here. Next, they have also asked us to determine the movement about A. Similar to the shear force, I have also I have written the diagram of the beam. [00:08:37.924] Which is to the left side of A and right side of A, that is this one. So left side of A what do we have? We have point load one point four and one point eight, both of them acting downwards. And at A we also have a reaction at A R A. [00:08:52.996] Then if you take to the right side of the section, again we have a point load or the reaction at V C that is acting, which is downwards, which is four point zero one, and the reaction at A that is acting. [00:09:04.484] So this is the free body diagram that I have written for the beam and we have determined the support reaction. Going forward to find out the shear force at V here, if I take the section to be acting to the [00:09:17.220] Left side or the beams acting to the left side, what do we get? We get 1.4 acting downwards, which is negative. 1.8 is also acting downwards to the left side of the section, so it is negative. Then we have [00:09:33.828] RA which is acting to the left side of the section upwards which is positive. So we get the summation of minus 1.4, minus 1.8, plus 7.21, that will be the shear force acting at B. [00:09:47.908] Whatever value that we get, that should be the same value what we get it when we consider to the right side of the section. So when we consider this, what do

we have is we have the point lot acting to the right side and it is downwards, so it is positive, and the value of BC which is 4.08.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.084] So whether we take it here or here, we should get the value as four point zero one itself. Let's see that. [00:10:12.964] Then coming to the bending moment, we take the bending movement at this point. So we will have 1.8 into 1.5 plus 1.6, which will produce a hogging bending movement, so that is negative. [00:10:26.944] Similarly for one point eight into one point eight, which also produces an argument, which is also negative. Now when we take uh the other side that is the right side of the beam, when we take the moment about here, we will get that as [00:10:39.680] 4.01 into this distance, which is equal to 1.8. So that would also be Hogging, so that will also be negative. So these two values also has to match. Now this is what is indicated here. Whatever I have discussed, the beam taken to the left side of the section B B. [00:10:59.652] Shear force at B will be equal to 1.8, which is acting downwards. Then we have 1.8 which is acting downwards, 1.4 is also acting downwards, both of them is indicated here. Plus RA which is acting upwards, so that is 7.21. So we get the value of 4.01 kilonotomer, which is a positive value. [00:11:19.204] Now considering to the left side of the section is what we get here. Considering to the right side of the section, that is this one, we get the beam acting to the right side and the point rod is acting downwards. So right side and downwards will also be positive. So shear force at B is equal to plus 4.01 kilogram time. [00:11:39.044] Next, to find out the bending moment, similarly, take the section to the left side from A. So movement will be 1.4, which will produce a hugging bonding moment, so it is negative. 1.6 is the distance, that is this distance, plus 1.5, so that is what we have written here. Minus [00:11:59.300] Uh one point eight, which will also be hanging. [00:12:02.336] into 1.6 so we get the value as 7.22. This is with respect to when we take the beam to the left side of the section. So we be if you take to the right side of the section what do we get? We get 4.01 into this distance of 1.8 which will also be Augin. So we will get the value as [00:12:19.488] Minus 7.219. So whether you take to the right side or to the left side, you should get the same value. That is what I have proved here for shear force as well as petting movement. I hope you have understood this problem. Next, let's go to the next problem, which is 5 bar 128. [00:12:38.980] The problem statements go like this. Draw the shear and bending movement diagrams for the diving board, which supports the ATKG man poised to dive. Specify the bending moment with the maximum magnitude. [00:12:57.796] So from the statement, what we can understand is we have to draw the shear force and bending movement diagram for the board, and the board is now the beam. Okay. Now on the board there is a 80 kg man which is who is acting, and if you see the diagram. [00:13:14.372] In the diagram, the man is acting at the free end of the board. And also we should specify the maximum winning moment. [00:13:21.540] Magnitude. Okay now the board if you observe very clearly at A it is hinged, so we can have a reaction upwards or downwards or right side and left side, whereas at B it is a roller support, so we can have only one reaction that is going to be acting upwards. [00:13:37.220] And also the distance between A and B is one point two meters, whereas the distance between A to the free end of the beam or the board where the person is standing is four meters. [00:13:49.476] Now to draw the free body diagram, replace the support by its reaction at A, it is replaced by a reaction R one, at B it is R two, then the weight of the man which is eighteen to nine point eight one newtons is acting. [00:14:06.180] So, first we have to find out the support reactions for which we will take the equations of equilibrium, considering summation of all the

moments acting at A to be equal to 0. [00:14:16.804] We will get when we take moment about A, we will get the weight of the person, which is 18 to 9.81, and the distance from A to the point where the mo where the person is is is uh [00:14:33.636] Is acting or the force of the person is acting is 4 meters, so that is 4 here. That would produce a clockwise movement. So we will take that as positive, whereas R2 is producing an anti-clockwise movement. [00:14:49.412] So we will take that as negative. So R2 into the distance is 1.2 meters. So that is 1.2 is equal to 0. The second convention used is given here, clockwise is positive. So that is what we have used. So from this equation we will get R2 which is equal to 2620 newtons.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:07.716] Then we will take summation of all the forces acting upwards to be equal to 0. So we get uh 2620 acting upwards, then the weight of the person is acting downwards, which is negative. Now here [00:15:23.908] Of course it is indicated that the reaction is acting downwards, but what is taken here is that it is acting upwards, that is why it is positive, which is equal to zero. But once we calculate the value, we get R1 to be equal to negative, and that is why this is indicated as downwards. [00:15:41.360] Remember that this R1 is positive because we are assuming it to be acting upwards. So we get a when we saw this we get a negative sign which is minus 1831 newtons. So that is acting downwards. Okay. [00:16:00.068] So I hope this is clear. Okay. Now to draw the shear force, there's one technique that we can use, very simple technique. What we can do is [00:16:14.116] You write down uh horizontal lines under the point loads plus the support reactions, that's what is indicated here. So you have drawn the lines. Then draw a datum, draw a datum line, which indicates a 0, 0. Okay. Then uh [00:16:34.308] You go down as the arrow of the support reaction is indicated, which is downwards of R1, then there is no load, so horizontal line. Then at B we have a reaction acting upwards, which is 26. [00:16:50.404] 50. You take it from the horizontal line above, so you get a value of 785. That is 2650 minus 1875. So then from uh [00:17:04.164] The point where the point load is acting to the edge of the beam, there is no load, so horizontal then go down by H and 5. So you get the shear force diagram. [00:17:13.060] For bending moment, determine the moment at B either taking the section to the left side or right side. So you get the reaction that is acting at R1 into the distance, so you get the maximum bending moment. So then from at the end A and the end points, the bending moment is going to be 0. So that is 0. So you can [00:17:33.284] And then draw the bending moment diagram. I hope all of you have understood. Now next we have asked you to find out the maximum bending moment in the beam, which will occur at the point where the support is acting at B that is equal to the R1 into the distance of [00:17:52.996] A to B. So that will be the maximum bending moment that is going to be acting. I hope all of you have understood this. We are going to be stopping this uh class here. Thank you very much. My name is Dr. Sv Mektesh from civil engineering and the department of [00:18:08.708] Science of Minnesota, PS University. Thank you. See you in the next class. That is class number 55. Thank you.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Full transcript: 3042 normalized words, 71 timestamped segments, 1097.3963 seconds duration. Low-confidence segments flagged: 0.

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12 UE25CV141A_Unit 3_Class 55_AV Summary

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Graphs and diagrams



Figure G000116. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 1.

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Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams Internal Effects

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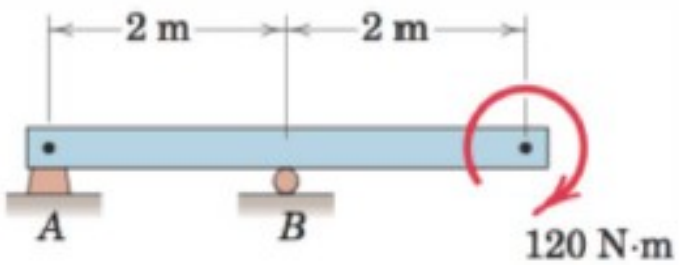
Figure G000117. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

5/127 Draw the shear and moment diagrams for the beam subjected to the end couple. What is the bending moment M at a section 0.5 m to the right of B?



Problem 5/127

Figure G000118. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 3.

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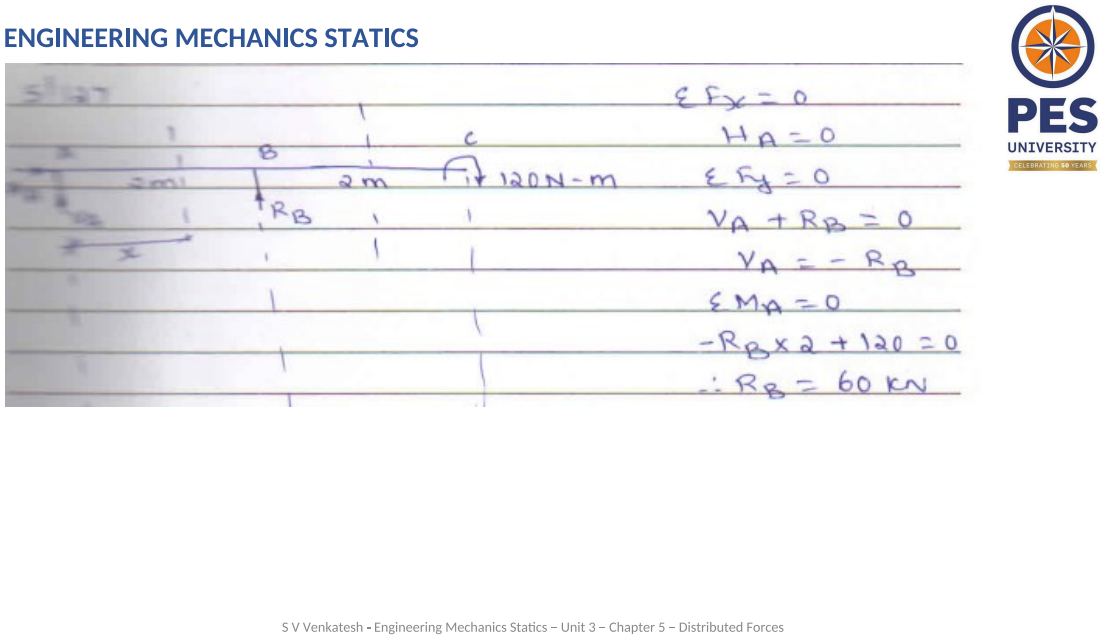


Figure G000119. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 4.

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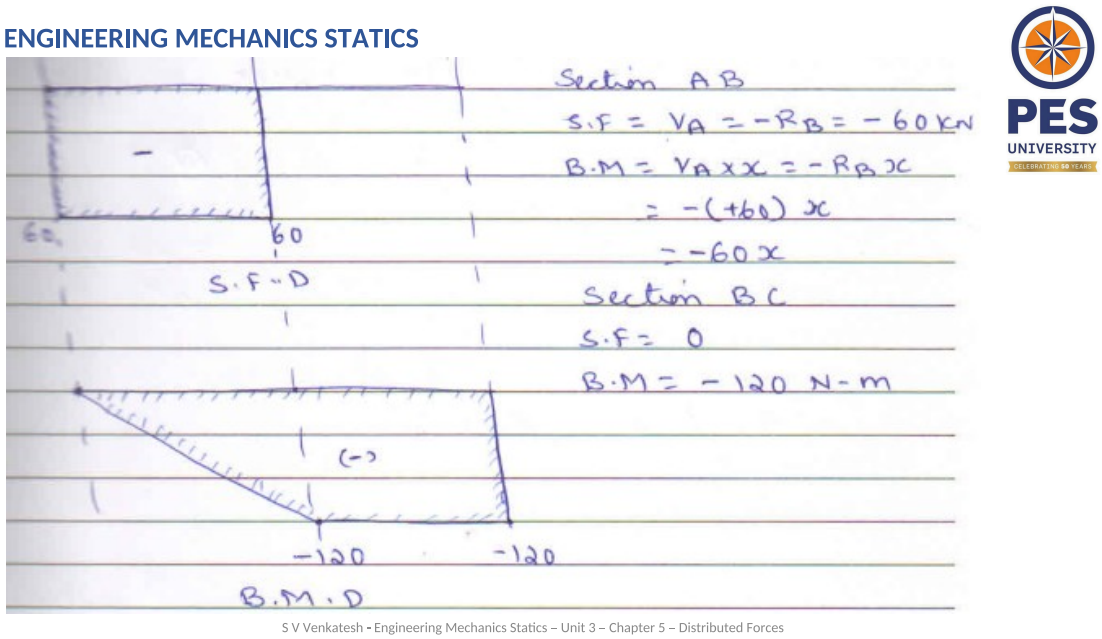


Figure G000120. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 5.

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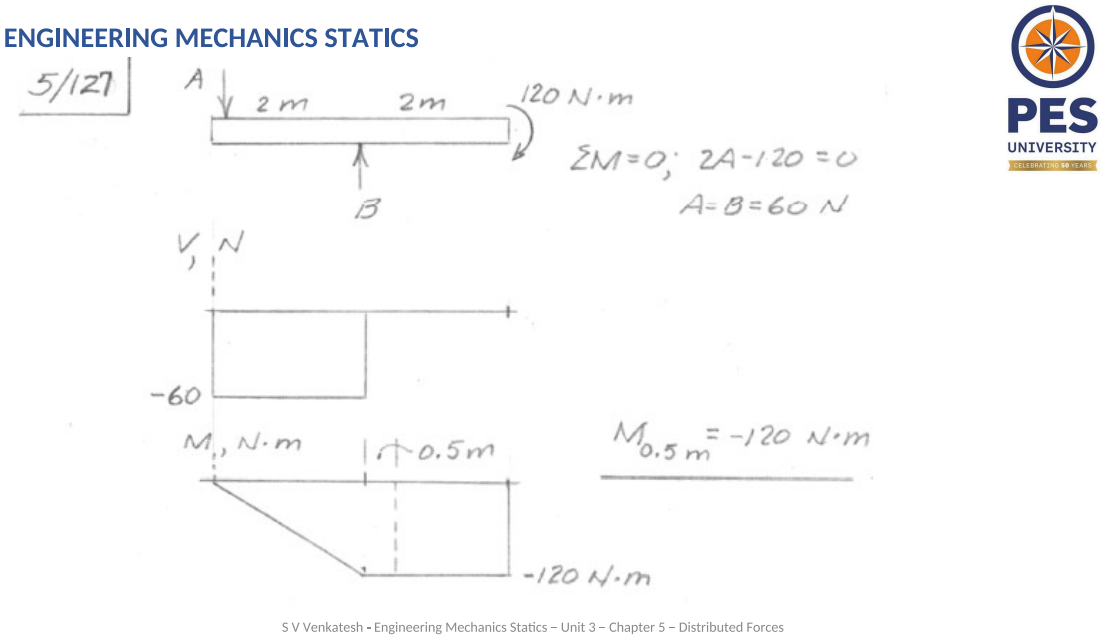


Figure G000121. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 6.

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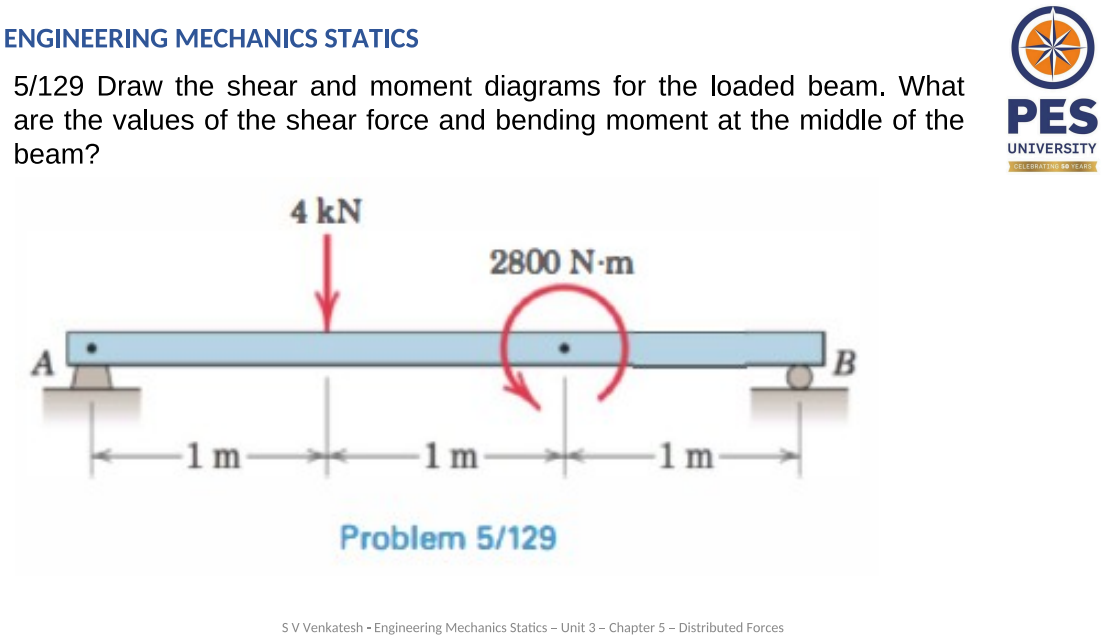


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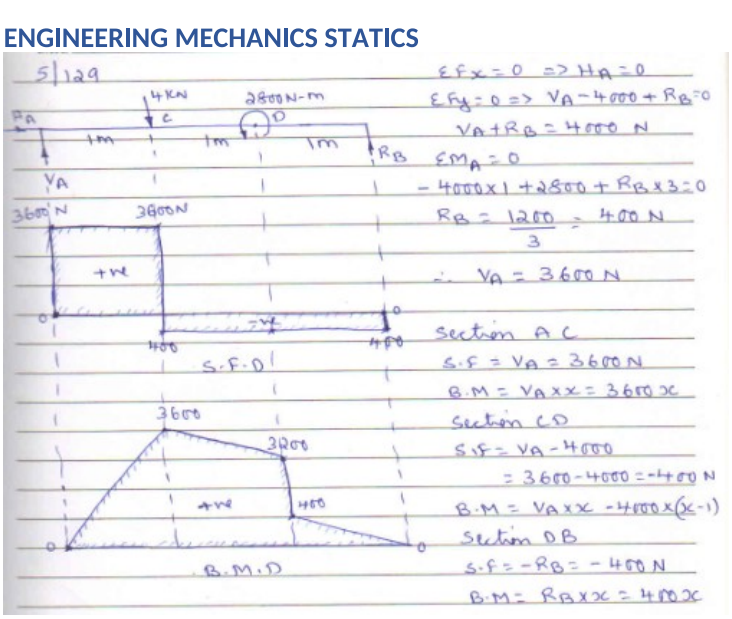


Figure G000123. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 8.

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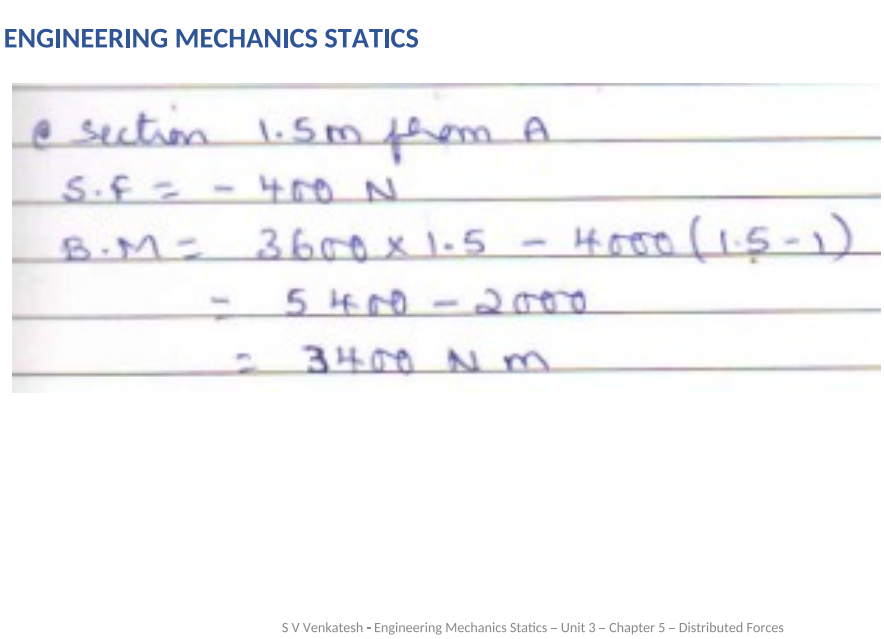


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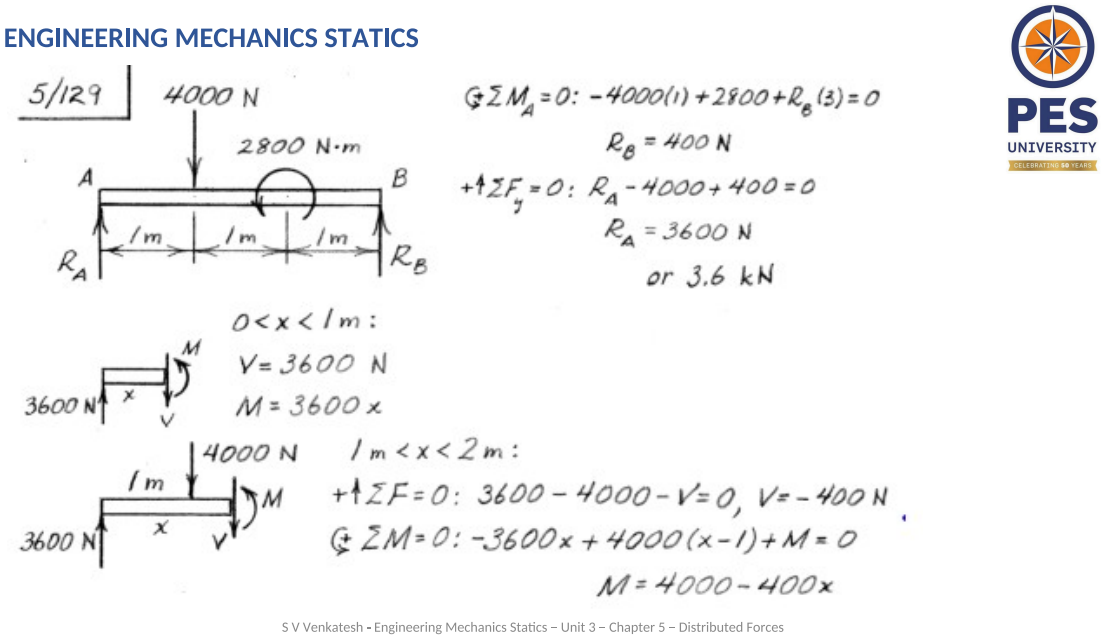


Figure G000125. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 10.

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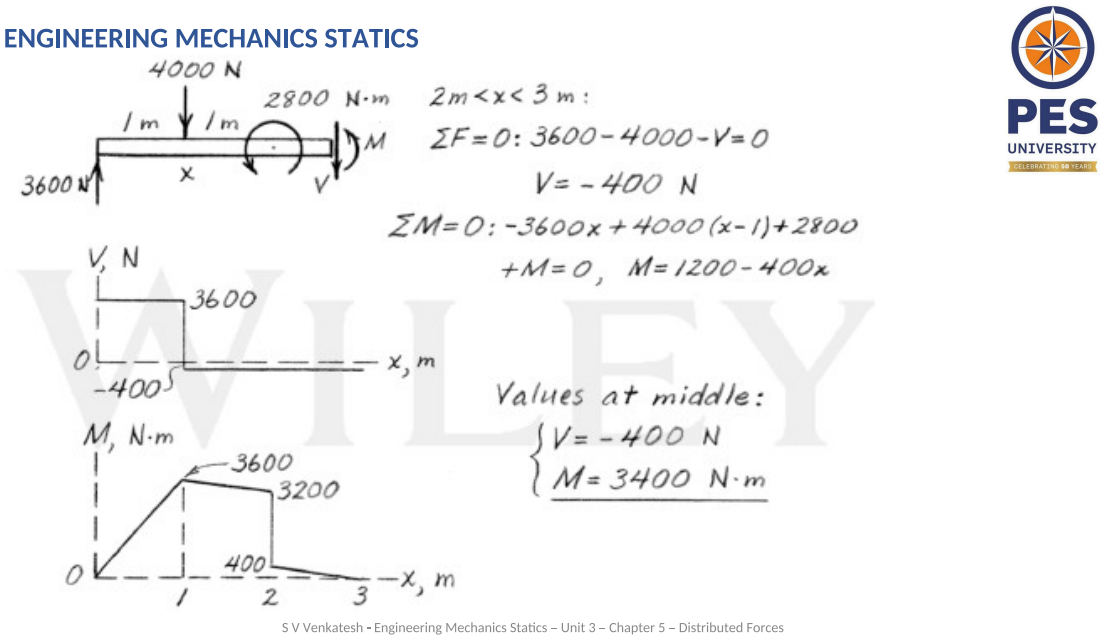


Figure G000126. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 11.

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THANK YOU

Dr. S V Venkatesh
Civil Engineering
Department of S & H

Figure G000127. UE25CV141A Unit 3 Class 55 Lecture PPT · source slide page 12.

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Important points and why

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2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams Internal Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000488 · EV0000489 · EV0000490
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 5/127 Draw the shear and moment diagrams for the beam subjected to the end couple. What is the bending moment M at a section 0.5 m to the right of B? S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000491 · EV0000492 · EV0000493
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 5 (OCR) ENGINEERINGMECHANICSSTATICS PES -R=-60KN VA UNIVERSITY B.M2YAXX2-RBOC CELEBRATING5OYEARS - (+b) S.F-D -150 S VVenkatesh-Engineering Mechanics Statics-Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 6 (OCR) ENGINEERINGMECHANICSSTATICS 5/127 120N.m 2m 2m PES M=O:2A-120=U UNIVERSITY CELEBRATING50YEARS A=B=6ON -60 =-12oNm M.N-m ↑0.5m 0.5m -120N.m S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000494 · EV0000495 · EV0000496
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000497 · EV0000498 · EV0000499

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000500 · EV0000501 · EV0000502
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS 5/129 Draw the shear and moment diagrams for the loaded beam. What are the values of the shear force and bending moment at the middle of the beam? S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000503
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000504
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 10 (OCR) ENGINEERINGMECHANICSSTATICS 5/129 4000N G2M=0:-4000(1)+2800+R(3) =0 PES =400N 2800N.m 8 UNIVERSITY CELEBRATING50YEARS +2F=0:R-4000+400=0 =3600N or3.6kN 0<x</m: V=3600N 3600N M=3600x 4000N /m<x<2m: /m +↑2F=0:3600-4000-V=0,V=-400N X 3600N G2M=0:-3600x+4000 (x-1)+M=D M=4000-400x S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 11 (OCR) ENGINEERINGMECHANICSSTATICS 4000N 2800N.m 2m<x<3m： PES /m 2F=0:3600-4000-V=0 UNIVERSITY X CELEBRATING50YEARS 3600 V=-400N ∑M=0:-3600x+4000(x-1)+2800 +M=0.M=1200-400x 3600 400 Yaluesatmiddle: N-m V=-400N 3600 M=3400N-m 3200 400 2 S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces ## Page / slide 12 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000505
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SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Dear students, welcome to engineer mechanics statics classes. My name is Dr. Sv. Civil Engineering from the Department of Science and Humanities, PS University. [00:00:14.436] This is class number fifty-five. In the last class, that is class number fifty-four, we had discussed about how to determine the shear force and bending movement for a beam which is loaded with different types of loadings. We will continue. [00:00:31.556] the same uh topic in this class also, and we will see how to solve some more problems which are loaded with different types of loadings. [00:00:41.796] And we are dealing with the topic uh under unit three, that is distributed forces, and in that chapter five, and the topic is beams internal effects. [00:00:53.604] And we have referred to the textbook, Engineer Mechanics, Statics, 8th edition, the authors are Miriam Craig and Bolton. [00:01:02.820] Okay. For today's class I have chosen this problem that is problem number 5 bar 127. The statement of the problem goes like this. Draw the shear and bending moment diagrams for the beam subjected to the end couple. [00:01:19.812] What is the bending moment M at a section 0.5 meter to the right of beam. So from the statement, what we need to understand is that we need to draw the shear force and bending

moment diagram for the horizontal beam. [00:01:36.996] Uh which is subjected to a couple at the end of the beam, that is to the right side of the beam. And also we need to find out what would be the value of the pending moment. [00:01:48.260] At a section 0.5 meters from the right side of the support B. Okay now look at the diagram that is given. So from the diagram we can make out that we have an horizontal beam. [00:02:03.300] Which is supported at point A by a pin or a hinge support and at B by a roller support. The distance between A and B is 2 meters. Similarly, the distance between B to the end of the beam is also 2 meters. At the end of the right side of the beam, [00:02:21.924] There is a couple or a moment that is acting in the clockwise direction of magnitude 120 newton meters. Okay. Now according to the statement of the problem, we need to draw the shear force diagram for this beam. [00:02:39.236] As well as we need to draw the bending movement diagram for the beam. [00:02:45.508] After drawing the shear force and bending moment diagram, we are supposed to find out what is the bending moment acting at a point right side of B at a distance of 0.5 meters from B. That is this distance, if you take a section here. [00:03:01.072] This distance has to be 0.5 meters and we need to find out what will be the bending moment at that point. Okay. Now since there is a moment that is acting at the end of the beam, that is right side end of the beam. [00:03:16.196] And it is the beam is supported at A by mean support under log support. First thing that we need to do to solve this problem is to determine the external support reactions acting at A and acting at B. [00:03:30.820] Now since the support at A is a pin connection or a hinge connection, there will be two reactions, one vertical and one horizontal. We will first draw the free body diagram by replacing the supports by their respective reactions. [00:03:46.448] Okay. So at A there will be a vertical reaction, we will assume it to be acting upwards. Then there is an horizontal reaction acting at A, we will assume it to be acting to the right. Then at B point we have a roller support. So at the roller support we will have a vertical reaction acting upwards only the sense of the [00:04:05.924] Reaction at B is upwards only. Whereas the sense or the reaction at A can be upwards or downwards, as well as it can be to the right or to the left because it's an hint support. Okay. Now once I do that, then we will take the movement about point A. [00:04:21.924] So when I take the algebraic sum of all the moments about point A and equal it to 0, I will get the reaction at B. So once I get the reaction at B then I will take algebraic sum of all the forces acting along the vertical direction to be equal to 0, then I will get the reaction at A, vertical reaction. [00:04:39.620] When I take the algebraic sum of all the forces acting along the x axis that is horizontal and equality to zero, then I will get the reaction at A along the horizontal which should be equal to zero because there is no horizontal force that is acting. [00:04:54.692] So, after getting this, we will take a section between A and B, find out the equation for shear force within this section, as well as the equation for bending moment within this section.

EV0000166 · EV0000167 · EV0000168 · EV0000169

Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.308] Similarly, we can take a section between B and the end of the beam that is to the right side end, and also we can get the equation for shear force and penny moment diagram. Then substituting the values between this. [00:05:21.696] section that is between A and B and B and C. We can get the different values for shear force and body moment at any particular point between A and B and B and end of the V. Okay. That's what is done here. We have I have written the free body diagram here. [00:05:38.916] The hint support is replaced by its reaction, vertical reaction acting upwards. I have assumed it to be VA and horizontal reaction acting at A, I have assumed it to be HA. Then the reaction at B

which is the because of the roller support and it will be acting upwards. [00:05:56.496] So that I have assumed it to be R B acting upwards. Then there is a movement acting at the free end of the beam, which is 120 newton meters, and that is what is shown. Now with this, we should be able to find out this support reaction by using the equations of equilibrium. [00:06:15.588] Equilibrium that is used R sigma M at A equals 0, as I have explained. Using that I have determined what should be the reaction that is going to be acting at B, which we will get it as 60 kilonewton. Then using the algebraic sum of all the forces acting [00:06:31.940] Uh along the vertical, we will find out what will be the reaction at A and we will find that the reaction at A is acting downwards BA. Then we will also find out what is the horizontal reaction at A by taking all the forces along X, but we found it to be 0. [00:06:50.148] Next, we will take a section in between A and B, okay, and that distance from A to be equal to X, that is what we will take. We will take all the forces acting to the left side of the section and we will determine the shear force equation. So when I take up [00:07:06.404] All the forces acting to the left, what we have is only the reaction acting at A, that is VA. [00:07:14.084] Which is equal to minus R B, which is equal to 60 kilonot minus R B and it is acting downwards. To the left and downwards, you will get it as negative minus 60, and that's what is indicated here. And that value will remain constant for the entire section between A and B. Right. [00:07:33.092] Next, if you take a section B C here, okay, any section the distance from B is X, if you take what will be the shear force. So if you take shear force or the section of the beam acting to the left side of the beam, then we will have B acting downwards, R B acting upwards. [00:07:52.240] The values are same, so we will get the shear force to be equal to zero. Using these two equations we can draw the shear force diagram as indicated here. Between A and B, it's a constant value and it is minus 60, so that is what is indicated. This is the datum line 00. [00:08:09.668] So below by 60, this will be a constant value that is 60. Then we have a 60 between this, which is a constant value, and then between the section B and the end of the beam. [00:08:23.396] Uh it is uh the shear force is zero, so we have we have drawn that shear force diagram. So we will close this. Okay, so this is negative and this is a shear force diagram that we get. [00:08:35.076] Similarly, if I want to find out the bending moment diagram or values for to draw the bending moment diagram, we'll write the equation for bending moment between the section B and A and B. We will take a section here. This distance is X. So we will take the left side of the beam. [00:08:52.804] Then we will write what will be the bending moment. So that will be equal to VA which is acting here V A into X [00:09:00.100] VA I have assumed it to be acting upwards. So V A into X will produce a uh saging bunny moment, so that should be positive, but since uh V A is equal to minus R B which is acting downwards, we will take it as negative. [00:09:15.396] Uh plus 60 into x, so we will get it as negative minus 60 into x because that will produce a hawking bending moment. So any value of x between this two points that is A and B, we will get the value of the bending movement. [00:09:30.532] Similarly, if I take a section between B and C, whereas this is C and this is B, this is X distance, and we will take all the moments acting or all the forces acting to the left of the beam and taking movement of all these forces, we will get bending moment to be equal to minus 120, that is because [00:09:49.732] Uh we will have this 60 into this distance that is 2 plus x, whereas this 60 into x we will get. So that will create a moment of movement which is equal to VA into 2, which will be 120. Or if I take the section to the right side of the beam.

EV0000170 · EV0000171 · EV0000172 · EV0000173

Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.680] We will get a movement which is equal to 120, which will produce a hawking bending movement, so it is minus 120. So drawing the bending movement diagram, this is the datum line. [00:10:17.508] But the support A and the free end it is 00. As we progress from A to B, as X increases, your bending moment also increases by 60 times X. So that would be equal to 120 at this point, and that is what is a straight line here. [00:10:33.732] And between B and C that is N of the beam, it's a constant value which is one twenty and it is negative, so it will get a constant value. So drawing this, we will get this diagram and this is negative because it is below the datum line and this is the bending moment diagram. [00:10:49.860] So we have now drawn the shear force and bending moment diagram. The next part of the problem is to determine what would be the bending moment value at a distance 0.5 meters from B to the right side, that is at this, this distance is 0.5. So we can easily identify here when I take this section. [00:11:08.996] that the bending moment is a constant value which is 120 that is negative bending moment. However, however we can get that. Okay. Now this indicates the total uh beam shear free body diagram here. [00:11:28.164] Then the shear force diagram and bending moment diagram together. So this is the free body diagram. We have determined the support reactions, which we have got it as 60 and 60 here, but at E it is acting downwards, at B it is acting upwards. [00:11:43.588] So we will we can write the shear force diagram as sixty and sixty here, that is what we have we know how to draw it, and the bending moment diagram also. Bending moment at point five meters, we will get it to be one twenty, that is what is indicated here. I hope all of you have understood this. We will move on and to the next. [00:12:00.752] Question I have considered a question 59. So the statement of the problem goes like this. Draw the shear and bending or draw the shear and movement diagrams for the loaded breed. [00:12:17.732] What are the values of the shear force and bending moment at the middle of the beam is what is ask. So according to the statement, we have to draw the shear force and moment diagram. [00:12:30.596] And after that, we have to determine what would be the value of the shear force and bending moment at the middle of the beam. Now look at the diagram. It's a simply supported beam. [00:12:41.604] At the left hand, that is at A, we have an inch support or a pin support. At the right hand, B we have a roller support. Then we have a point load acting downwards of magnitude 4 kiloneeton at a distance of 1 meter from A, which we will take it as point C. [00:12:58.948] Then we also have a counterclockwise moment of magnitude 2800 newton meters acting at a this point, which we will take it as D, which is at a distance of 2 meters from A, whereas 1 meter from C and as well as 1 meter from B to the left side. [00:13:18.756] So, again in this problem, if I want to draw the shear force on bending moment diagram, first thing that we need to do is to draw or to draw the free body diagram and then to determine the support reaction's value. So at here we have [00:13:34.628] Hinch so two reaction, one vertical and one horizontal, but B there is roller, always it is going to be acting upwards because the roller as you can see it is indicated like this. So reaction will be upwards. So that will be taken as RB. [00:13:47.108] So using the equations of equilibrium, we can find out what is the reaction at A, what is the reaction at B. So going forward, so here you can see the free body diagram. I have replaced the [00:14:01.956] Heat support by two reactions, vertical and horizontal, then roller support at B by one reaction RB and I have indicated the load that is acting on the beam. [00:14:12.676] Then first equations of equivalent that I have taken is sigma fx is equal to 0, with which I will get movement or the horizontal force at A to be equal to 0. [00:14:23.812] Then the next equation taken as algebraic sum of all the vertical forces to be equal to 0. So from this we will

get the value of VA and which is acting upwards, so it is positive. 4 kilonut which I have converted into newtons which is 4000 acting downwards, so negative. [00:14:42.212] Then we have R B acting upwards or positive, R B equals 0. So we get VA plus V B equals 4000 un. [00:14:50.820] Now, taking algebraic sum of all the movements about point A to be equal to 0, that is at this point we will take the movement. So what we will get is we will get 4000, that is because of this point load into 1 meter.

EV0000174 · EV0000175 · EV0000176 · EV0000177

Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.348] Which is clockwise, so negative. Then we have this movement which is anti-clockwise, so 2800 is acting in anticlockwise direction, so positive. Then R B will also produce a counterclockwise movement, so that is also positive. So R B into the total distance is 3. [00:15:23.780] From this equation we will get the value of R B which is equal to 400 newtons. We will substitute this 400 newtons in this equation and get the value of VA, which will be equal to 3600 newtons. So from this we have got the values of the support reactions at A as well as B. [00:15:43.204] Now reaction at A horizontal reaction is 0. Reaction at A vertical reaction is 3600 newtons. Vertical reaction at B is 400 newtons. And both the reactions at A and B which are vertical are acting upwards itself. [00:16:00.548] Now to determine the shear force and bending moment, we will take the section AC, that is between this A and C. [00:16:11.940] So shear force is equal to Va because there is only one reaction that is acting to the left of the beam that is VA, so that's a constant value which we will get as 3600 as indicated here. [00:16:25.572] Similarly, we will also take up the bending moment when we take the section here, we will get the bending moment as this force into this distance, if I take it as x, that would be equal to V A into X, which is equal to 3600, which will produce a sagging bending moment, so that is positive. [00:16:42.212] Next, we will take a section between C and D, that is here. If you take the left side of the section, we will get shear force equals 400, that is VA, which is acting upwards, 3600, and 4000 newtons, which is acting downwards, so negative. So we will get the value of minus 400 newtons. [00:17:02.212] between the section C and D. Similarly for wending moment we will get it as VA into X. From A is X that we have taken. Minus 4000 into X minus 1. Because I have taken the X distance from A to this point as X, it is [00:17:20.228] X minus 1, which will be this distance. Okay. I hope this is clear. Similarly, we can take a section between D and B, that is here. Okay. Now if you take the beam acting to the [00:17:34.384] right side you will get only one reaction that is r b to the right side and upwards it will be positive so shear force is equal to 400 until it is positive whereas bending moment will be r b into x which will be uh sagging bending moment so it is positive. [00:17:49.988] So now substituting the value for X within that section, we'll get the value. So if you draw the shear for diagram, [00:17:58.820] This is the datum line 00. Between this section, it's a constant value of 3600. So that is what is indicated here. Then coming to this section, again, shear force is a constant value which is minus 400. So that is what is indicated. Then till the [00:18:15.904] C to B, it is constant itself because there is no vertical force acting between this. So that is a constant value. So now joining all these points, we get this as the shear force. This is a shear force diagram. Above datum line is positive, below datum line is negative. [00:18:33.072] Similarly, if I take the bending moment values, V a into X, it is positive. So as x increases, [00:18:40.932] The bending movement value is also increased, and if I put x is equal to 1, I get it as 3600, that is a point B, so it is a linear line because it's a point load. Then between uh C and D [00:18:56.672] Also we have this equation substituting the value of x, anything more than one and less than two, we will get the variation at x is equal to zero, we get it as 3600 and

when x is equal to 2 we get the value of 3200. So on that is a linear line again. [00:19:16.100] Then considering the moment we will get it to be 400 itself, and that is what we have indicated here. When we take the section B and D we get it as 400 when X is equal to [00:19:30.372] 0 we will get it this to be 400 and that is what is indicated and when x is equal to 0 we will get x to be 0 when x is equal to 1 meter because we have taken it to the right side we will get this to be 4 [00:19:45.952] So joining all this, we get this as the bending movement diagram. Okay, and since it is above the random line, it is positive. [00:19:53.796] Then he has asked you to determine what will be the value of the shear force embedding moment at the middle of the beam, which will be at a distance of 1.5 meters from A as well as 1.5 meters from B. From the shear force embeddement moment, if I take it.

EV0000178 · EV0000179 · EV0000180 · EV0000181

Transcript block 5 (source interval 001200-001499 seconds): [00:20:08.016] I the shear force value is four hundred, whereas the bending moment value we have to compute. It is varies from three thousand six hundred to three thousand two hundred. So the difference between this is [00:20:20.132] Or 3600 minus 3200, which is 400, and since it is the middle, it will be 200 mm, which is lesser than the value at B and 200 m 200 newton millimeters greater than the value at C. So that would be equal to your [00:20:36.784] 3400 and that's what is computed here at section 1.5 meters from here, shear force is minus 400, pending moment is equal to 3600 into 1.5 minus 4000 into 1.5 minus 1, we get this to be 3400 newton millimeters. [00:20:55.396] So we have taken two and in this you can look at what will be the uh values that we get whether we when we take it to the left side of the section for different values, whatever I have explained and when we take the section between [00:21:14.052] uh B and C how to get the shear force and bending movement diagrams. I'll explain this clearly. Okay so we will end this uh class here. Thank you very much. My name is Dr. Svintesh from Civil Engineering under the Department of Science and Humanities PS University. Let's [00:21:32.068] Discuss in the next class. Thank you very much.

EV0000182 · EV0000183 · EV0000184 · EV0000185

Full transcript: 3730 normalized words, 86 timestamped segments, 1294.741 seconds duration. Low-confidence segments flagged: 0.

EV0000166 · EV0000167 · EV0000168 · EV0000169

13 UE26ME142A_Unit 3_Class 44.1_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 3_Class 44.1_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000186 · EV0000187

Graphs and diagrams



Figure G000128. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 1.

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Engineering Mechanics

Statics

EIGHTH Edition
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J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams Internal Effects

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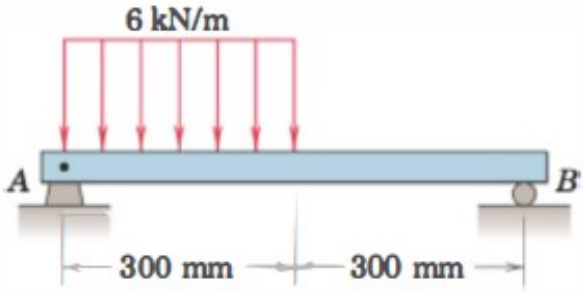
Figure G000129. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

5/130 Determine the shear force V and bending moment M at a section of the loaded beam 200 mm to the right of A.



Problem 5/130

Figure G000130. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 3.

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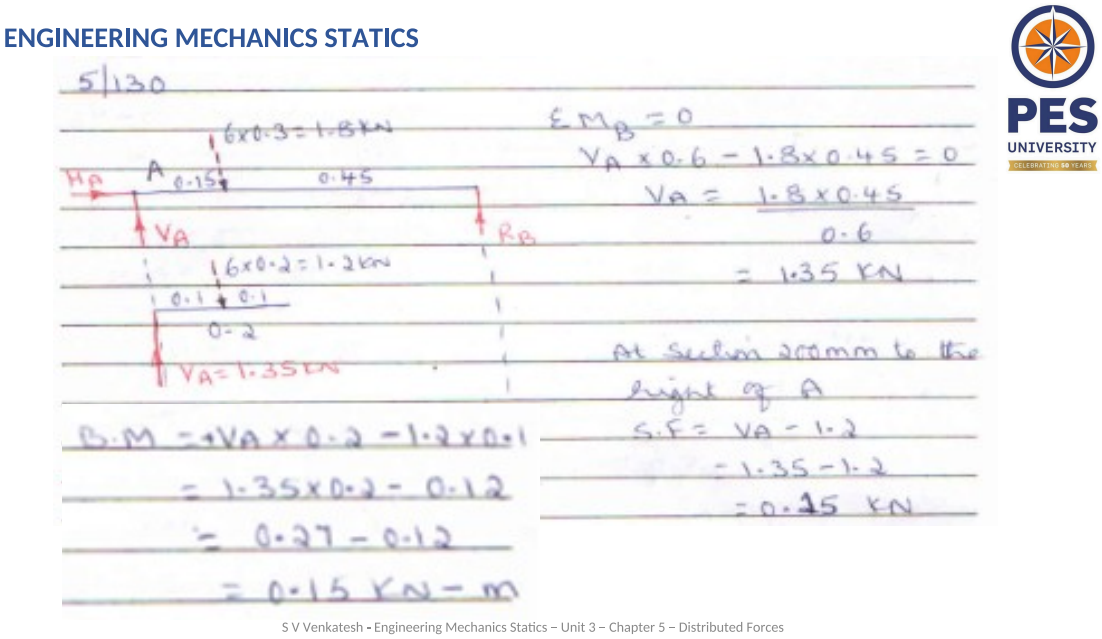


Figure G000131. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 4.

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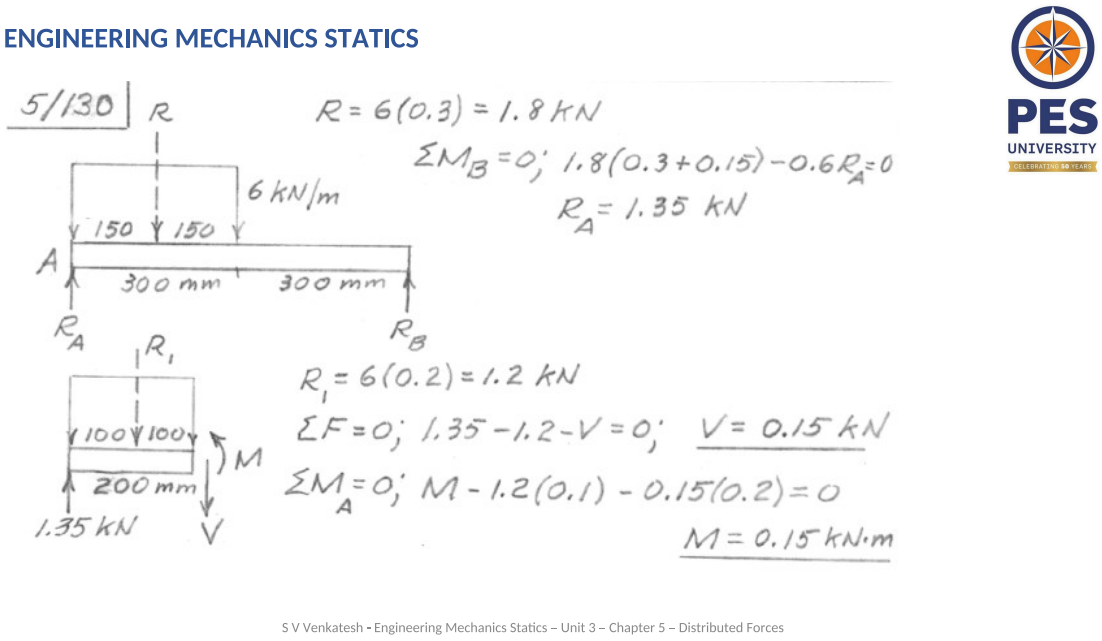


Figure G000132. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 5.

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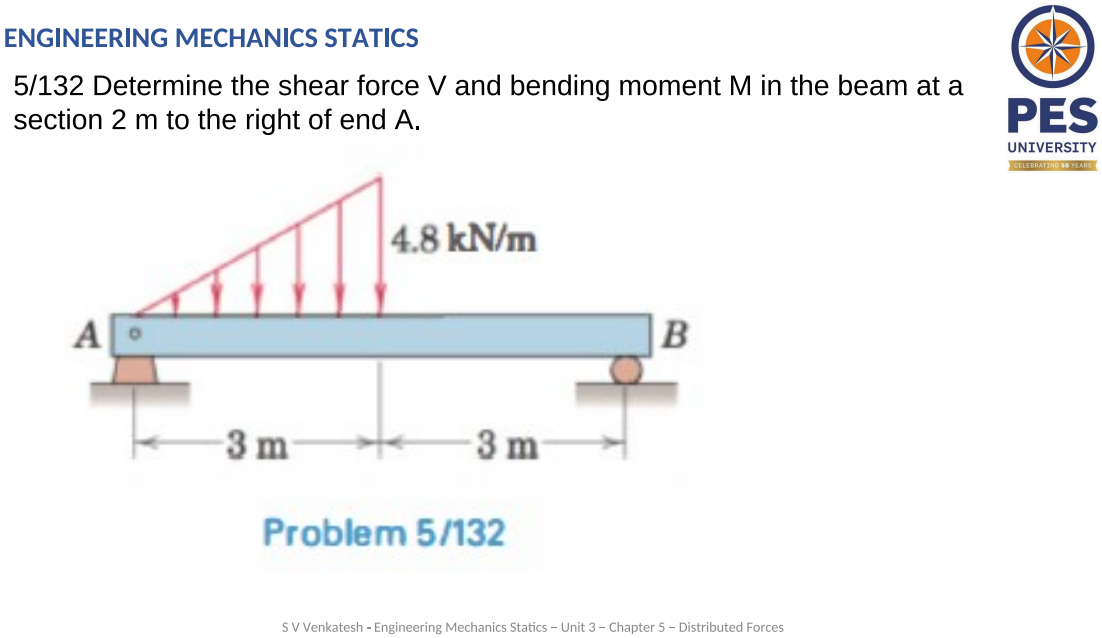


Figure G000133. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 6.

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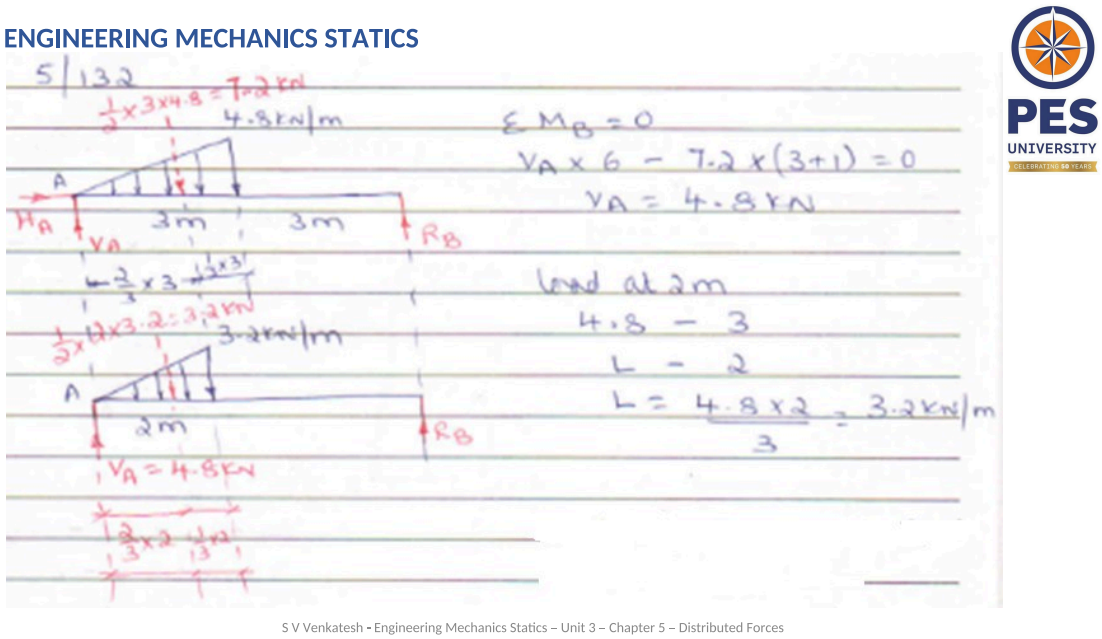


Figure G000134. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 7.

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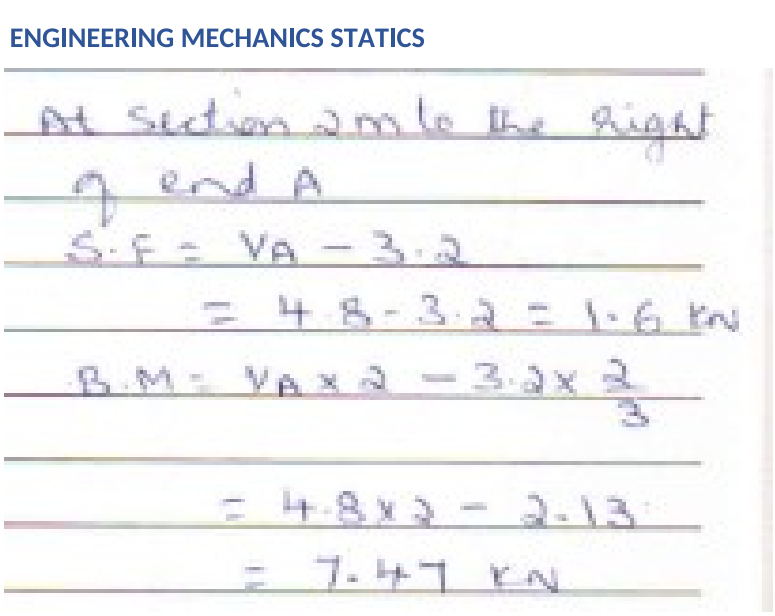


Figure G000135. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 8.

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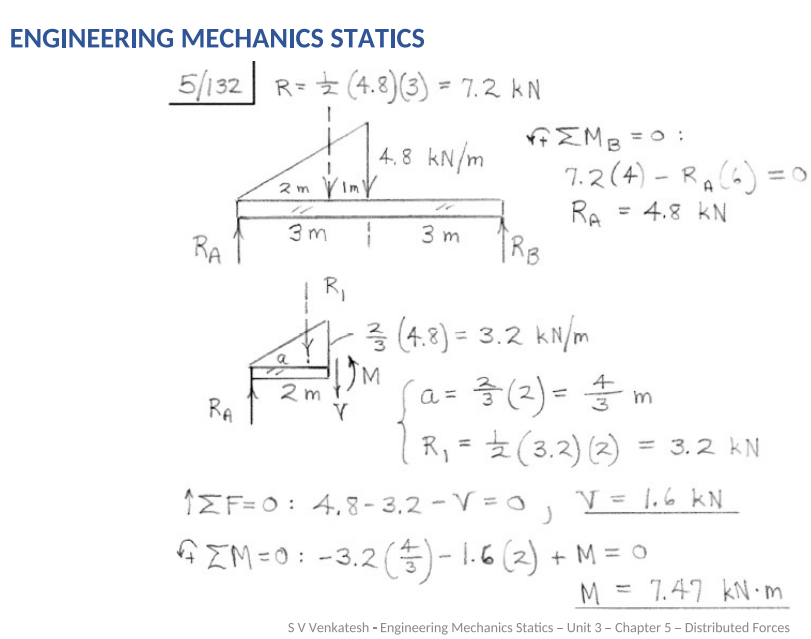


Figure G000136. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 9.

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Figure G000137. UE26ME142A Unit 3 Class 44.1 Lecture PPT · source slide page 10.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000509 · EV0000510 · EV0000511
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams Internal Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000512 · EV0000513 · EV0000514
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 5/130 Determine the shear force V and bending moment M at a section of the loaded beam 200 mm to the right of A. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 4 (OCR) ENGINEERINGMECHANICSSTATICS PES GX03AL-B1N UNIVERSITY CELEBRATING50YEARS 16x0.281.2 A303STON S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000515 · EV0000516 · EV0000517
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000518 · EV0000519 · EV0000520
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000521 · EV0000522 · EV0000523

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS 5/132 Determine the shear force V and bending moment M in the beam at a section 2 m to the right of end A. S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 7 (OCR) ENGINEERINGMECHANICSSTATICS 132 x3x4.8=7 PES 4.82Nm VAx6-7.x(3+1)=0 UNIVERSITY CELEBRATING5OYEARS 4.8Y2 YA RP Ndaam 48 — 1x3.2:3.2m 8X2 3.2K RB VA=H-8EN S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000524 · EV0000525 · EV0000526
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000527
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000528
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 10 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000529
10. Source slide page 10: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000530

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Dear students, welcome to Engineer Mechanics Statics classes. My name is Dr. SV Venkitesh, a civil engineer from the Department of Science and Humanities, PS University. This is class number 56. [00:00:16.388] Uh in the previous class we had discussed few examples on internal effects of beams. We'll be continuing to solve few more problems in today's class. And uh we are discussing about the [00:00:30.756] Unit 3 distributed forces in that chapter 5, beams internal effect, and we are referring to the engineer mechanics statics book written by Miriam and Craig, and it is the 8th edition. So for today's class, I have chosen this problem. The problem number is 5 bar 130. [00:00:51.076] The statement of the problem goes like this. Determine the shear force V and bending moment M at a section of the loaded beam 200 mm to the right of A. And the diagram that is given here is the diagram that is given. It's a simply supported beam. [00:01:10.628] At the left hand, that is at the point A, you have a hint support or a pin connection or pin support, and then at the right end, that is at point B, we have a roller support. [00:01:24.516] And to this beam, there is an uniformly distributed load or a rectangle loading of intensity 6 kN per meter for a distance of 300 mm from the left support that is from A. [00:01:38.212] The length of the load is three hundred mm, whereas the total length of the beam is three hundred mm plus three hundred mm, totally it is six hundred mm. [00:01:50.212] Okay, and according to the problem definition, what we are supposed to determine is the shear unbending moment at a distance of 200 mm from the right support that is A from here. [00:02:04.016] 200 mm will be somewhere here. So we need to find out what will be the shear force and bending moment at this point, which is at 200 mm from A. So if I want to determine the shear force and bending moment, first thing that I need to do is to find out what is the support reactions. To determine the support reactions. [00:02:23.300] I have to draw the free body diagram. [00:02:26.756] To draw the free body diagram I have to replace the support by its respective supports. So at A, I'll replace this support reaction by the

reactions. That is one vertical reaction at A and one horizontal reaction, vertical reaction BA and horizontal reaction H A. [00:02:46.916] Then at B I will replace this by one reaction that is R B which is acting vertically upwards. Then the uniformly distributed load will be replaced by a point load of magnitude equal to the [00:03:01.732] Length of the load that is 0.3 mm because the intensity of the loading is given as 6 kN per meter to have consistent in units, I'll convert this distance 300 mm into 0.3 meters. [00:03:16.704] So 6 into 0.3 will be 1.8 kilonute tons, which will be acting at the center of the load that is at 0.15 meters from A. [00:03:26.116] So once I draw this free body diagram, next I'll apply the equations of equilibrium, all the three equations of equilibrium, to determine the vertical reaction at A, horizontal reaction at A, and then vertical reaction at B. [00:03:41.892] Once I get these reactions, then I can consider the section at [00:03:47.364] uh 200 mm from a I can take either left side of the section or the light side of the section. If I consider the left side of the section then I will write down the reactions acting at A and then for this 200 mm beam [00:04:04.672] Whatever is the load that is acting, I will consider that and replace the UDL by a point load. So this is how you can see that I have written the free body diagram of the entire beam, replacing the support at A by its reaction that is V and H A. [00:04:23.364] Then reaction at B by R B and then the UDL by a point load of magnitude 1.8 kilonewtons, which is acting at a distance of 0.15 meters from A. [00:04:36.196] So, for this I will apply the equations of equilibrium, that is summation of moments about P equals 0, that is at this point I am taking the moment to be equal to 0. So what do I get? I get V A into this total distance, which is 0.6 meters, which will produce a clockwise movement. [00:04:53.988] Whereas this one would produce an anti-clockwise moment, so it is minus 1.8 into the distance from this to this is 0.6 minus 0.15, which is 0.14, that's what is mentioned here. So that should be equal to 0. So from this equation

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.532] We will get V A to be equal to 1.35 kilon and we got a positive value. So that means the sense of VA is acting upwards. Next, once I get this, of course I can determine the reaction at B by using [00:05:27.204] The algebraic sum of all the vertical forces acting to be equal to zero. But however, since I have to determine the uh shear force and bearing moment at 200 mm from A, I need not [00:05:41.252] I have to determine RB. Instead I can take the section of the beam to the left side and then solve it easily. So that's why I have not determined the reaction at B or the reaction at HA of obviously H A is going to be zero because there is no horizontal force that is acting. [00:06:00.196] Now you write the free body diagram of the beam considering from the section that is a 200 mm from A, left side of the beam only. So that's what is written here. So to the left side of [00:06:14.884] The section 200 mm. What do we have? We have VA which is acting vertically upwards. Then very importantly, what we need to draw here is whatever is the load that is acting till 200 mm is what we need to take. So here there will be a [00:06:34.084] UDL which is acting of my intensity 6 kN per meter and for a distance of 0.2 meters. So converting that into a point load I'll get it as 6 into 0.2 which is 1.2 kilonewtons only. Very important. Earlier when we consider the entire beam. [00:06:54.852] The point load was 6 into 0.3. Remember this. It's the total load that we are taking, whatever is the load that is acting on the beam. [00:07:03.428] But when we are taking a section to find out the shear force embedding moment, which is at point two meters from A, we have to take only that load. So that is why this is 6 into 0.2 only, and this would be acting at the center of 0.2, which is 0.1 m. [00:07:21.380] I hope this is very clear. So once I do this, now I will take or find out

what is the shear force acting at this point. So at section 200 mm, to the right of a shear force will be shear force at this point, to the left of the section. [00:07:40.836] The force is acting upward that is V A, so it is positive, which has a magnitude of one point three five kilonewton. [00:07:47.364] Then to the left of the section and downwards, so it would be negative of the value 1.2, so that is what is mentioned here. So we get the value as 0.15 kN. That is the shear force that is acting at the section 200 mm to the right of A. [00:08:06.852] Similarly to determine the bending movement, so bending movement at this point will be equal to V VA into point two, which is the moment, and that causes a sagging bending movement because the beam is going to bend like this. [00:08:23.620] So which is a sagging bending movement, so it is positive. Then because of this load, the beam is going to bend like this, which is a hogging bending movement. So that would be equal to minus 1.2 into the [00:08:38.340] The distance 0.1, which is mentioned here. So that would also give me the moment as 0.15 kilonewton meters. So in this problem, they have asked you to determine only the shear force value and bendy moment value at a distance of 200 mm from the [00:08:58.244] Uh support A and that's what we have done. Now we have the entire solution in one diagram here. You can see the free body diagram of the beam. The UDL has been converted into a point load and represented as R. Then [00:09:14.084] What is the value of R is given here, that is 6 into 0.3, which is 1.8 kilonewton. Taking moment about B we get RA, which is a value 1.35 kilonewton. Now taking the section to the left of [00:09:29.476] Where we want the shear force of minimum that is 200 mm from A. This is the diagram. So this is the UDL, converting that in UDL into a point load, it is R1. Okay, R1 will be 6 into 0.2, which is 1.2 kilonet times. Now take [00:09:46.852] The sum of all the forces acting unequality to zero, so you get the shear V is equal to 0.15 kilonaut time, then takes algebraic sum of all the movements acting at A to be equal to 0. So you get the movement about at point.

EV0000190 · EV0000191 · EV0000192 · EV0000193

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.156] But the section 200 mm from A, which is equal to 0.15 kilonewton per meter. I hope all of you have understood this problem. Okay. So next we will move on and this is the next problem that I have considered. Problem number 5 bar 132. [00:10:20.004] The problem statement goes like this. Determine the shear force V and bending moment M in the beam at a section 2 meter to the right of end A. Now the previous problem also we had a similar kind of problem where [00:10:39.780] The beam was subjected to a UDL, whereas in this problem, the beam is a simply supported beam again, of span 6 meters. The left support is a hinge support, the right support is a roller support. [00:10:56.132] But here the loading is a uniformly varying load or a triangular load with zero intensity at point A and maximum intensity of 4.8 kN per meter at a distance of 3 meters from A. Okay? [00:11:13.668] So this is what is given similarly as we did in the previous problem, draw a free body diagram, replace the supports by its reaction, replace the UVL with the point load, okay, and then determine the reaction at A. Once you get the reaction at A. [00:11:30.320] Consider a section 2 meters from A. Okay. Draw a separate diagram. Consider all the forces that is acting in this 2 meter span of the beam, and then find out the shear force and bending movement at that point. Okay. So again here. [00:11:49.860] I have drawn the free body diagram of the beam, replacing the support by its reaction B A and H A at point A where there was hinge. Then the triangular loading [00:12:01.892] Okay is replaced by a point load. The magnitude of that would be half of 3, which is the span of the loading into the intensity of loading, which is 4.8. So that we will get as 7.2 kilonewton as a point load magnitude. [00:12:18.308] The point of application of this is going to

be at a distance of two thirds from A. [00:12:24.324] Okay, which I have mentioned it here, two thirds of three, which will be two meters from here, one meter from here. Then at point P the roller support is replaced by a reaction R B, vertical reaction R B. Now taking moment about B, we will determine the [00:12:41.508] uh vertical reaction at a which is equal to four point uh [00:12:49.700] Four point eight kilonewtons what we have got. Next we will draw only the [00:12:57.316] Forces that is acting at 2 meters from V A. Okay. Considering the left side of the support, so this is what we get. We get a vertical reaction VA, which is of 4.8 kilonewtons. For 2 meters a triangular loading. Now here [00:13:14.692] When we have to convert this triangular loading into a point load, we should find out what is the intensity of the load at two meters from A. So for that, use the similar triangles. This is one triangle, and then this is another triangle, they are similar. [00:13:34.756] So, for a distance of 3 meters, the load intensity is 4.8. So, for a distance of 2 meters, what will be the load intensity is what I have calculated here. So we get the load as 3.2 kN per meter, which will be acting at a distance of 2 meters from here, that is what's 3.2 kilonewton per meter, which is mentioned here. [00:13:54.884] Now to convert this triangular loading into a point load, again half of the length two meters into the intensity of loading that is three point two, which we will get it as three point two kilonewtons itself and that would be acting at a distance of two thirds. [00:14:12.324] of 2 from A, which I have mentioned it here. Okay? 2 thirds of 2 that is 4 by 3. So once I get this [00:14:23.012] Now I can find out what will be the shear that is going to be acting at 2 meters and what will be the bending moment that is acting at 2 meters. So [00:14:32.580] But section 2 meters from the right of End A, shear force will be we have VA to the left of the section and is acting upwards, so it is positive, whereas 3.2 is to the left and downwards, it is negative. So that's what is mentioned here. So we get a value of the shear as 1.6 kN. [00:14:52.068] Coming to bending moment, considering the moment at two meters from point A. [00:14:58.756] We have VA into A which will cause a saging bending moment, so it's positive, whereas 3.2 will cause an hogging bending movement, so it is negative, so it's 3.2 into the distances 2 by 3. Okay, that is one third of 2 from the 2 meters.

EV0000194 · EV0000195 · EV0000196 · EV0000197

Transcript block 4 (source interval 000900-001199 seconds): [00:15:16.112] section. So from this replacing VA by 4.8, we get the value as 7.47 kilonewton meters. Sorry, I have left out the kilonewton meters here. Please kindly make that correction. Okay. So with this we have determined the shear force [00:15:33.440] and acting at a point two meters from A. Now this uh slide gives you the entire picture or entire solution of the problem so the free body diagram of the problem [00:15:49.924] Replacing the triangular loading by a point load, then determining what is the reaction at RA, which is 4.8 kilon, then considering only the portion of the beam to the left side of the section that is at 2 meters from here. So this is what is the diagram. So [00:16:09.508] Whatever is the loading that is acting within this pan, that is for two meters, this is a triangle. We have EA acting here. [00:16:22.436] So this intensity of loading is two thirds of four point eight, which is three point two kilone per meter. Then determining the distance at which this R1 is acting, which he has considered as A. [00:16:40.004] Which is 2 thirds of 2, which is 4 by 3 meters. Then R1 is half of 3.2 into 2, that is 3.2 kilonewtons. Using this, we can find out the shear force considering all the vertical forces acting on the beam. So we get [00:17:00.132] 4.8 is the reaction RA minus 3.2 minus V equals 0. So we get V to be equal to this value of 1.6 kilonewton. Then taking movement about this point and equating it to 0, we get [00:17:20.292] Minus 3.2 into 4 by 3 which is this force. [00:17:31.780] Then minus one point six into two. [00:17:38.916] We are equal to

zero. We are equal to zero. [00:17:44.708] So we get the movement to be equal to 7.47 kilonewton meters as I explained earlier, both of them are same. So this is how we get the values of shear force and bending movement. I hope everybody is clear with this. [00:18:07.524] So we will stop today's class here. If you have any doubts, you can always contact me. I am Dr. S V Vength, Civil Engineering from Department of Science and Humanities, PS University. Thank you. See you in the next class.

EV0000198 · EV0000199 · EV0000200 · EV0000201

Full transcript: 2817 normalized words, 72 timestamped segments, 1104.747 seconds duration. Low-confidence segments flagged: 0.

EV0000186 · EV0000187 · EV0000188 · EV0000189

14 Source slide deck: Centroid

This entry preserves the complete slide deck titled Centroid as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_03, to avoid an unsupported match. EV0000202 · EV0000203

Graphs and diagrams

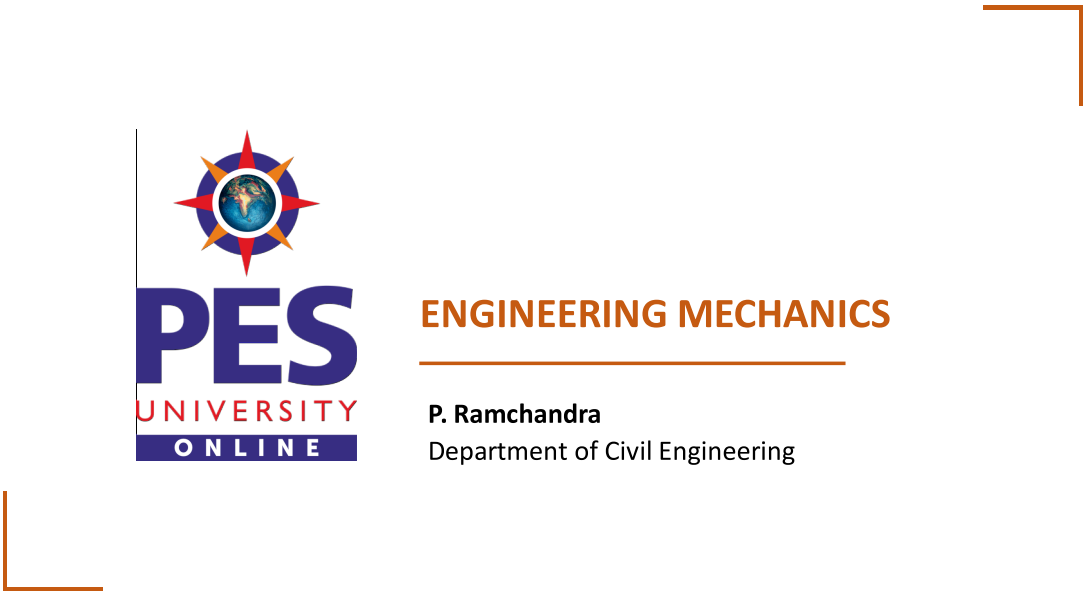


Figure G000001. Centroid · source slide page 1.

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ENGINEERING MECHANICS

Centroid

P. Ramchandra
Department of Civil Engineering

Figure G000002. Centroid · source slide page 2.

EV0000206

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ENGINEERING MECHANICS
Centroid

5/47 Determine the coordinates of the centroid of the trapezoidal area shown.

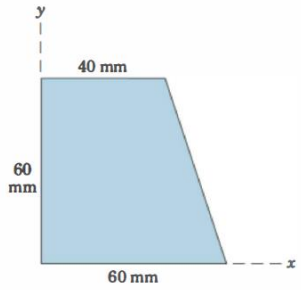


Figure G000003. Centroid · source slide page 3.

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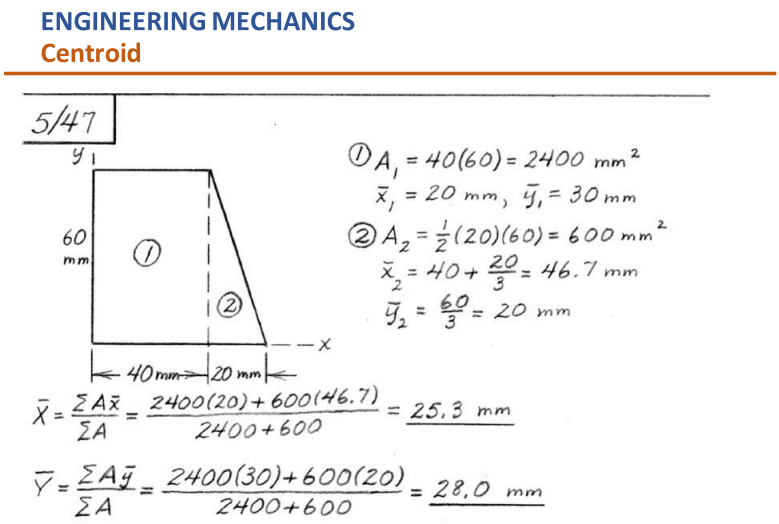


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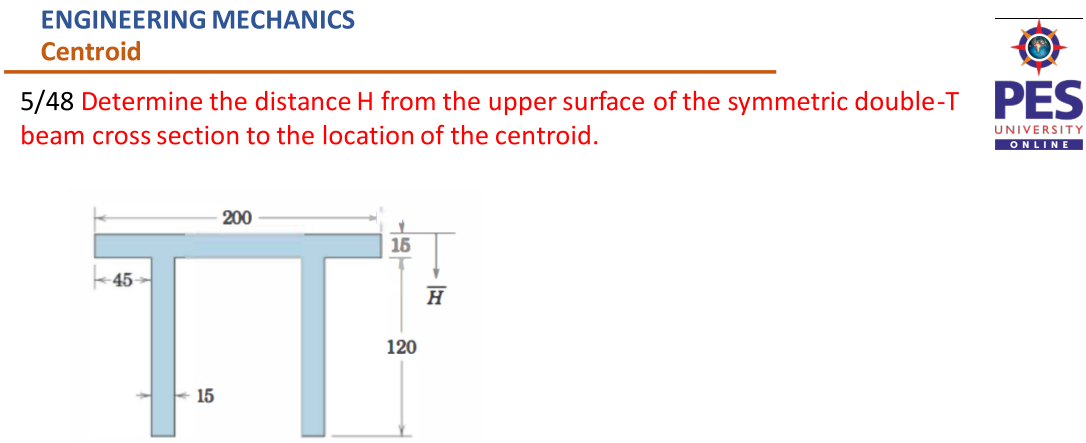


Figure G000005. Centroid · source slide page 5.

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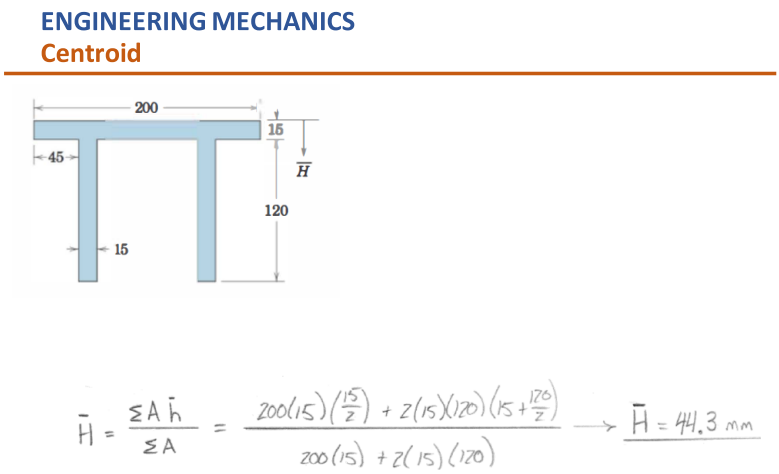


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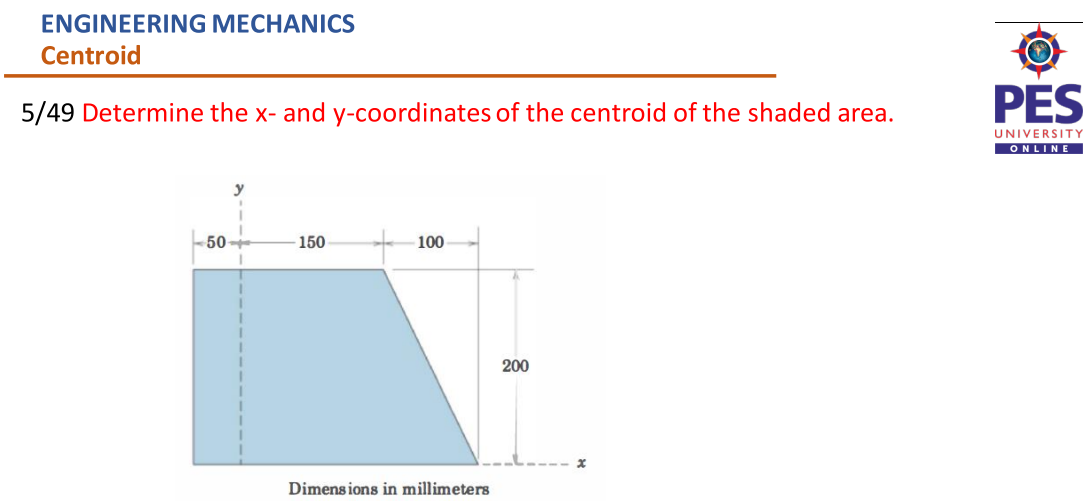


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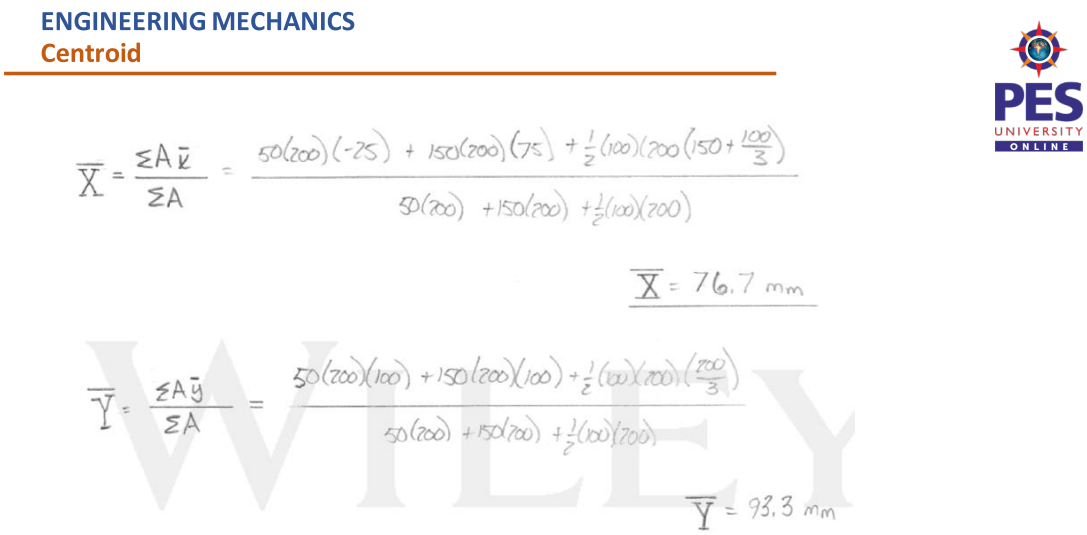


Figure G000008. Centroid · source slide page 8.

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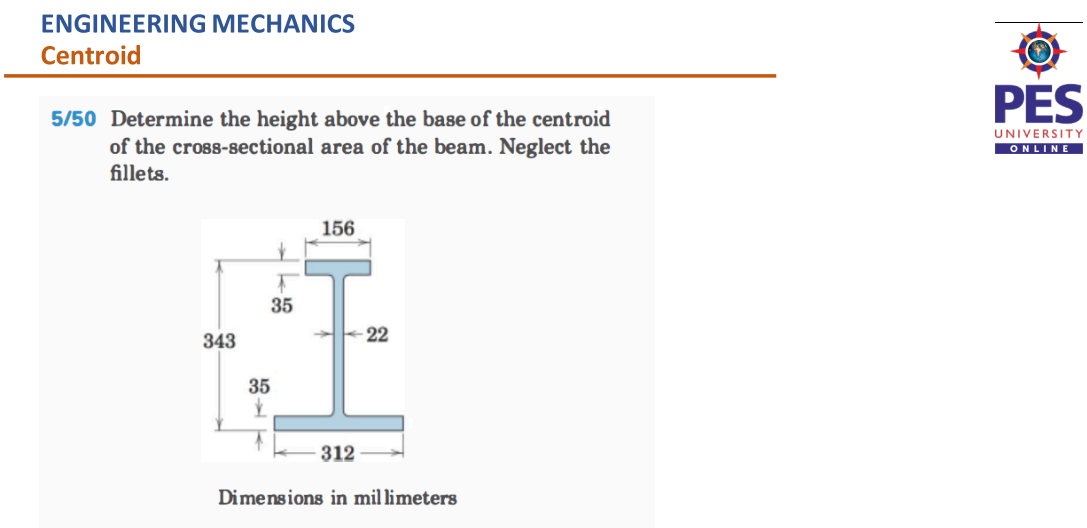


Figure G000009. Centroid · source slide page 9.

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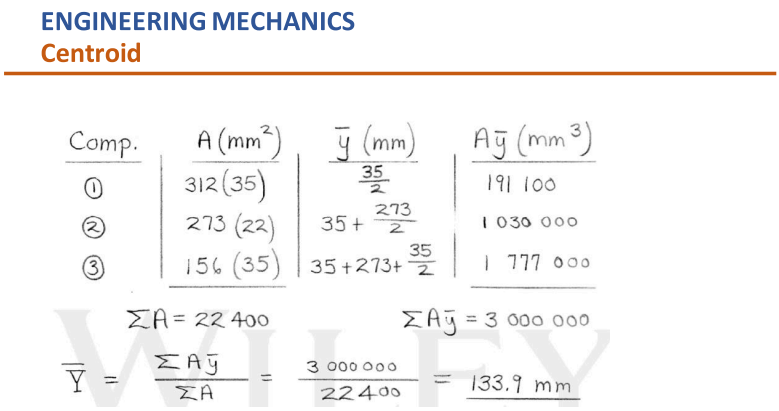


Figure G000010. Centroid · source slide page 10.

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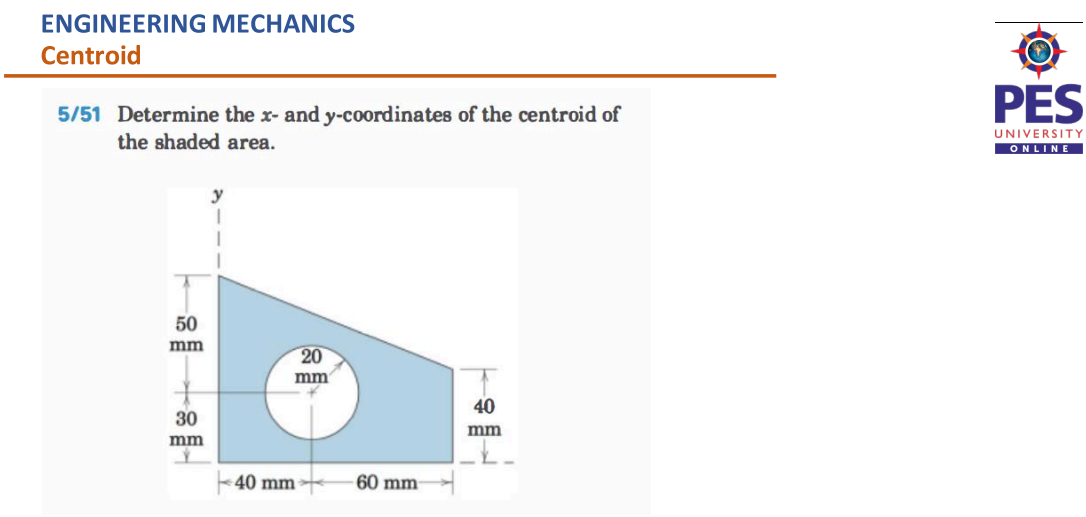
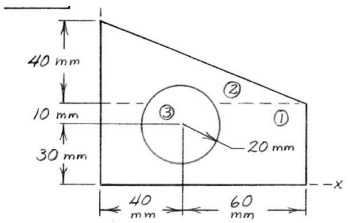


Figure G000011. Centroid · source slide page 11.

EV0000224

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ENGINEERING MECHANICS
Centroid



Part	A (mm^2)	$\bar{x}$ (mm)	$\bar{y}$ (mm)	$A\bar{x}$ (mm^3)	$A\bar{y}$ (mm^3)
1	4000	50	20	$200(10^3)$	$80(10^3)$
2	2000	100/3	$40+\frac{40}{3}$	$66.7(10^3)$	$106.7(10^3)$
3	$-\pi(20^2)$	40	30	$-50.3(10^3)$	$-37.7(10^3)$
Totals	4740			$216(10^3)$	$149.0(10^3)$

$$\bar{X} = \frac{\Sigma A\bar{x}}{\Sigma A} = \frac{216(10^3)}{4740} = 45.6 \text{ mm}$$

$$\bar{Y} = \frac{\Sigma A\bar{y}}{\Sigma A} = \frac{149.0(10^3)}{4740} = 31.4 \text{ mm}$$

Figure G000012. Centroid · source slide page 12.

EV0000225

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ENGINEERING MECHANICS
Centroid

5/55 Determine the y-coordinate of the centroid of the shaded area.

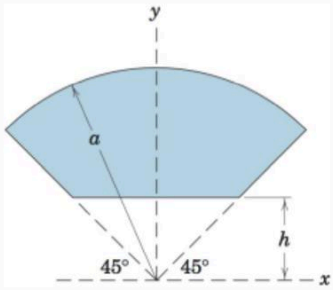


Figure G000013. Centroid · source slide page 13.

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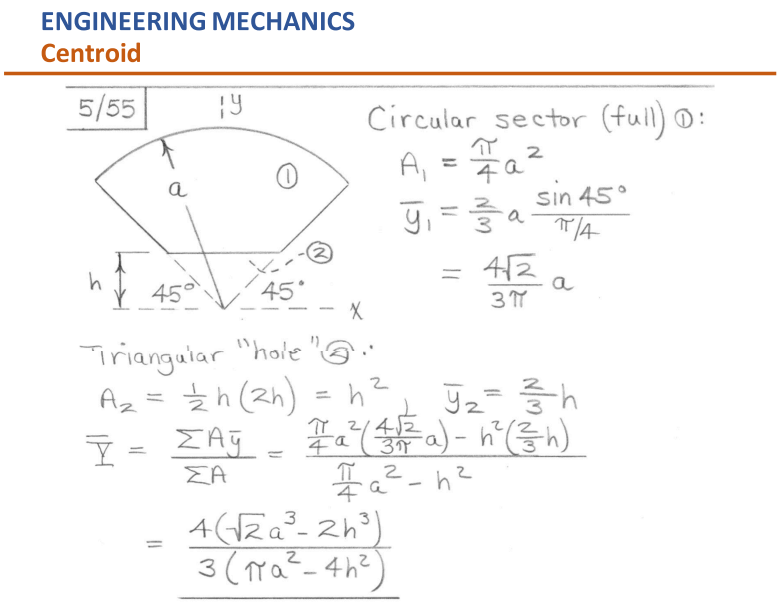


Figure G000014. Centroid · source slide page 14.

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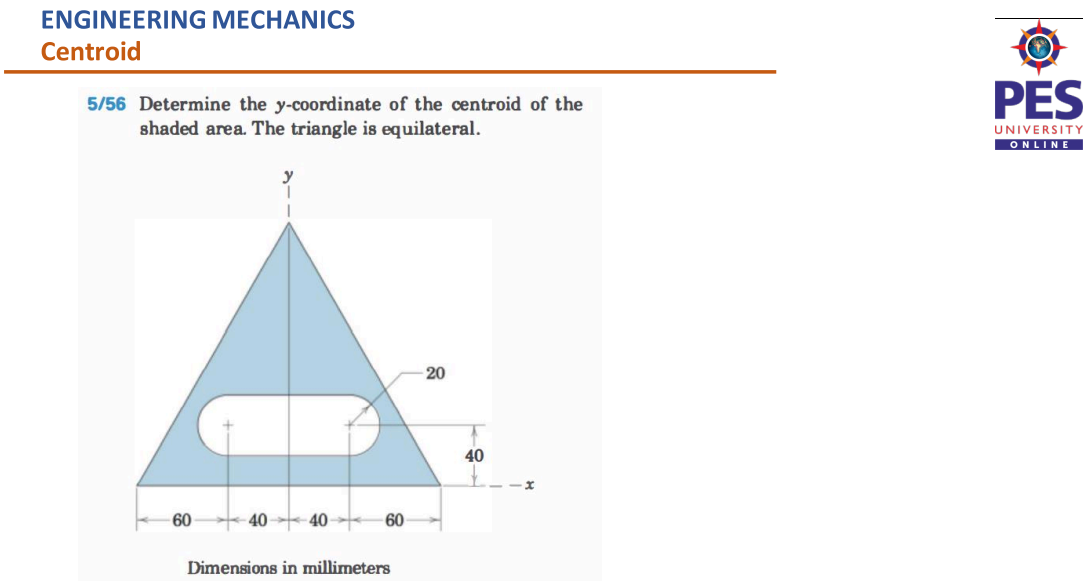


Figure G000015. Centroid · source slide page 15.

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ENGINEERING MECHANICS
Centroid

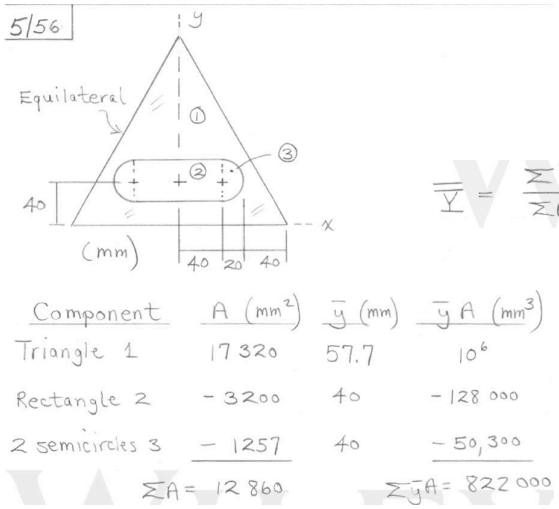


Figure G000016. Centroid · source slide page 16.

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ENGINEERING MECHANICS
Centroid

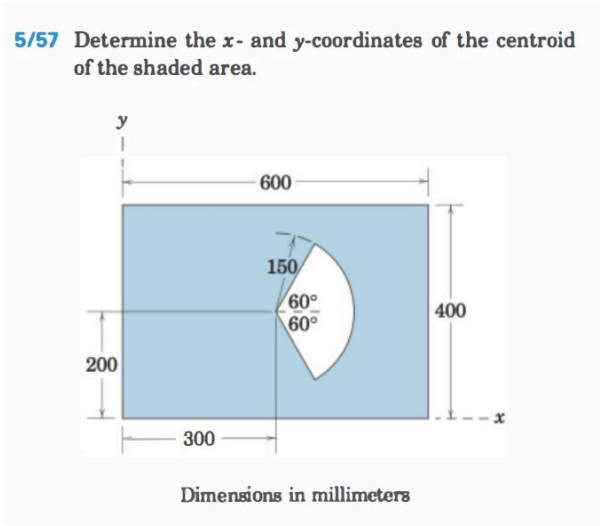


Figure G000017. Centroid · source slide page 17.

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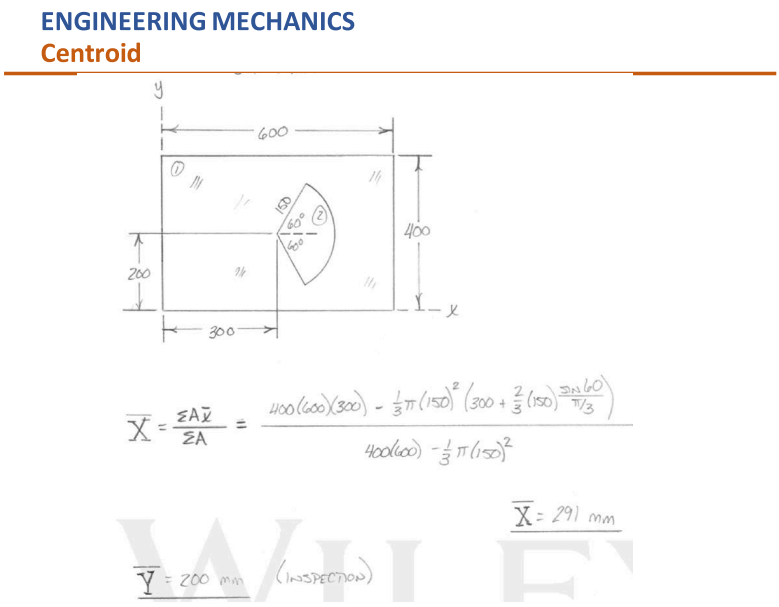


Figure G000018. Centroid · source slide page 18.

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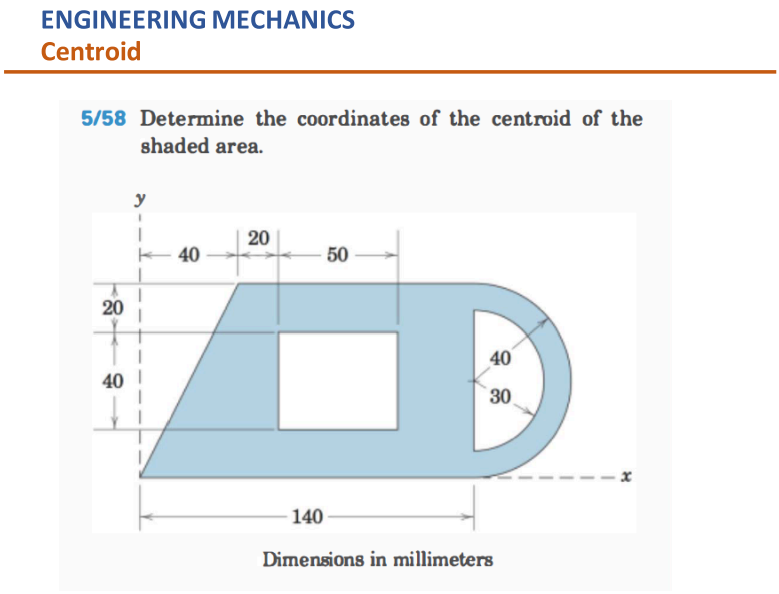


Figure G000019. Centroid · source slide page 19.

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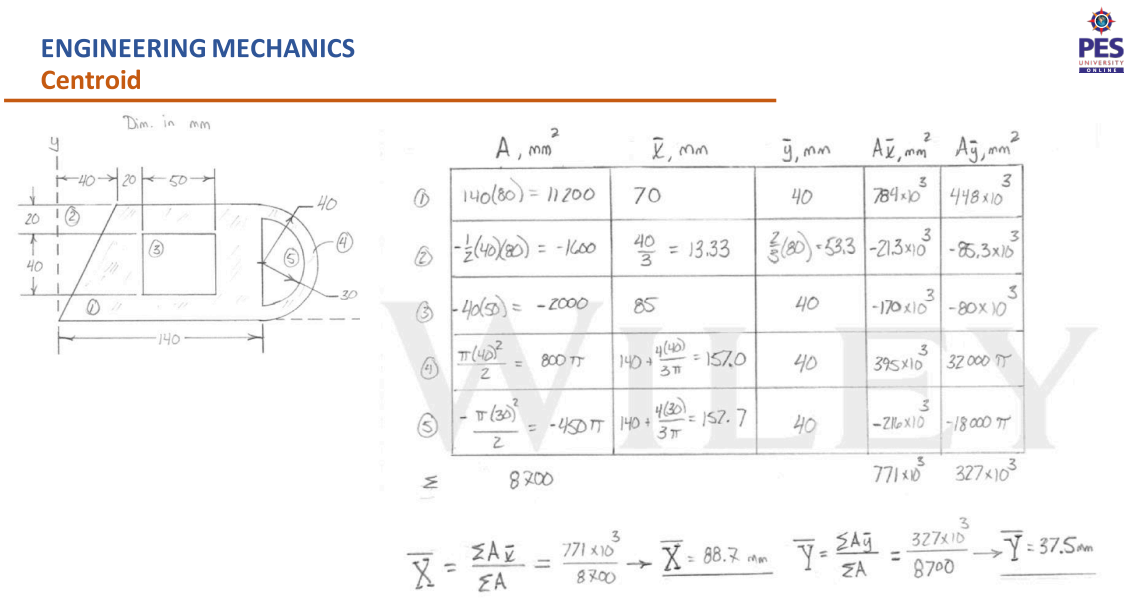


Figure G000020. Centroid · source slide page 20.

EV0000233

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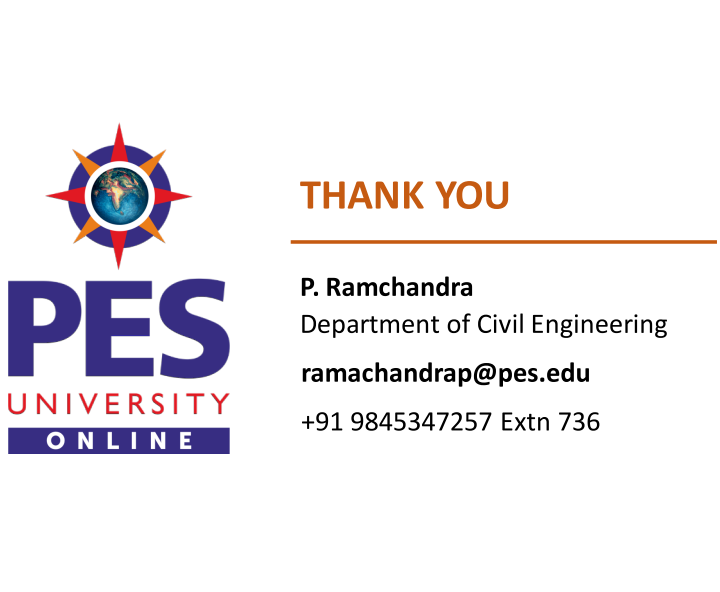


Figure G000021. Centroid · source slide page 21.

EV0000234

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS P. Ramchandra Department of Civil Engineering EV0000202 · EV0000203 · EV0000204
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS Centroid P. Ramchandra Department of Civil Engineering EV0000205 · EV0000206 · EV0000207
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS Centroid 5/47 Determine the coordinates of the centroid of the trapezoidal area shown. ## Page / slide 4 (OCR) ENGINEERING MECHANICS Centroid PES 5/47 UNIVERSITY ①A,=40(60)=2400mm ONLINE y 2 x,=20mm， g,=30mm 60 ②A2=(20)(60)=600mm 20=46.7mm mm x=40+ 3 2 60 =20mm 3 40mm>20mm 25.3mm EA 2400+600 =A=2400(30)+600(20) 28.0 mm ∑A 2400+600 EV0000208 · EV0000209 · EV0000210
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000211 · EV0000212 · EV0000213
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS Centroid 5/48 Determine the distance H from the upper surface of the symmetric double-T beam cross section to the location of the centroid. ## Page / slide 6 (OCR) ENGINEERING MECHANICS Centroid PES 200 16 UNIVERSITY ONLINE 45 120 15 20015)(# + 2(1520)(5+2) ∑Ah H = 4.3 mm H ∑A 200(15) + Z(15)(120) EV0000214 · EV0000215 · EV0000216
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000217 · EV0000218 · EV0000219

7. Source slide page 7 (extracted text): ENGINEERING MECHANICS Centroid 5/49 Determine the x- and y-coordinates of the centroid of the shaded area. ## Page / slide 8 (OCR) ENGINEERING MECHANICS Centroid PES 50 (200)(-25)+ 150 (20(5)+(100(20(50+) UNIVERSITY EAE ONLINE X EA 5D (200)+150 (200)+ (100X200) x=76.7mm 50 (200 (100)+1501200100)+ (00X00) 700 EAg EA 50 (200)+1501700)+ (00700) =93.3mm ## Page / slide 9 (OCR) ENGINEERINGMECHANICS Centroid PES 5/50 Determine the height above the base of the centroid UNIVERSITY of the cross-sectional area of the beam. Neglect the ONLINE fillets. 156 35 ←22 343 35 312 Dimensions in mil limeters ## Page / slide 10 (OCR) ENGINEERING MECHANICS Centroid PES Comp. 2 Au mm UNIVERSITY mm mm ONLINE 312(35) 35 191100 2 273 273 (22) 35+ 1030000 2 156(35) 35 35+273+ 777 2 MA=22400 MAg =3 000 000 MA 3000000 133.9 mm ZA 22400 ## Page / slide 11 (OCR) ENGINEERINGMECHANICS Centroid PES 5/51 Determine the x-and y-coordinates of the centroid of theshadedarea. UNIVERSITY ONLINE 50 mm 20 mm 40 30 mm mm 40mm 60mm ## Page / slide 12 (OCR) ENGINEERING MECHANICS Centroid PES UNIVERSITY ONLINE 40mm ZAx 10mm 216(103) X 45.6 ZA mm 20mm 4740 30mm =ZAg 149.0(103) 40 60 =31.4 mm mm mm A 4740 Part A g Ax Ag (mm²) (mm) (mm) (mm3) (mm3) / 4000 50 20 200(10) 80(103) 100/3 40+ 40 2 2000 66.7(103) 106.7(103) 3 -(20²) 40 30 -50.3(10) -37.7(103) Totals 4740 216(103)149.0(103) ## Page / slide 13 (OCR) ENGINEERINGMECHANICS Centroid PES 5/55 Determine they-coordinateof thecentroid of the UNIVERSITY ONLINE shaded area. ## Page / slide 14 (OCR) ENGINEERING MECHANICS Centroid PES 5/55 Circular sector (full) ①: UNIVERSITY A 2 ONLINE 4a Sin 45° y a 3 T/4 A a 45 3T Trianguiar "hole A=h(2h) 2/4J2 a)-h (h) MAy_= ∝ MA 4(a²-2h3 ↑a²_4h² 3 ## Page / slide 15 (OCR) ENGINEERING MECHANICS Centroid PES 5/56 Determine they-coordinateof thecentroidof the shaded area.The triangle isequilateral. UNIVERSITY ONLINE 20 40 60 4040 60 Dimensionsinmillimeters ## Page / slide 16 (OCR) ENGINEERING MECHANICS Centroid PES 5/56 UNIVERSITY ONLINE Equilateral MyA 822000 + 63.9mm ZA 12860 (mm 40 20 4o Component A mm y(mm) A mm Triangle 17320 57.7 106 Rectangle 3200 40 -128000 2semicircles3 1257 50,300 MA=12860 MyA=822000 ## Page / slide 17 (OCR) ENGINEERINGMECHANICS Centroid PES 5/57 Determine the x-and y-coordinates of the centroid UNIVERSITY oftheshadedarea. ONLINE 600 150 60° 400 60° 200 300 Dimensionsinmillimeters ## Page / slide 18 (OCR) ENGINEERING MECHANICS Centroid PES 600 UNIVERSITY ONLINE 60 200 300 SN60 400 (G00(30) (150)(300+ (50) T/3 EAY X EA 400l00) (150)² X=291mm Z00mm INSPECnON ## Page / slide 19 (OCR) ENGINEERINGMECHANICS Centroid PES 5/58 Determine the coordinates of the centroid of the UNIVERSITY shaded area. ONLINE 20 40 50 20 40 40 30 140 Dimensions in millimeters ## Page / slide 20 (OCR) ENGINEERING MECHANICS PES UNIVERSITY Centroid ONLINE Dim.inmm A,mm 2 y ,mm Ay,mm 2 2 g,mm Ag mm 4020K50 3 3 140 (80)=11200 70 40 784x0 448×10 HO # 20 (40(0)=-1600 40 (80-53 3 3 3 4 =13.33 -21.3x10 -85.3x10 3 40 3 3 40(5)=-2000 8S 40 -170x10 -80x10 140 4(40) =157.0 3 800π 140+ 40 395×10 32000T 2 3T (30 4(30) 3 =157.7 140+ 40 -4S0T - Zlx10 -18000 3π 2 3 771x10 327×10 3 800 M 3 3 Ag 327×10 771×10 =37.Smm A X =88.7mm ZA 8700 EA 800 EV0000220
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9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000222

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13. Source slide page 13: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000226
14. Source slide page 14: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000227
15. Source slide page 15: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000228
16. Source slide page 16: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000229
17. Source slide page 17: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000230
18. Source slide page 18: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000231
19. Source slide page 19: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000232
20. Source slide page 20: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000233
21. Source slide page 21 (extracted text): THANK YOU P. Ramchandra Department of Civil Engineering ramachandrap@pes.edu +91 9845347257 Extn 736 EV0000234

SHORT-NOTE RECAP

Source deck preserved: Centroid. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is score-based; this index does not claim a manual content review.

EV0000202

15 Source slide deck: UE26ME142A Unit 3 Class 33 Lecture PPT

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Graphs and diagrams



Figure G000022. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 1.

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Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Centroid

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Figure G000023. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 2.

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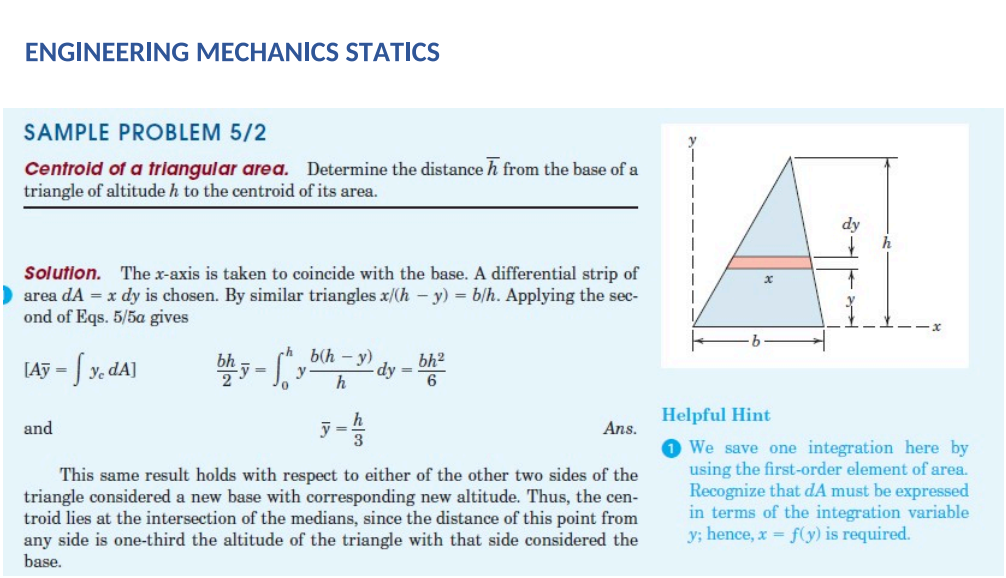
ENGINEERING MECHANICS STATICS

- Center of mass
- Center of gravity
- Principle or Body diagonal
- Diagonal
- Lamina
- Centroid - $\dot{x}$, $\bar{y}$
- Centroidal axis – x_o, y_o
- Reference axis – x, y

Figure G000024. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 3.

EV0000242

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ENGINEERING MECHANICS STATICS

SAMPLE PROBLEM 5/2

Centroid of a triangular area. Determine the distance $\bar{h}$ from the base of a triangle of altitude h to the centroid of its area.

Solution. The x-axis is taken to coincide with the base. A differential strip of area $dA = x\,dy$ is chosen. By similar triangles $x/(h - y) = b/h$. Applying the second of Eqs. 5/5a gives

$$[A\bar{y} = \int y_c\,dA] \qquad \frac{bh}{2}\bar{y} = \int_0^h y\frac{b(h-y)}{h}dy = \frac{bh^2}{6}$$

and $$\bar{y} = \frac{h}{3} \qquad \textit{Ans.}$$

This same result holds with respect to either of the other two sides of the triangle considered a new base with corresponding new altitude. Thus, the centroid lies at the intersection of the medians, since the distance of this point from any side is one-third the altitude of the triangle with that side considered the base.

Helpful Hint

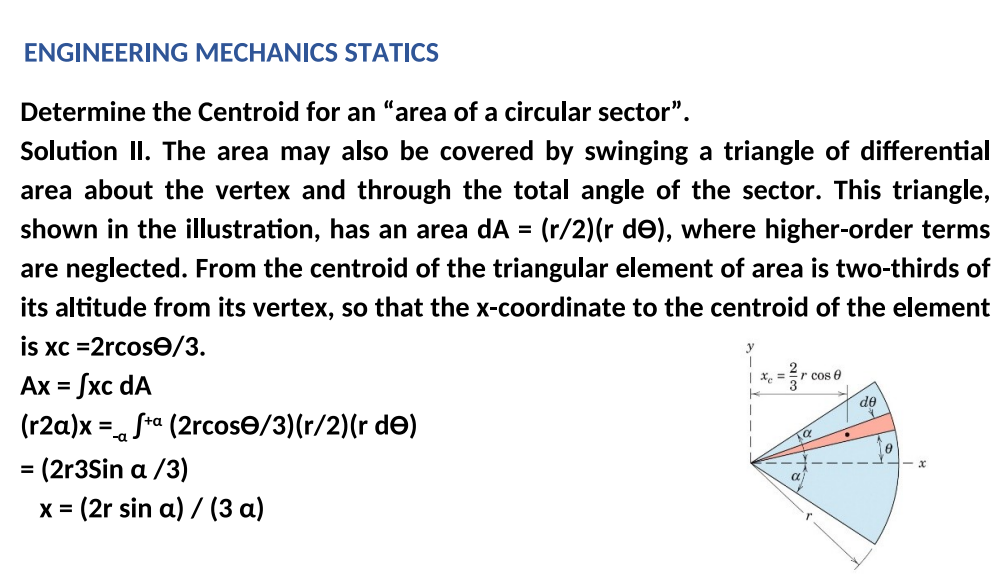
1. We save one integration here by using the first-order element of area. Recognize that dA must be expressed in terms of the integration variable y; hence, $x = f(y)$ is required.

S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Figure G000025. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS

Determine the Centroid for an "area of a circular sector".

Solution II. The area may also be covered by swinging a triangle of differential area about the vertex and through the total angle of the sector. This triangle, shown in the illustration, has an area dA = (r/2)(r dΘ), where higher-order terms are neglected. From the centroid of the triangular element of area is two-thirds of its altitude from its vertex, so that the x-coordinate to the centroid of the element is xc =2rcosΘ/3.

Ax = ∫xc dA

(r2α)x = $\int_{-\alpha}^{+\alpha}$ (2rcosΘ/3)(r/2)(r dΘ)

= (2r3Sin α /3)

x = (2r sin α) / (3 α)

S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Figure G000026. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 5.

EV0000248

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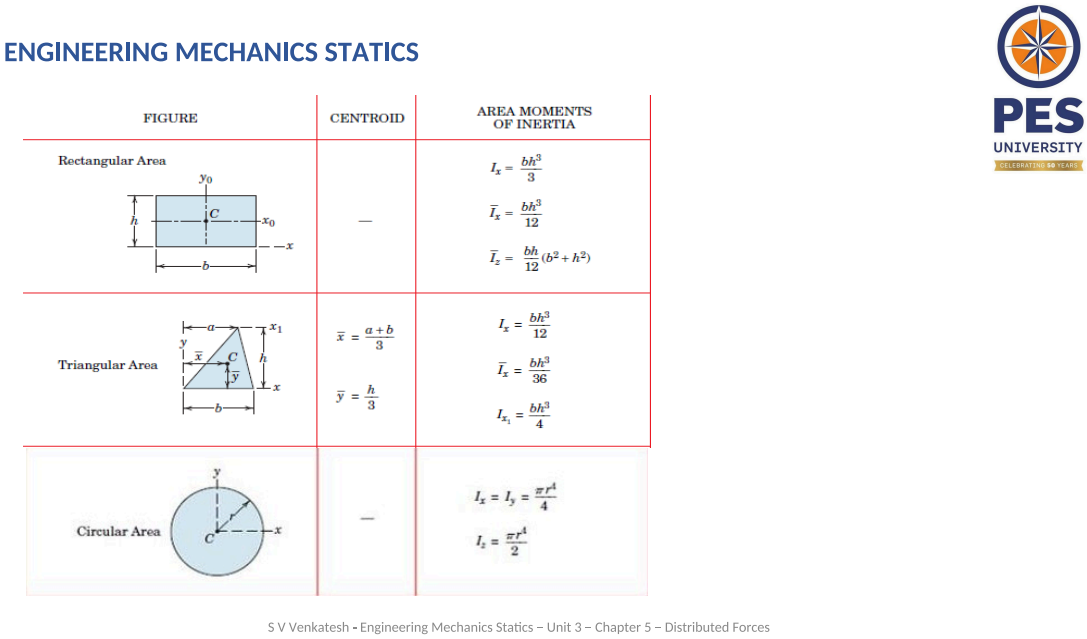


Figure G000027. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 6.

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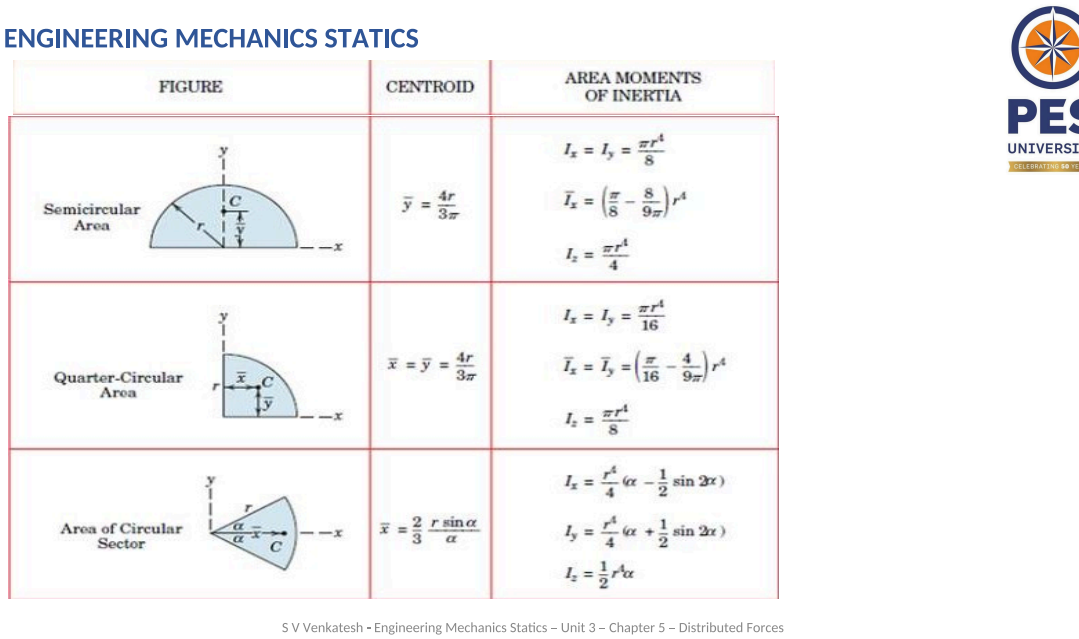


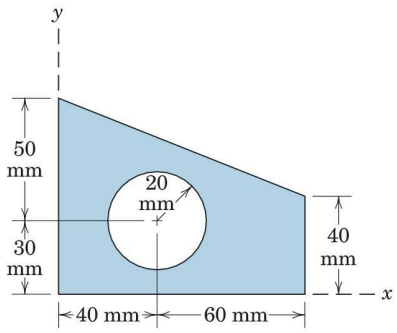
Figure G000028. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 7.

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ENGINEERING MECHANICS STATICS

Determine the x and y - coordinates of the centroid of the shaded area.



$\Sigma A = 4000 + 2000 - \prod(20^2) = 4740$

$\Sigma A \, \tilde{x} = 200(10^3) + 66.74(10^3) - 50.3(10^3) = 216(10^3)$

$\Sigma A \, \bar{y} = 80\,(10^3) + 106.7(10^3) - 37.7(10^3) = 149.0(10^3)$

$\tilde{x} = \Sigma A \, \tilde{x} / \Sigma A = 216(10^3) / 4740 = 45.60$ mm

$\bar{y} = \Sigma A \, \bar{y} / \Sigma A = 149.0\,(10^3) / 4740 = 51.4$ mm

S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Figure G000029. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 8.

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ENGINEERING MECHANICS STATICS

S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces

Figure G000030. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 9.

EV0000260

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Figure G000031. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 10.

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Figure G000032. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 11.

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Figure G000033. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 12.

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Figure G000034. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 13.

EV0000272

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Figure G000035. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 14.

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Figure G000036. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 15.

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Figure G000037. UE26ME142A Unit 3 Class 33 Lecture PPT · source slide page 16.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000235 · EV0000236 · EV0000237
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Centroid Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000238 · EV0000239 · EV0000240
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Center of mass Center of gravity Principle or Body diagonal Diagonal Lamina Centroid - $\dot{x}$, $\bar{y}$ Centroidal axis – xo, yo Reference axis – x, y S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000241 · EV0000242 · EV0000243
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000244 · EV0000245 · EV0000246

5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Determine the Centroid for an “area of a circular sector”. Solution II. The area may also be covered by swinging a triangle of differential area about the vertex and through the total angle of the sector. This triangle, shown in the illustration, has an area dA = (r/2)(r dΘ), where higher- order terms are neglected. From the centroid of the triangular element of area is two-thirds of its altitude from its vertex, so that the x-coordinate to the centroid of the element is xc =2rcosΘ/3. Ax = ∫xc dA (r2α)x =-α ∫+α (2rcosΘ/3)(r/2)(r dΘ) = (2r3Sin α /3) x = (2r sin α) / (3 α) S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000247 · EV0000248 · EV0000249
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000250 · EV0000251 · EV0000252
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000253 · EV0000254 · EV0000255
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS rmine the x and y - coordinates of the centroid of the shaded area. ΣA = 4000 + 2000 - ∏(202) = 4740 ΣA $\bar{X}$ = 200(103) + 66.74(103) – 50.3(103) = 216(103) ΣA $\bar{y}$ = 80 (103) + 106.7(103) – 37.7(103) = 149.0(103) $\bar{X}$ = ΣA $\bar{X}$ / ΣA = 216(103) / 4740 = 45.60 mm $\bar{y}$ = ΣA $\bar{y}$ / ΣA = 149.0 (103) S V Venkatesh - Engineering Mechanics Statics/ 4740– Unit 3 – Chapter 5 – Distributed Forces EV0000256 · EV0000257 · EV0000258
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000259 · EV0000260 · EV0000261
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000262 · EV0000263 · EV0000264
11. Source slide page 11 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000265 · EV0000266 · EV0000267
12. Source slide page 12 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000268 · EV0000269 · EV0000270
13. Source slide page 13 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000271 · EV0000272 · EV0000273
14. Source slide page 14 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces EV0000274 · EV0000275 · EV0000276
15. Source slide page 15 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 16 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000277

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16 Source slide deck: UE20CV101 UNIT 4 SESSION 2

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Graphs and diagrams

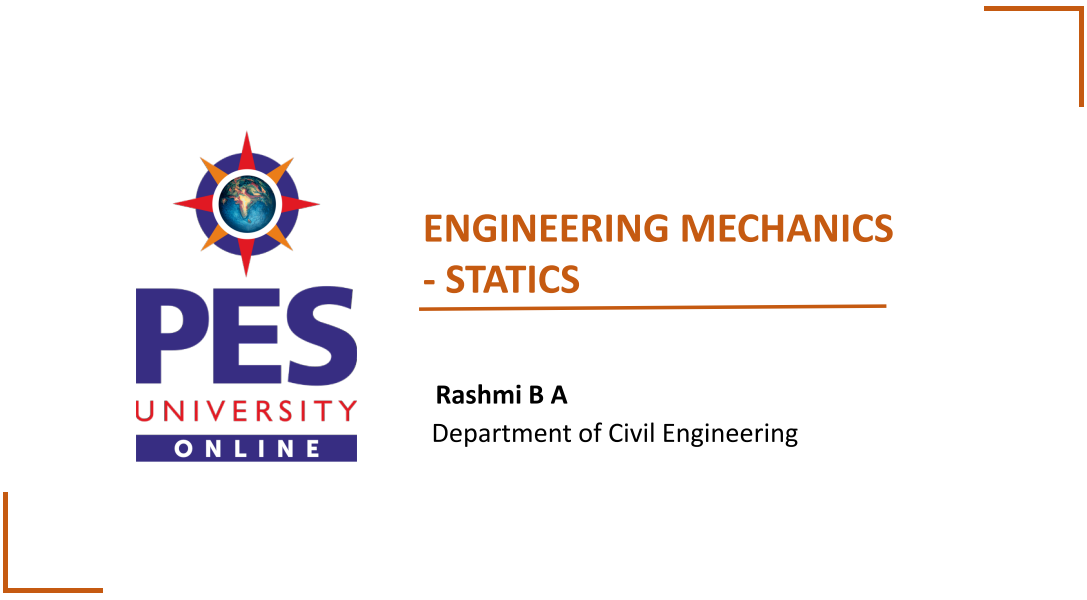


Figure G000038. UE20CV101 UNIT 4 SESSION 2 · source slide page 1.

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ENGINEERING MECHANICS - STATICS

DISTRIBUTED FORCES

Session- 2

Rashmi B A
Department of Civil Engineering

Figure G000039. UE20CV101 UNIT 4 SESSION 2 · source slide page 2.

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ENGINEERING MECHANICS - STATICS
Centroid

Centroid of the area of a circular sector:

Due to symmetry $\bar{y} = 0$ & $\bar{x} = \frac{\int x_c dA}{A}$ ----(1)

Consider an element at a distance r_o from the centre O of a sector, radial width being dr_o & bound by radii at θ & $(\theta + d\theta)$

$dA = r_o d\theta\, dr_o \qquad x_c = r_o \cos\theta \qquad A = R^2\alpha$

Substituting in equation(1)

$$\bar{x} = \frac{\int r_o \cos(\theta)\, r_o d\theta\, dr_o}{A} = \frac{\int_{-\alpha}^{+\alpha}\int_0^R r_o^{\,2}\, dr_o \cos\theta\, d\theta}{A} = \frac{\frac{R^3}{3}\, 2\sin\alpha}{A}$$

$$= \frac{\frac{R^3}{3}\, 2\sin\alpha}{R^2\alpha} = \frac{2R\sin\alpha}{3\alpha}$$

Here α is in radians.

$$\boxed{\bar{x} = \frac{2R\sin\alpha}{3\alpha} \;\&\; \bar{y} = 0}$$

Figure G000040. UE20CV101 UNIT 4 SESSION 2 · source slide page 3.

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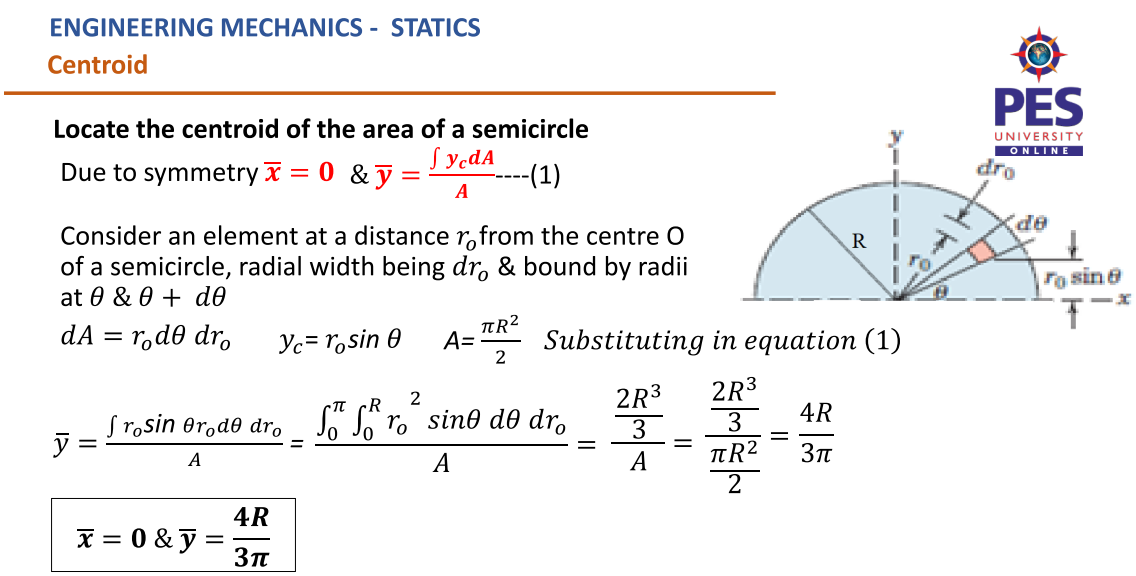


Figure G000041. UE20CV101 UNIT 4 SESSION 2 · source slide page 4.

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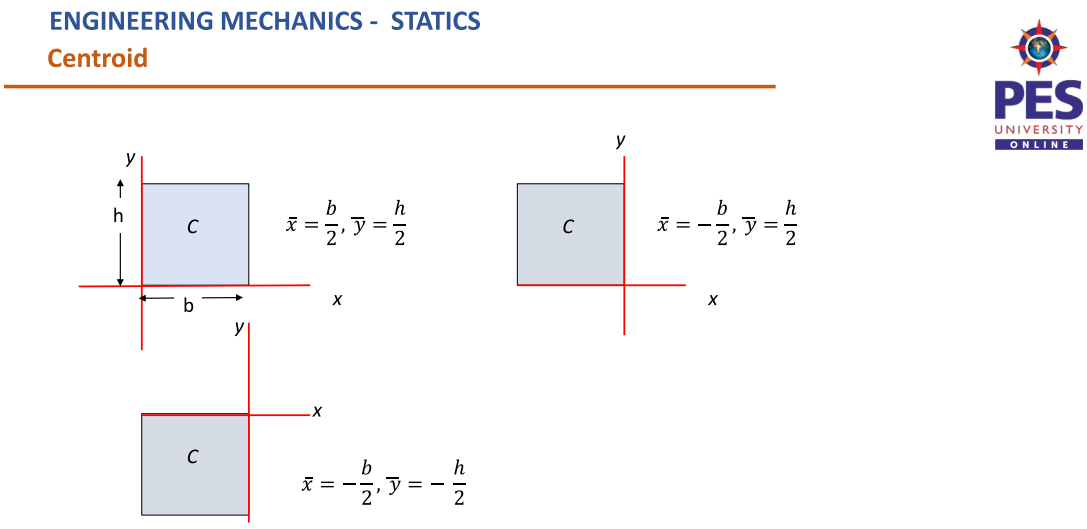


Figure G000042. UE20CV101 UNIT 4 SESSION 2 · source slide page 5.

EV0000292

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

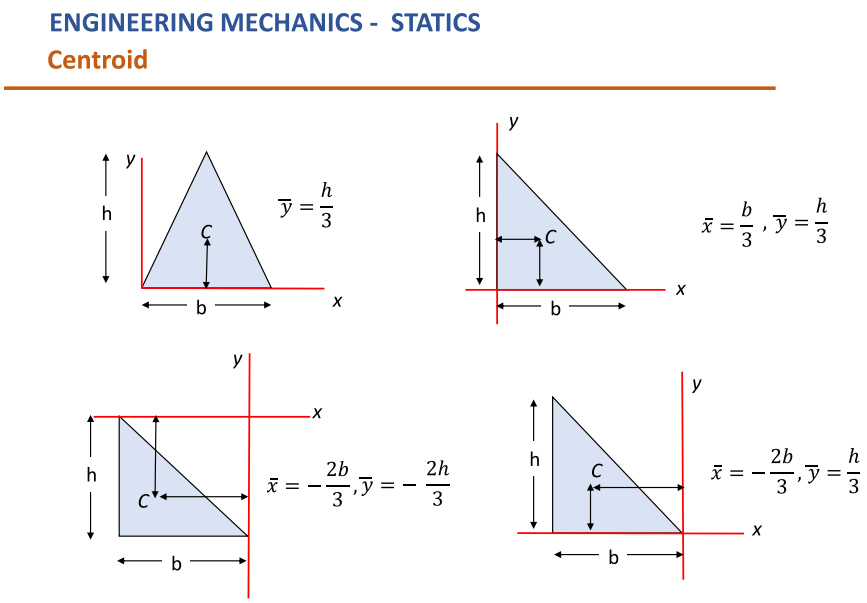


Figure G000043. UE20CV101 UNIT 4 SESSION 2 · source slide page 6.

EV0000295

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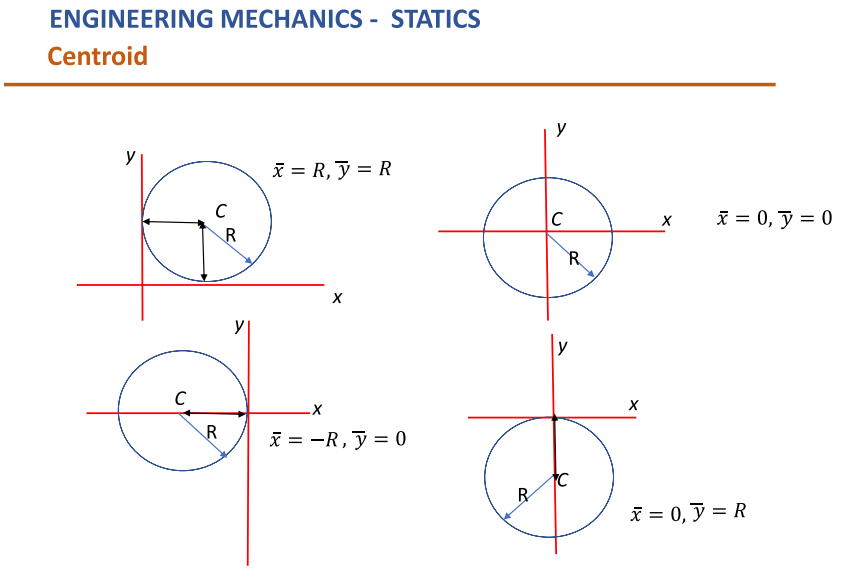


Figure G000044. UE20CV101 UNIT 4 SESSION 2 · source slide page 7.

EV0000298

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

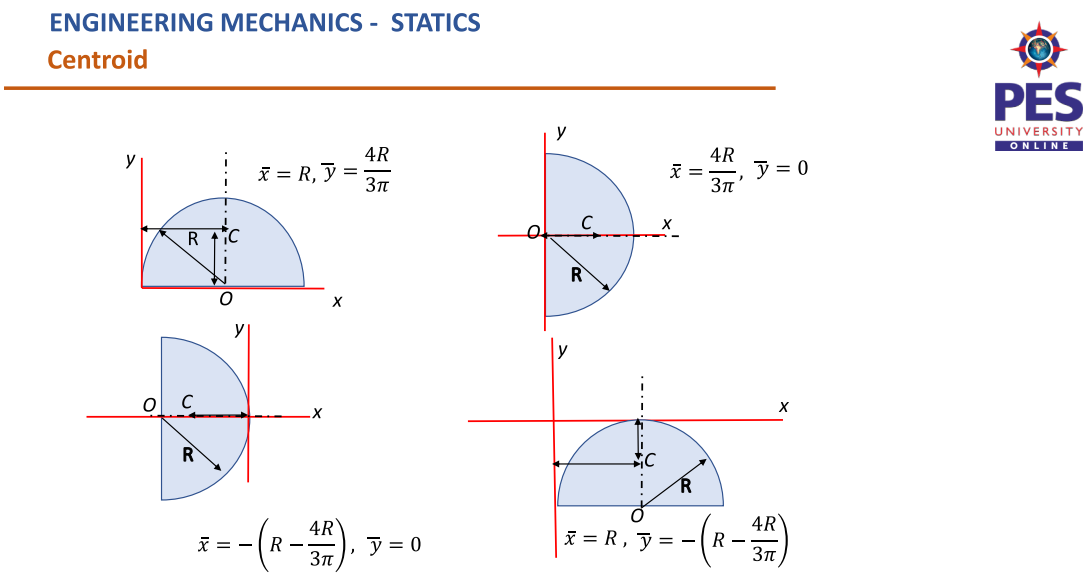


Figure G000045. UE20CV101 UNIT 4 SESSION 2 · source slide page 8.

EV0000301

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

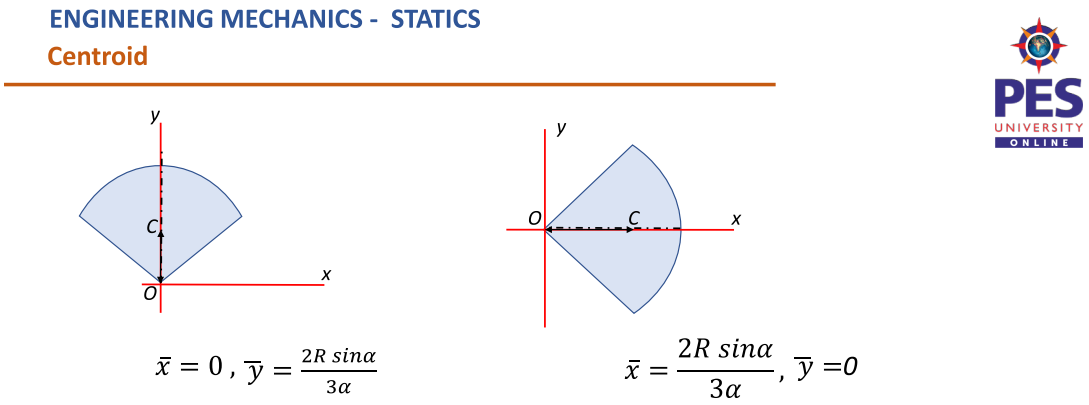


Figure G000046. UE20CV101 UNIT 4 SESSION 2 · source slide page 9.

EV0000304

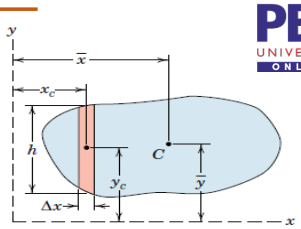
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS - STATICS

Centroid

Composite Bodies And Figures; Approximations

The area is divided into strips of width *dx* and variable height h. The area A of each strip, such as the one shown in red, is h *dx* and is multiplied by the coordinates x_c and y_c of its centroid to obtain the moments of the element of area. The sum of the moments for all strips divided by the total area of the strips will give the corresponding centroidal coordinate. A systematic tabulation of the results will permit an orderly evaluation of the total area $\sum A$, the sums $\sum Ax_c$ and $\sum Ay_c$, and the centroidal coordinates



$$\bar{x} = \frac{\sum_{i=1}^{i=n} a_i x_i}{\sum_{i=1}^{i=n} a_i}, \bar{y} = \frac{\sum_{i=1}^{i=n} a_i y_i}{\sum_{i=1}^{i=n} a_i} \quad \sum_{i=1}^{i=n} a_i = \mathrm{A}$$

Figure G000047. UE20CV101 UNIT 4 SESSION 2 · source slide page 10.

EV0000307

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THANK YOU

Rashmi B A
Department of Civil Engineering
rashmiba@pes.edu

Figure G000048. UE20CV101 UNIT 4 SESSION 2 · source slide page 11.

EV0000310

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS - STATICS Rashmi B A Department of Civil Engineering EV0000279 · EV0000280 · EV0000281
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS - STATICS DISTRIBUTED FORCES Session- 2 Rashmi B A Department of Civil Engineering EV0000282 · EV0000283 · EV0000284
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - STATICS Centroid Centroid of the area of a circular sector: ☒ $xcdA$ Due to symmetry y= **0**& x= ----(1) A Consider an element at a distance $ro$from the centre O of a sector, radial width being dro& bound by radii at θ& (θ+ $d\theta$) $R2\alpha dA$= $rod\theta dro$ xc= $rocos\theta$ A= $Substitutinginequation$1 2 +α $R3$ ☒ rocos(θ) $rod\theta dro$ 2 $sin\alpha$ ☒☒ Rro $drocos\theta d\theta$ x= = −α 0 3 A = A A $R3$ 2 $sin\alpha$ $2Rsin\alpha$ 3 = = $R2$ α 3α **2Rsinα** Here $\alpha$is in radians. x= & y= **0 3α** EV0000285 · EV0000286 · EV0000287
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS - STATICS Centroid Locate the centroid of the area of a semicircle ☒ $ycdA$ Due to symmetry x= **0** & y= ----(1) A Consider an element at a distance $ro$from the centre O of a semicircle, radial width being dro& bound by radii at θ& θ+ $d\theta$ $\pi R2$ dA= $rod\theta dro$ yc= rosin θ A= $Substitutinginequation$1 2 2 2$R3$ π 2$R3$ 4R ☒ rosin $\theta rod\theta dro$ ☒☒Rro $sin\theta d\theta dro$ 3 0 0 3 =y= = = = A $\pi R2$ 3π A A 2 **4R** x= **0**& y= **3π** EV0000288 · EV0000289 · EV0000290
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - STATICS Centroid y y b h h h C x= , y= C x= −b , y= 2 2 2 2 x x b y x C x= −b , y= −h 2 2 EV0000291 · EV0000292 · EV0000293
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - STATICS Centroid y y h y= b h h h 3 , y= x= C C 3 3 x x b b y y x h h C x= −2b , y= h x= −2b , y= −2h 3 3 C 3 3 x b b EV0000294 · EV0000295 · EV0000296
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - STATICS Centroid y y x= R, y= R C C x x= 0, y= 0 R R x y y C x x R x= −R , y= 0 C R y= R x= 0, EV0000297 · EV0000298 · EV0000299
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS - STATICS Centroid y 4R 4R y x= R, y= x= y= 0 3π 3π, C x O R C O x y y O C x x C O x= R, y= −R−4R x= −R−4R y= 0 3π 3π, EV0000300 · EV0000301 · EV0000302
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS - STATICS Centroid y y O C x C x O 2$Rsin\alpha$ 2$Rsin\alpha$ x= 0 , y= x= , y=0 3α 3α EV0000303 · EV0000304 · EV0000305
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS - STATICS Centroid Composite Bodies And Figures; Approximations The area is divided into strips of width dx and variable height h. The area A of each strip, such as the one shown in red, is h dx and is multiplied by the coordinates $xc$and $yc$of its centroid to obtain the moments of the element of area. The sum of the moments for all strips divided by the total area of the strips will give the corresponding centroidal coordinate. A systematic tabulation of the results will permit an orderly evaluation of the total area σ A, the sums σ $Axc$and σ Ayc, and the centroidal coordinates σi=1i=$nai$$xi$ σi=1i=$nai$$yi$ x= y= σi=1i=nai= A σi=1i=nai, σi=1i=nai EV0000306 · EV0000307 · EV0000308

11. Source slide page 11 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000309 · EV0000310 · EV0000311

SHORT-NOTE RECAP

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EV0000279

17 Source slide deck: UE24CV141A Unit 3 Beams

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Graphs and diagrams



Figure G000067. UE24CV141A Unit 3 Beams · source slide page 1.

EV0000367

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Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams external Effects

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Figure G000068. UE24CV141A Unit 3 Beams · source slide page 2.

EV0000370

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ENGINEERING MECHANICS STATICS

Types of Beams

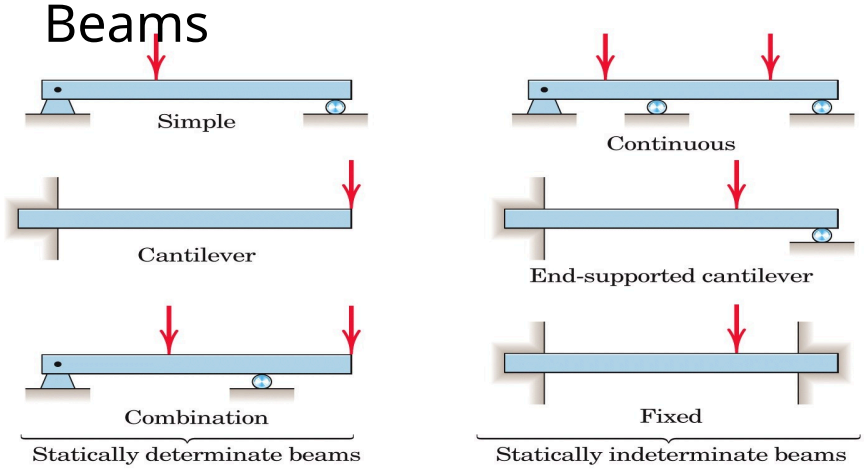


Figure 5-18
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Figure G000069. UE24CV141A Unit 3 Beams · source slide page 3.

EV0000373

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

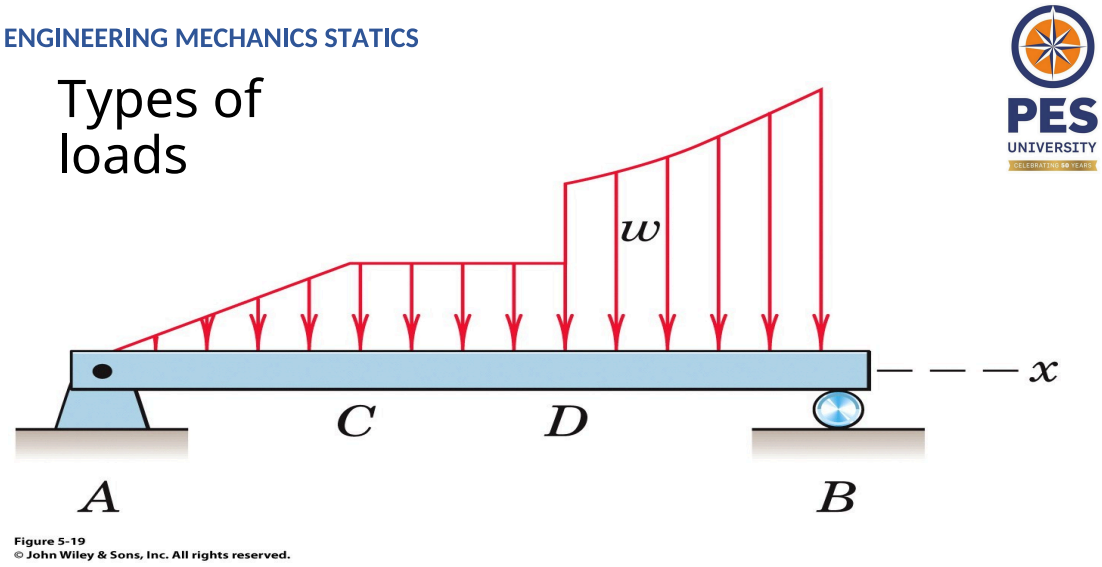


Figure G000070. UE24CV141A Unit 3 Beams · source slide page 4.

EV0000375

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

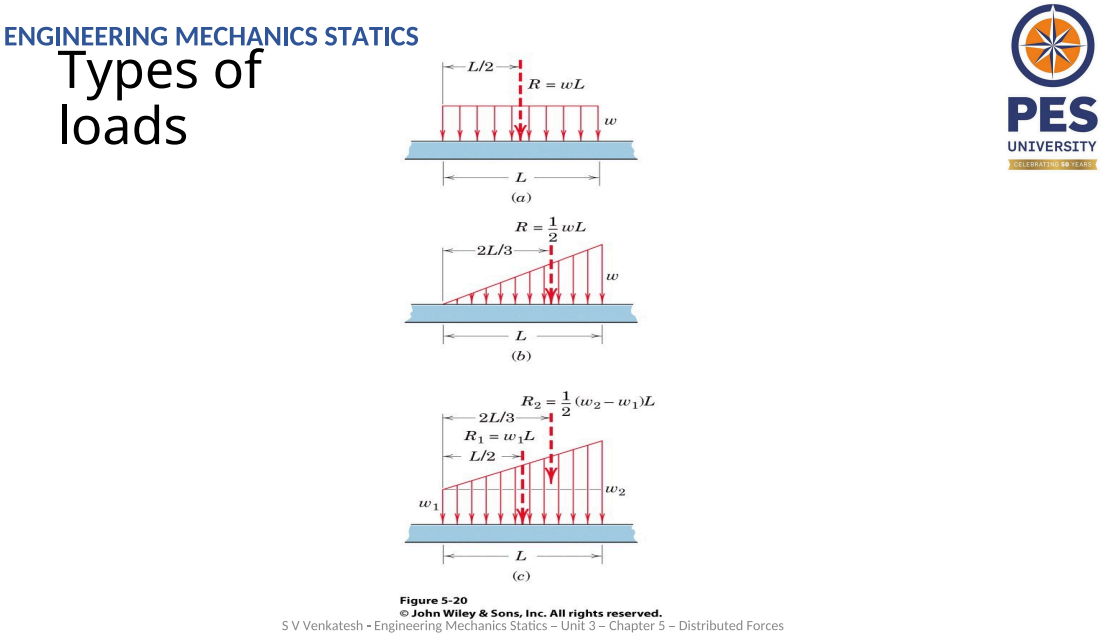


Figure G000071. UE24CV141A Unit 3 Beams · source slide page 5.

EV0000376

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000072. UE24CV141A Unit 3 Beams · source slide page 6.

EV0000377

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000366 · EV0000367 · EV0000368
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams external Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000369 · EV0000370 · EV0000371
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Types of Beams S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 4 (OCR) ENGINEERINGMECHANICSSTATICS Types of PES loads UNIVERSITY CELEBRATING50YEARS w C D A B Figure5-19 JohnWiley&Sons,Inc.Allrightsreserved. S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000372 · EV0000373 · EV0000374
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000375
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Types of loads S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000376

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000377

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EV0000366

18 Source slide deck: UE25CV141A Unit 3 Class 48 Lecture PPT

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Graphs and diagrams



Figure G000073. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 1.

EV0000379

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 3
DISTRIBUTED FORCES
Chapter 5
Beams external Effects

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Figure G000074. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 2.

EV0000382

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

Types of Beams

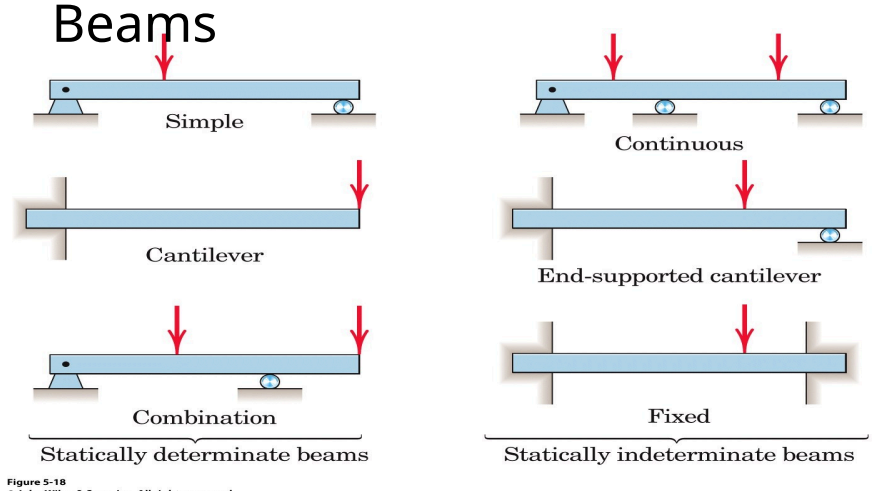


Figure 5-18
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Figure G000075. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 3.

EV0000385

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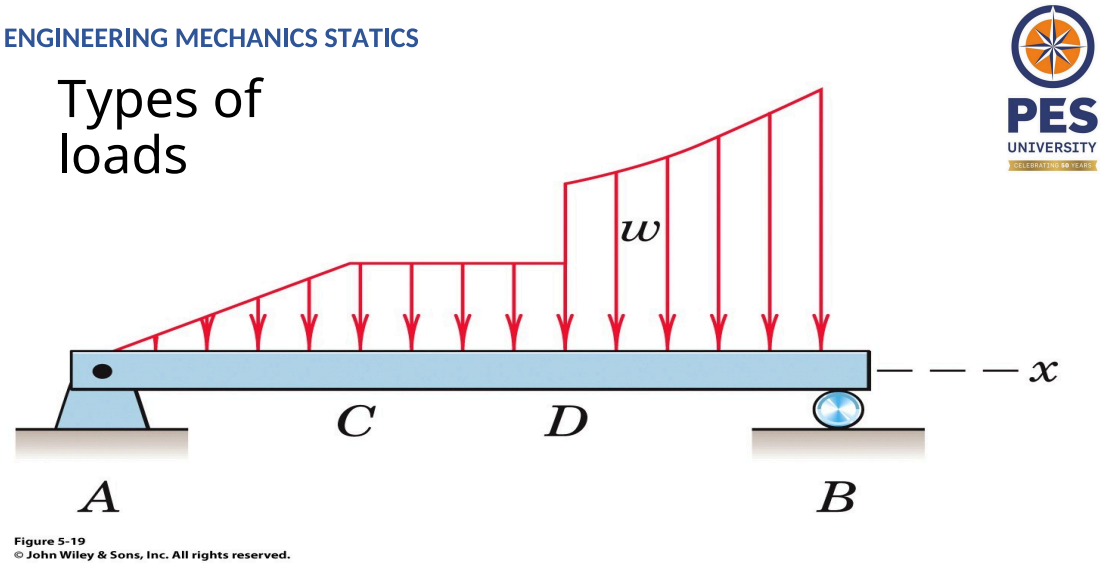


Figure G000076. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 4.

EV0000387

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

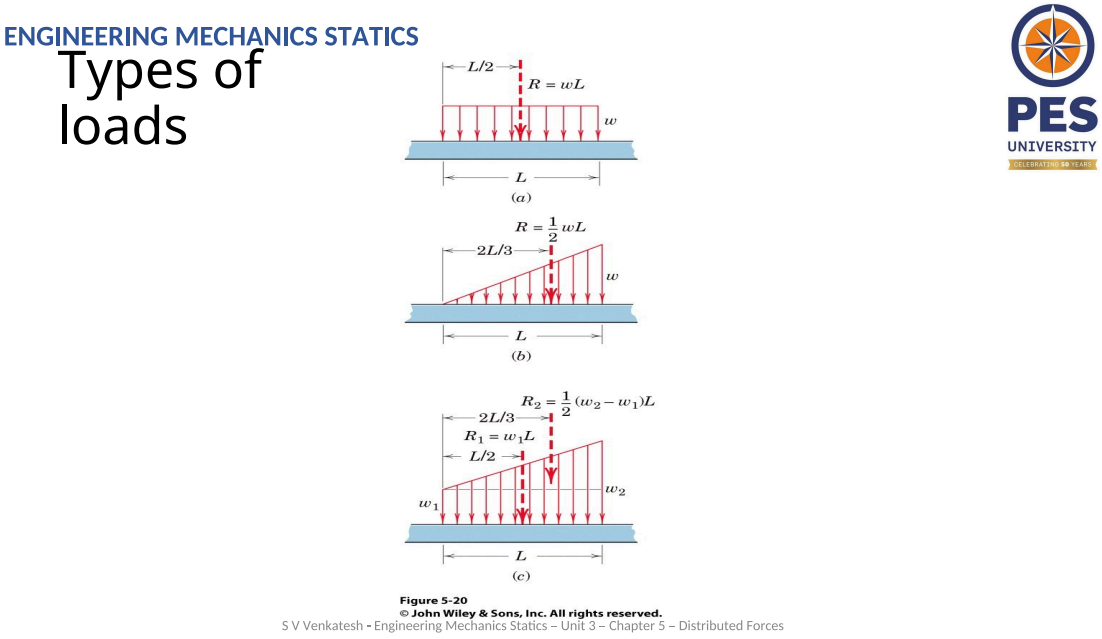


Figure G000077. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 5.

EV0000388

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Figure G000078. UE25CV141A Unit 3 Class 48 Lecture PPT · source slide page 6.

EV0000389

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000378 · EV0000379 · EV0000380
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 3 DISTRIBUTED FORCES Chapter 5 Beams external Effects Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics – Unit 3 – Chapter 5 – Distributed forces EV0000381 · EV0000382 · EV0000383
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Types of Beams S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 4 (OCR) ENGINEERINGMECHANICSSTATICS Types of PES loads UNIVERSITY CELEBRATING50YEARS w C D A B Figure5-19 JohnWiley&Sons,Inc.Allrightsreserved. S V Venkatesh - Engineering Mechanics Statics - Unit 3 - Chapter 5 - Distributed Forces EV0000384 · EV0000385 · EV0000386
4. Source slide page 4: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000387
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Types of loads S V Venkatesh - Engineering Mechanics Statics – Unit 3 – Chapter 5 – Distributed Forces ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000388

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000389

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EV0000378

19 Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 8

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Graphs and diagrams

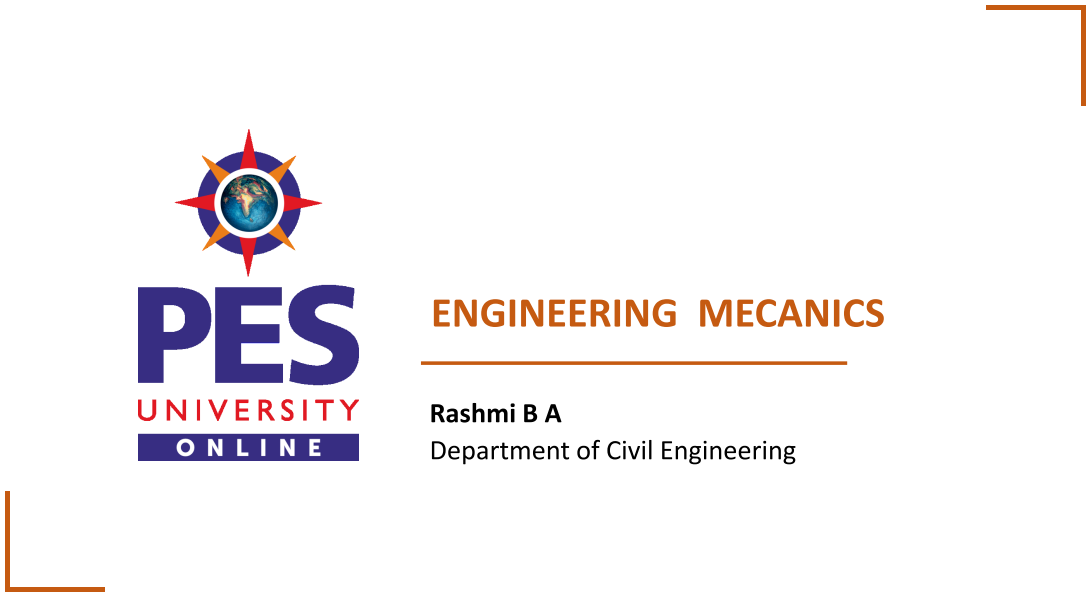


Figure G000079. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 1.

EV0000391

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECANICS

Beams

Rashmi B A
Department of Civil Engineering

Figure G000080. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 2.

EV0000394

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS

5/101 Determine the reactions at *A* and *B* for the beam subjected to the uniform load distribution.

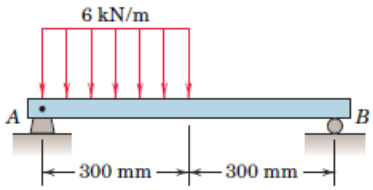


Figure G000081. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 3.

EV0000397

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

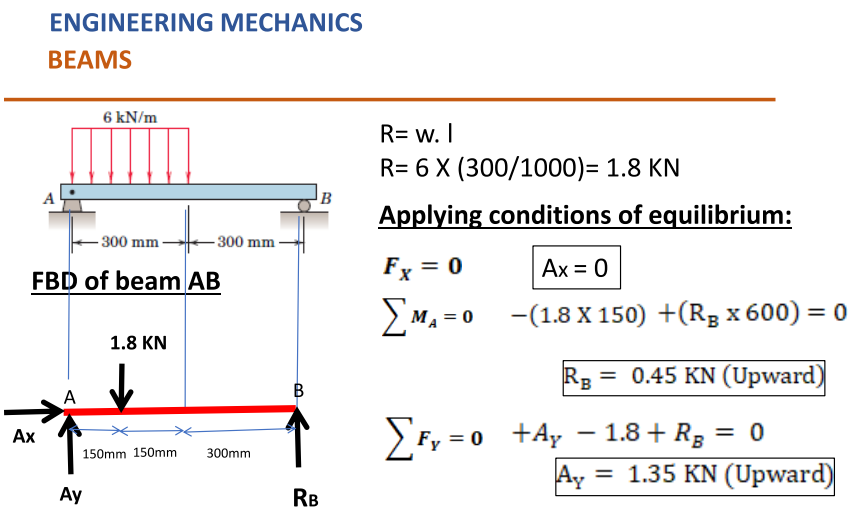


Figure G000082. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 4.

EV0000400

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

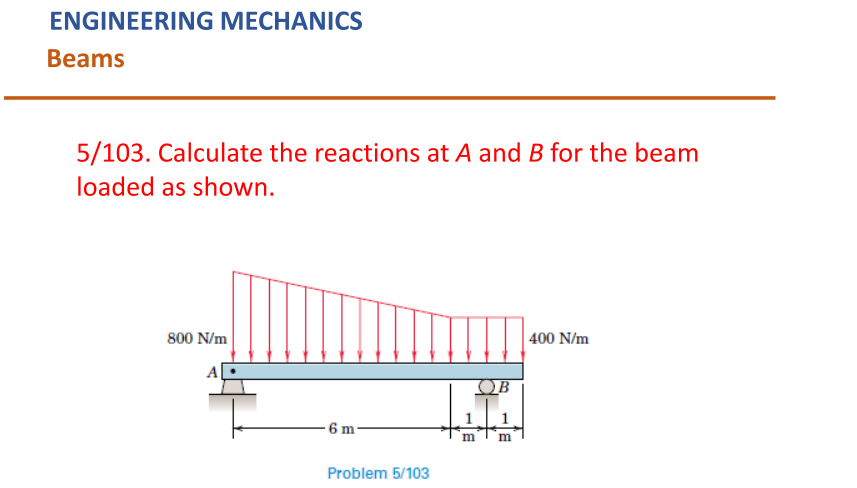


Figure G000083. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 5.

EV0000403

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

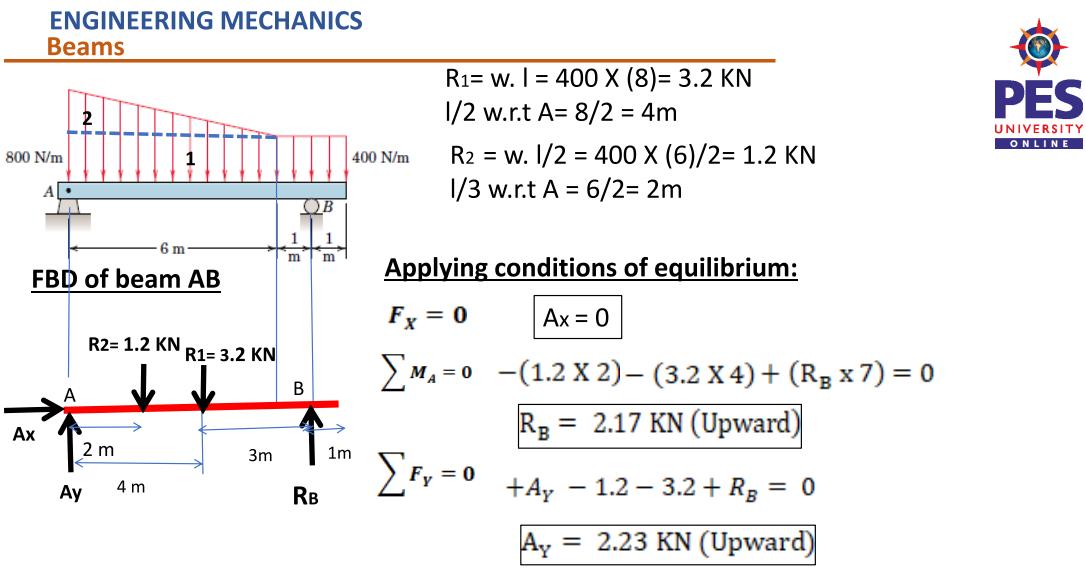


Figure G000084. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 6.

EV0000406

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Figure G000085. UE20CV101 UNIT 3 ONLINE SESSION 8 · source slide page 7.

EV0000409

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECANICS Rashmi B A Department of Civil Engineering EV0000390 · EV0000391 · EV0000392
2. Source slide page 2 (extracted text): ENGINEERING MECANICS Beams Rashmi B A Department of Civil Engineering EV0000393 · EV0000394 · EV0000395
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS BEAMS 5/101 Determine the reactions at A and B for the beam subjected to the uniform load distribution. EV0000396 · EV0000397 · EV0000398
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS BEAMS R= w. l R= 6 X (300/1000)= 1.8 KN Applying conditions of equilibrium: Ax = 0 FBD of beam AB 1.8 KN B A Ax 150mm 150mm 300mm Ay RB EV0000399 · EV0000400 · EV0000401
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS Beams 5/103. Calculate the reactions at A and B for the beam loaded as shown. EV0000402 · EV0000403 · EV0000404
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS Beams R1= w. l = 400 X (8)= 3.2 KN 2 l/2 w.r.t A= 8/2 = 4m 1 R2 = w. l/2 = 400 X (6)/2= 1.2 KN l/3 w.r.t A = 6/2= 2m Applying conditions of equilibrium: FBD of beam AB Ax = 0 R2= 1.2 KN R1= 3.2 KN B A Ax 2 m 3m 1m 4 m Ay RB EV0000405 · EV0000406 · EV0000407
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000408 · EV0000409 · EV0000410

SHORT-NOTE RECAP

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EV0000390

20 Source slide deck: UE20CV101 UNIT 3 ONLINE SESSION 9

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Graphs and diagrams

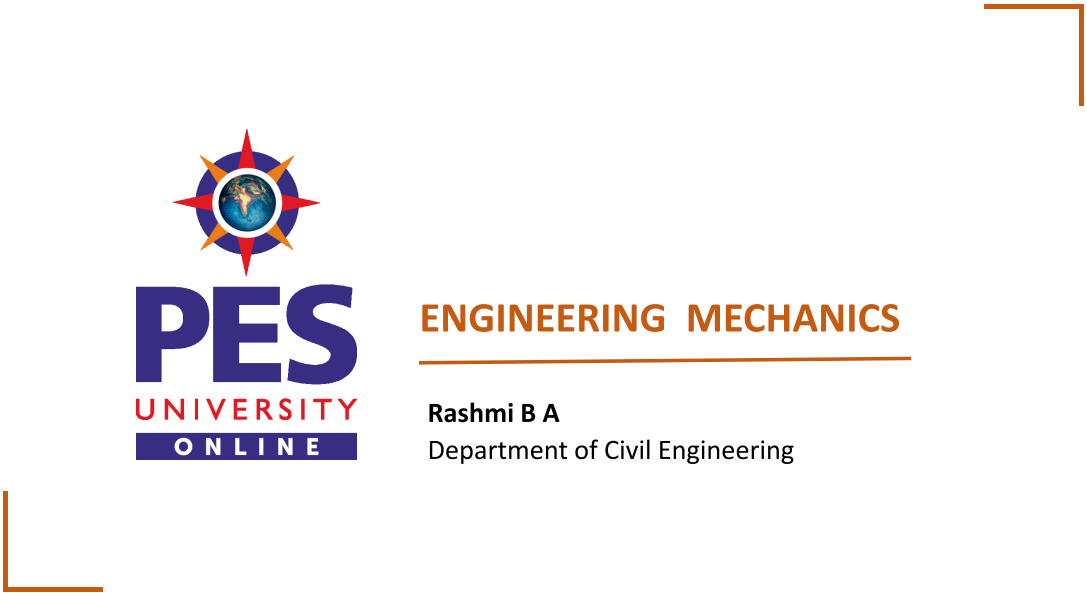


Figure G000086. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 1.

EV0000412

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS

STRUCTURES

Rashmi B A
Department of Civil Engineering

Figure G000087. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 2.

EV0000415

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS

5/105. Find the reaction at A due to the uniform loading and the applied couple.

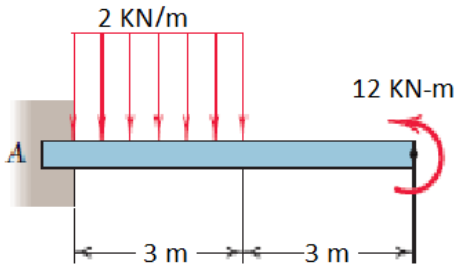
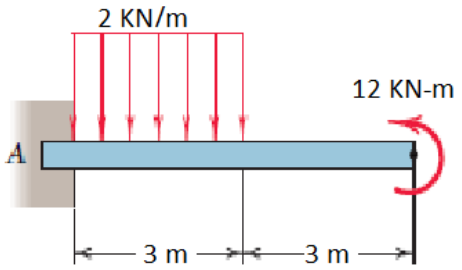


Figure G000088. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 3.

EV0000418

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS



R_A= 6 KN Up
M_A= 3 KN.m CW

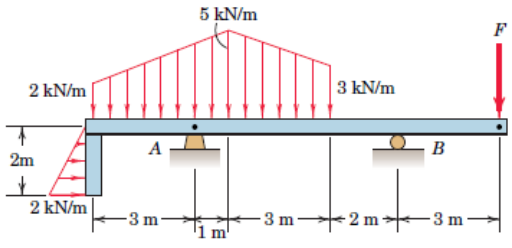
Figure G000089. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 4.

EV0000421

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS

5/116. For the beam and loading shown, determine the magnitude of the force *F* for which the vertical reactions at *A* and *B* are equal. With this value of *F*, compute the magnitude of the pin reaction at *A*.



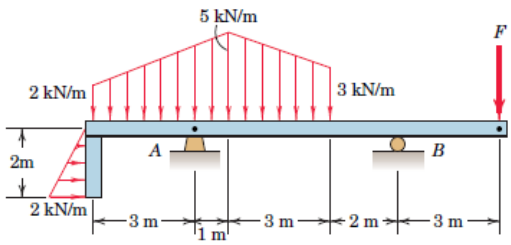
Problem 5/116

Figure G000090. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 5.

EV0000424

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS



Problem 5/116

F = 10.36 KN
R_A= 18.29 KN

Figure G000091. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 6.

EV0000427

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

THANK YOU

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Figure G000092. UE20CV101 UNIT 3 ONLINE SESSION 9 · source slide page 7.

EV0000430

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS Rashmi B A Department of Civil Engineering EV0000411 · EV0000412 · EV0000413
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STRUCTURES Rashmi B A Department of Civil Engineering EV0000414 · EV0000415 · EV0000416
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS BEAMS 5/105. Find the reaction at A due to the uniform loading and the applied couple. EV0000417 · EV0000418 · EV0000419
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS BEAMS RA= 6 KN Up MA= 3 KN.m CW EV0000420 · EV0000421 · EV0000422
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS BEAMS 5/116. For the beam and loading shown, determine the magnitude of the force F for which the vertical reactions at A and B are equal. With this value of F, compute the magnitude of the pin reaction at A. EV0000423 · EV0000424 · EV0000425
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS BEAMS F = 10.36 KN RA= 18.29 KN EV0000426 · EV0000427 · EV0000428
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000429 · EV0000430 · EV0000431

SHORT-NOTE RECAP

Source deck preserved: UE20CV101 UNIT 3 ONLINE SESSION 9. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is score-based; this index does not claim a manual content review.

EV0000411

21 Source slide deck: UE26ME142A UNIT 3 ONLINE SESSION 41

This entry preserves the complete slide deck titled UE26ME142A UNIT 3 ONLINE SESSION 41 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_03, to avoid an unsupported match. EV0000432 · EV0000433

Graphs and diagrams

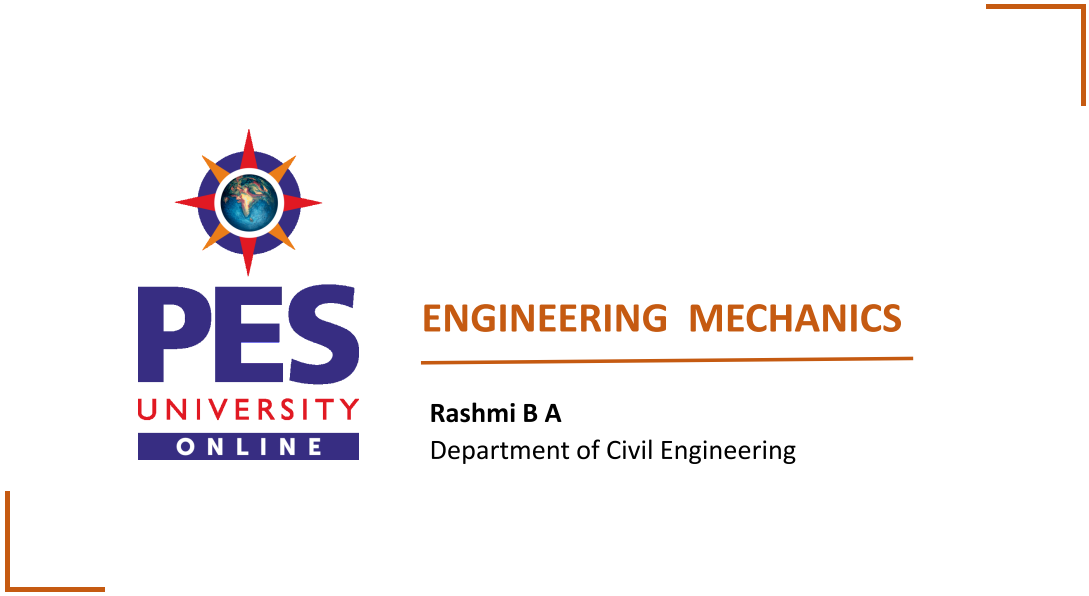


Figure G000093. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 1.

EV0000433

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS

STRUCTURES

Rashmi B A
Department of Civil Engineering

Figure G000094. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 2.

EV0000436

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
Beams

5/107 Determine the reactions at A and B for the beam loaded as shown.

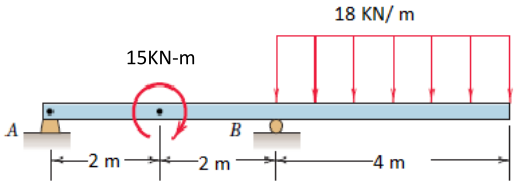
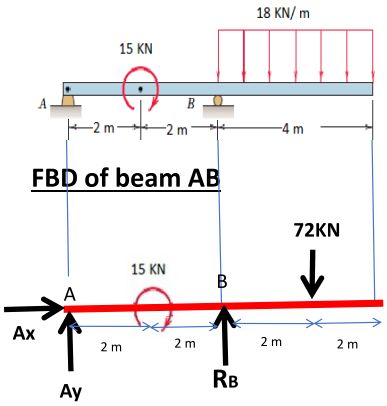


Figure G000095. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 3.

EV0000439

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
BEAMS



R= w. l
R= 18 X (4)= 72 KN

Applying conditions of equilibrium:

$F_X = 0$ Ax = 0

$\sum M_A = 0$ - 15 + (R_B x 4) - (72 x (2+2+2)) = 0

R_B = 111.75 KN

$\sum F_Y = 0$ + A_Y + R_B -72 = 0

A_Y = 39.75KN

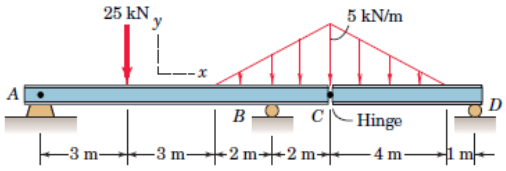
Figure G000096. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 4.

EV0000442

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
Beams

5/109. Determine the reactions at *A, B,* and *D* for the pair of beams connected by the ideal pin at *C* and subjected to the concentrated and distributed loads.



Problem 5/109

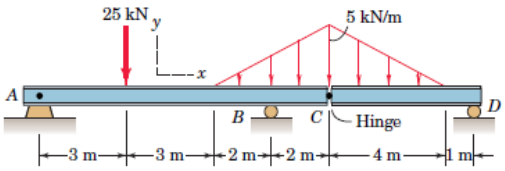
Figure G000097. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 5.

EV0000445

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS

Beams



Problem 5/109

$A_X = 0$

$A_Y = 12.96KN$

$B_Y = 29.4KN$

$D_Y = 2.67KN$

Figure G000098. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 6.

EV0000448

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

THANK YOU

Rashmi B A
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Figure G000099. UE26ME142A UNIT 3 ONLINE SESSION 41 · source slide page 7.

EV0000451

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS Rashmi B A Department of Civil Engineering EV0000432 · EV0000433 · EV0000434
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STRUCTURES Rashmi B A Department of Civil Engineering EV0000435 · EV0000436 · EV0000437
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS Beams 5/107 Determine the reactions at A and B for the beam loaded as shown. 15KN-m EV0000438 · EV0000439 · EV0000440
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS BEAMS R= w. l R= 18 X (4)= 72 KN Applying conditions of equilibrium: Ax = 0 FBD of beam AB - 15 + (RB x 4) - (72 x (2+2+2)) = 0 72KN RB = 111.75 KN B A + + -72 = 0Ax AY RB 2 m 2 m 2 m 2 m = 39.75KN RB AY Ay EV0000441 · EV0000442 · EV0000443
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS Beams 5/109. Determine the reactions at A, B, and D for the pair of beams connected by the ideal pin at C and subjected to the concentrated and distributed loads. EV0000444 · EV0000445 · EV0000446
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS Beams AX = 0 BY = 29.4KN AY = 12.96KN DY = 2.67KN EV0000447 · EV0000448 · EV0000449
7. Source slide page 7 (extracted text): THANK YOU Rashmi B A Department of Civil Engineering rashmiba@pes.edu EV0000450 · EV0000451 · EV0000452

SHORT-NOTE RECAP

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EV0000432

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.

Source	Path	Use
CLASS 33 Introduction to Centroid transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 33 Introduction to Centroid transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/033_CLASS_33_Introduction_to_Centroid__67dd96fa80cc7e7c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 34 Location of the Centroid of a Sector & Semicircular Lamina transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/034_CLASS_34_Location_of_the_Centroid_of_a_Sector_Semicircular_Lamina__3bf50905d5d6aee5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 36 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/035_CLASS_36_Centroid__Numerical__8d2e6c9175d9d883.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 35 Centroid_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 35 Centroid_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/036_CLASS_35_Centroid__Numerical__6bf222208dbdadf5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 29 Introduction to Beams transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 29 Introduction to Beams transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/039_CLASS_29_Introduction_to_Beams__719328349228b241.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
CLASS 30 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/040_CLASS_30_Beams_Numerical__eba902d8efe49ef0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 31 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE25CV141A_Unit 3_Class 53_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 53_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/042_UE25CV141A_Unit_3_Class_53_AVSummary__fb2a14ea9c91ccb3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 54_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE25CV141A_Unit 3_Class 54_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/043_UE25CV141A_Unit_3_Class_54_AV_Summary__912c9dc62b19aae7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE25CV141A_Unit 3_Class 55_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE25CV141A_Unit_3_Class_55_AV_Summary__292b04ba1e28ea04.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Centroid	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Centroid	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Centroid (converted PDF; original hash 6f51d7bc8a1daab8830750f6f5293f7b5339619f4152cedcb407c7e55a2b64f5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 3 Class 33 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A Unit 3 Class 33 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 3 Class 33 Lecture PPT (converted PDF; original hash 21e5fd6df4dee659c3d04ed9736319637d645f4506f5cdc7a06008eb470938d4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 2 (converted PDF; original hash 2637a5f5acfd6be279cbdc1ada5c369beec1a971e612312a6303ceb42b71ded8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 3 (converted PDF; original hash 855d90c138ebd09ccd87444c3ca92bc80298d14585d21a759dab47f34f50f340)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 4 (converted PDF; original hash 98aff97f5c1596c1d7d2a9a246bd2617a83cb4eb8293cdac8a7732f46224d2e5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE24CV141A Unit 3 Beams	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE24CV141A Unit 3 Beams	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE24CV141A Unit 3 Beams (converted PDF; original hash ebb61d8de2a89dc7df0dd92e8f1b503458f063b51cbf37478c9aebffc2651e41)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 48 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 48 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 48 Lecture PPT (converted PDF; original hash ebb61d8de2a89dc7df0dd92e8f1b503458f063b51cbf37478c9aebffc2651e41)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 3 ONLINE SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE20CV101 UNIT 3 ONLINE SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 3 ONLINE SESSION 8 (converted PDF; original hash 1ba5fc810a2d6b1882d2198242bbf43c152b21e3b65ae248aa8423782827ef48)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 3 ONLINE SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 3 ONLINE SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 3 ONLINE SESSION 9 (converted PDF; original hash 0b4358b69167fcc42fb2884667d9299882cabd80425a2a68535a12554f7ac094)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A UNIT 3 ONLINE SESSION 41	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A UNIT 3 ONLINE SESSION 41	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A UNIT 3 ONLINE SESSION 41 (converted PDF; original hash d50b5f40928a41599890eee7efc577bfe0f2f9c52818f0c722a071108930895e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 53 Lecture PPT (converted PDF; original hash 764f7c39a17dfafc7dee861f13a669ecec990a216b3bc40402cf6930874a3d6b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 54 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE25CV141A Unit 3 Class 54 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 54 Lecture PPT (converted PDF; original hash a2b0aaf2ea90e653a4f79c8bbbf71b30fabe0cca15e221d3cdbee88b44728b9d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE25CV141A Unit 3 Class 55 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE25CV141A Unit 3 Class 55 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE25CV141A Unit 3 Class 55 Lecture PPT (converted PDF; original hash ab718cce2de3498ea99fc0ee34a3028cc61a5fd54dd151458ff8a2936eac7ef2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 3 Class 44.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A Unit 3 Class 44.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 3 Class 44.1 Lecture PPT (converted PDF; original hash 2ef2625de901226124ab0b819e8dc0e27969ca67ff246d04970156ac6728792a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.

9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0085. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0066 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0098. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000023	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000024	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000025	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000026	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000027	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000034	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0083. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000037	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00006/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000039	SRC00007/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00008/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0086. Sidecar variants compare at similarity 0.9984; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000041	SRC00009/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000048	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000050	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000055	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000062	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000069	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000070	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000071	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000072	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000076	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000082	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000083	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000084	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0071. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000085	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000086	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000087	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000088	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000089	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000090	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000091	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000097	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000103	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 04	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 05	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 06	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 07	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 08	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 09	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 10	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001111	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001112	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001113	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001114	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001115	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001116	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0067. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001117	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000118	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000119	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000120	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 25	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 26	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 27	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 28	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 29	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 30	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 31	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000132	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0067. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000133	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0067. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000134	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000135	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000139	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000142	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001146	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001147	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001148	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0076. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001149	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001150	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001151	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001152	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000153	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000154	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000155	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000160	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0071. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00041/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000167	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000170	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000174	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 81	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 82	SRC00042/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 83	SRC00043/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 84	SRC00044/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0086. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 85	SRC00045/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 86	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 87	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000188	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000189	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000190	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000191	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 95	SRC00047/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 96	SRC00048/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 97	SRC00049/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 98	SRC00046/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00047/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00048/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0072. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00049/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000202	SRC00050/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000203	SRC00052/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000204	SRC00051/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000205	SRC00050/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000206	SRC00052/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000207	SRC00051/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000208	SRC00050/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000209	SRC00052/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000210	SRC00051/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000211	SRC00050/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000212	SRC00052/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000213	SRC00051/p4	covered	Alternate extracted slide text page 4; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000214	SRC00050/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000215	SRC00052/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000216	SRC00051/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000217	SRC00050/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000218	SRC00052/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000219	SRC00051/p6	covered	Alternate extracted slide text page 6; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000220	SRC00052/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000221	SRC00052/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000222	SRC00052/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000223	SRC00052/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002 24	SRC00052/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 25	SRC00052/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 26	SRC00052/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 27	SRC00052/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 28	SRC00052/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 29	SRC00052/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002230	SRC00052/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002231	SRC00052/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002232	SRC00052/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002233	SRC00052/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002234	SRC00052/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_Centroid__9f0e55503970de0e.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002235	SRC00053/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000236	SRC00055/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000237	SRC00054/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000238	SRC00053/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000239	SRC00055/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000240	SRC00054/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000241	SRC00053/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000242	SRC00055/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000243	SRC00054/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000244	SRC00053/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000245	SRC00055/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000246	SRC00054/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000247	SRC00053/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000248	SRC00055/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000249	SRC00054/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000250	SRC00053/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000251	SRC00055/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000252	SRC00054/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000253	SRC00053/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 54	SRC00055/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 55	SRC00054/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 56	SRC00053/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 57	SRC00055/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 58	SRC00054/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 59	SRC00053/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 60	SRC00055/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 61	SRC00054/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 62	SRC00053/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 63	SRC00055/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 64	SRC00054/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 65	SRC00053/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 66	SRC00055/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 67	SRC00054/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 68	SRC00053/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 69	SRC00055/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 70	SRC00054/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 71	SRC00053/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000272	SRC00055/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000273	SRC00054/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000274	SRC00053/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000275	SRC00055/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000276	SRC00054/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000277	SRC00055/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002 78	SRC00055/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 79	SRC00056/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 80	SRC00058/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 81	SRC00057/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 82	SRC00056/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 83	SRC00058/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002284	SRC00057/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002285	SRC00056/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002286	SRC00058/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002287	SRC00057/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00002288	SRC00056/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002289	SRC00058/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002290	SRC00057/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002291	SRC00056/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002292	SRC00058/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002293	SRC00057/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002294	SRC00056/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002295	SRC00058/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002296	SRC00057/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 97	SRC00056/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 98	SRC00058/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 99	SRC00057/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 00	SRC00056/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 01	SRC00058/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 02	SRC00057/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 03	SRC00056/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 04	SRC00058/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 05	SRC00057/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 06	SRC00056/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 07	SRC00058/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 08	SRC00057/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000309	SRC00056/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000310	SRC00058/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/034_UE20CV101_UNIT_4__SESSION-2__70f5373bd98843ba.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00057/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000312	SRC00059/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00061/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00060/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003315	SRC00059/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003316	SRC00061/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003317	SRC00060/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003318	SRC00059/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003319	SRC00061/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003320	SRC00060/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 21	SRC00059/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 22	SRC00061/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 23	SRC00060/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 24	SRC00059/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 25	SRC00061/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 26	SRC00060/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003327	SRC00059/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003328	SRC00061/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003329	SRC00060/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003330	SRC00059/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003331	SRC00061/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003332	SRC00060/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003333	SRC00059/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003334	SRC00061/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003335	SRC00060/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003336	SRC00059/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003337	SRC00061/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003338	SRC00060/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 39	SRC00059/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 40	SRC00061/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/035_UE20CV101_UNIT_4__SESSION-3__50fa7f321184f351.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 41	SRC00060/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 42	SRC00062/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 43	SRC00064/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 44	SRC00063/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 45	SRC00062/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 46	SRC00064/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 47	SRC00063/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 48	SRC00062/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 49	SRC00064/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 50	SRC00063/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 51	SRC00062/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 52	SRC00064/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 53	SRC00063/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 54	SRC00062/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 55	SRC00064/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 56	SRC00063/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 57	SRC00062/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 58	SRC00064/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 59	SRC00063/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 60	SRC00062/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 61	SRC00064/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 62	SRC00063/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 63	SRC00062/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 64	SRC00064/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/036_UE20CV101_UNIT_4__SESSION-4__f39bbd7029add5e8.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 65	SRC00063/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 66	SRC00065/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 67	SRC00067/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 68	SRC00066/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 69	SRC00065/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 70	SRC00067/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 71	SRC00066/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 72	SRC00065/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 73	SRC00067/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 74	SRC00066/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 75	SRC00067/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 76	SRC00067/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003 77	SRC00067/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 78	SRC00068/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 79	SRC00070/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 80	SRC00069/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 81	SRC00068/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 82	SRC00070/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 83	SRC00069/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000384	SRC00068/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000385	SRC00070/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000386	SRC00069/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000387	SRC00070/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000388	SRC00070/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00070/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003390	SRC00071/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003391	SRC00073/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003392	SRC00072/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003393	SRC00071/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003394	SRC00073/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003395	SRC00072/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 96	SRC00071/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 97	SRC00073/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 98	SRC00072/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 99	SRC00071/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00073/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00072/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000402	SRC00071/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000403	SRC00073/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000404	SRC00072/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000405	SRC00071/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000406	SRC00073/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00072/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 08	SRC00071/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 09	SRC00073/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/040_UE20CV101_UNIT-3__ONLINE_SESSION-_8__215a671018614683.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 10	SRC00072/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 11	SRC00074/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 12	SRC00076/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 13	SRC00075/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000414	SRC00074/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00076/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00075/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000417	SRC00074/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000418	SRC00076/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00075/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 20	SRC00074/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 21	SRC00076/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 22	SRC00075/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 23	SRC00074/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 24	SRC00076/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 25	SRC00075/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 26	SRC00074/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 27	SRC00076/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 28	SRC00075/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 29	SRC00074/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 30	SRC00076/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE20CV101_UNIT-3__ONLINE_SESSION-_9__1c0ade18519c2896.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 31	SRC00075/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000432	SRC00077/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00079/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00078/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000435	SRC00077/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000436	SRC00079/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00078/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000438	SRC00077/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00079/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00078/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00077/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00079/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00078/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000444	SRC00077/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00079/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00078/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000447	SRC00077/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00079/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00078/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 50	SRC00077/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 51	SRC00079/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/041_UE26ME142A_UNIT-3__ONLINE_SESSION-_41__78ba18f903784e2b.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 52	SRC00078/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 53	SRC00080/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 54	SRC00082/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 55	SRC00081/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000456	SRC00080/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00082/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00081/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000459	SRC00080/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000460	SRC00082/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000461	SRC00081/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000462	SRC00080/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00082/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00081/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000465	SRC00082/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000466	SRC00082/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00082/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000468	SRC00083/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000469	SRC00085/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00084/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000471	SRC00083/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000472	SRC00085/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00084/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 74	SRC00083/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 75	SRC00085/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 76	SRC00084/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 77	SRC00083/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 78	SRC00085/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 79	SRC00084/p4	covered	Alternate extracted slide text page 4; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 80	SRC00085/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 81	SRC00085/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 82	SRC00085/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 83	SRC00085/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 84	SRC00085/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00004 85	SRC00086/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 86	SRC00088/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 87	SRC00087/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00004 88	SRC00086/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 89	SRC00088/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 90	SRC00087/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000491	SRC00086/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000492	SRC00088/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000493	SRC00087/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000494	SRC00086/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000495	SRC00088/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000496	SRC00087/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 97	SRC00086/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 98	SRC00088/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 99	SRC00087/p5	covered	Alternate extracted slide text page 5; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00005 00	SRC00086/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 01	SRC00088/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 02	SRC00087/p6	covered	Alternate extracted slide text page 6; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 03	SRC00088/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 04	SRC00088/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 05	SRC00088/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 06	SRC00088/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 07	SRC00088/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000508	SRC00088/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000509	SRC00089/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000510	SRC00091/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000511	SRC00090/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000512	SRC00089/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000513	SRC00091/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000514	SRC00090/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000515	SRC00089/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000516	SRC00091/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00090/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000518	SRC00089/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000519	SRC00091/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000520	SRC00090/p4	covered	Alternate extracted slide text page 4; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005521	SRC00089/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005522	SRC00091/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005523	SRC00090/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00005524	SRC00089/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005525	SRC00091/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005526	SRC00090/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 27	SRC00091/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 28	SRC00091/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 29	SRC00091/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 30	SRC00091/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/Slides/044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.ppt; visual page included as vector figure. Human layout/content review is pending.

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